

MARS RC Robot System

Operation, Safety, Service, and Maintenance Manual

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Start here

1.1 Before first use

This machine can move unexpectedly and uses hazardous UV-C light. A trained adult must supervise it. Keep the physical emergency stop within reach. UV-C can injure eyes and skin; operate the lamps only after establishing and controlling a field exclusion zone, posting warnings, confirming remote supervision, and verifying every installed safety device.

1.1.1 What a non-technical operator must know

- **Hold** means the robot is commanded not to move.
- **Manual** means a handheld joystick controls movement.
- **Auto** means the navigation controller follows an uploaded route.
- **RTK** is the precise GPS state needed for accurate navigation.
- **Arm** prepares UVC control; it must not turn the lamps on.
- **All Off** and **Disarm** are required before anyone enters the UV-C area.

1.1.2 First-use checklist

1. Read every component chapter before operating the system.
2. Identify the physical emergency stop and test it with UVC power disconnected.
3. Attach every radio and GPS antenna before switching equipment on.
4. Test driving with the wheels raised or the robot otherwise restrained.
5. Test the UVC relay with safe indicator loads before connecting UV-C lamps.
6. Run the first automatic mission in a large empty area at low speed.

1.2 Robot system overview

The handheld controller, mobile robot, RTK GPS, navigation controller, power hub, UVC relay, inverters, ballasts, and lamps work together as shown in Figure 1.1.

Component

1.2.1 RC controllers

Handheld manual control, robot status, and Dual Joystick Remote Controller RTK-correction reception.

Component

1.2.2 Mobile robot and UVC

Drive, steering, batteries, CAN relay, inverters, ballasts, and four lamps.

Component

1.2.3 RTK GPS

Fixed base plus rover receiver on the robot.

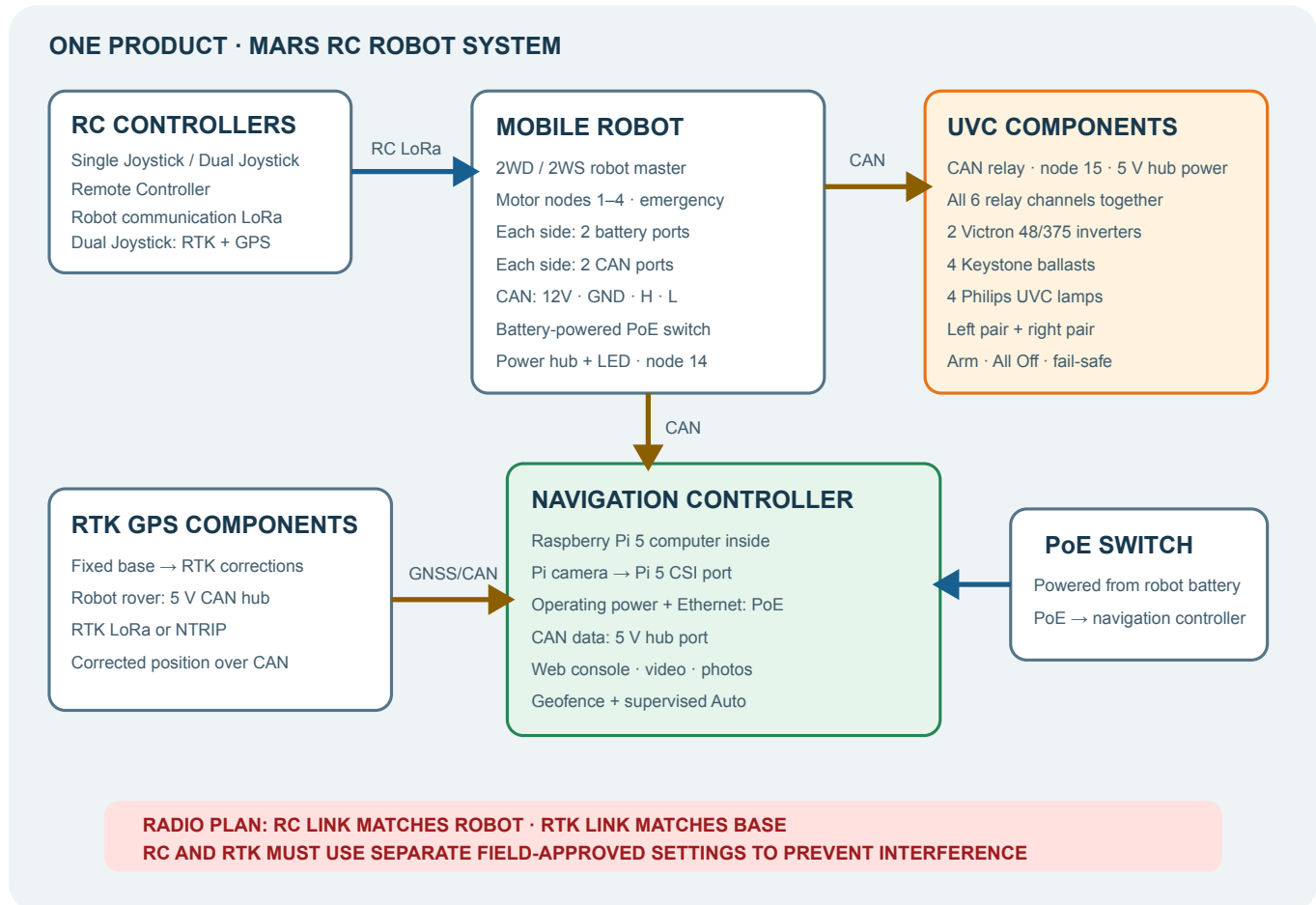


Figure 1.1: The controller commands the robot. The GPS base corrects the robot rover. The navigation controller uses corrected position and its camera for supervised automatic navigation. The robot master controls motion and all six UVC relay channels.

Component

1.2.4 Navigation controller

Raspberry Pi 5 computer, CSI-connected Pi camera, PoE power module, web console, and route planning.

1.2.5 Robot views

Figures 1.2–1.4 identify the robot orientation, major external components, and service compartments from the rear, front, and side.

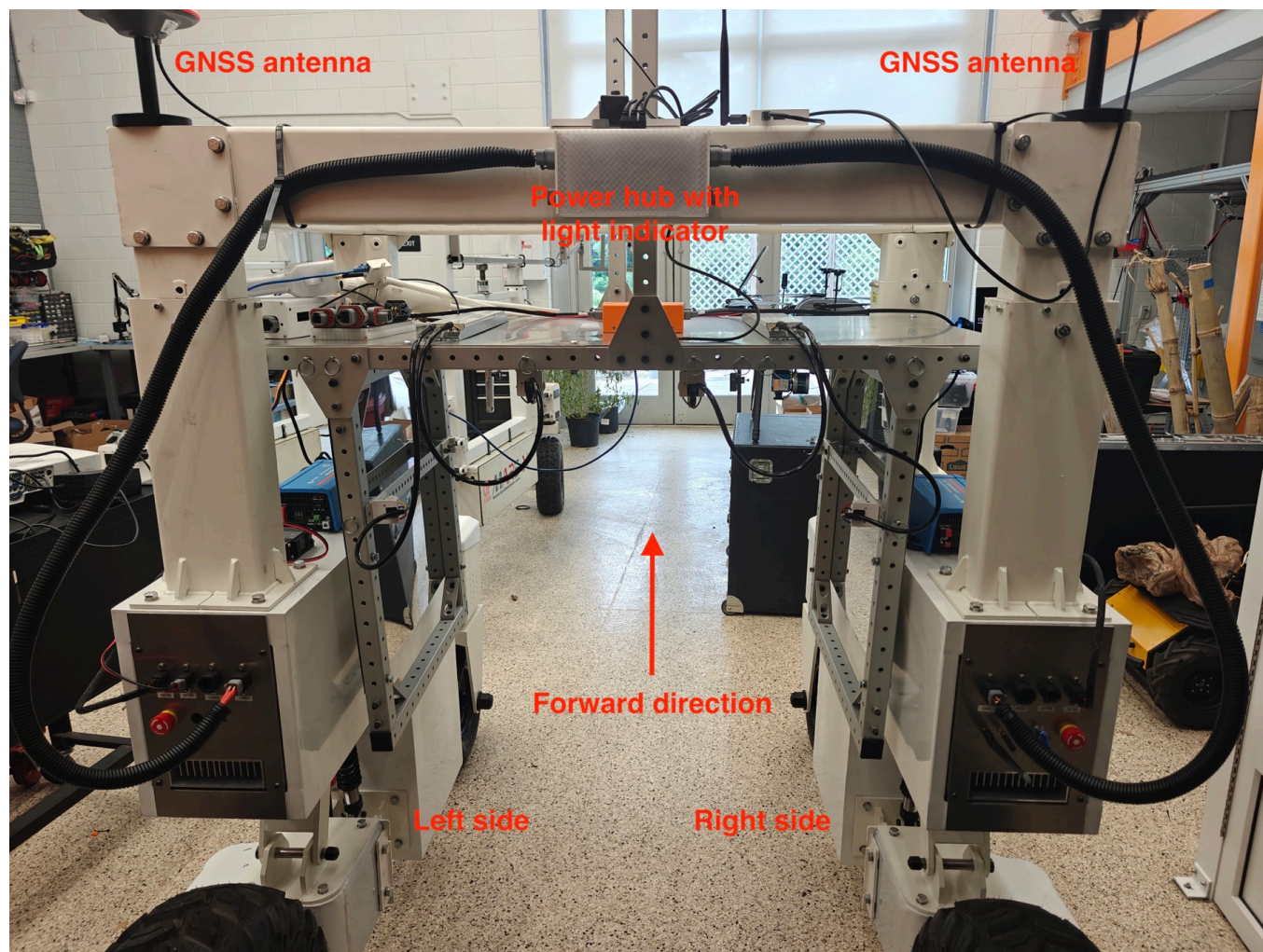


Figure 1.2: Rear view: the labels identify the left and right sides, forward direction, two GNSS antennas, and the power hub with its light indicator.

1.2.6 Inside the controller compartments

Figures 1.5 and 1.6 identify the control, switching, and high-current components inside the controller compartments.

Controller compartments contain high-current electrical equipment. Operators should keep the covers closed. Qualified service personnel must switch the robot off, isolate both batteries, verify the contactor is open, and follow the site lockout procedure before opening or touching internal components.

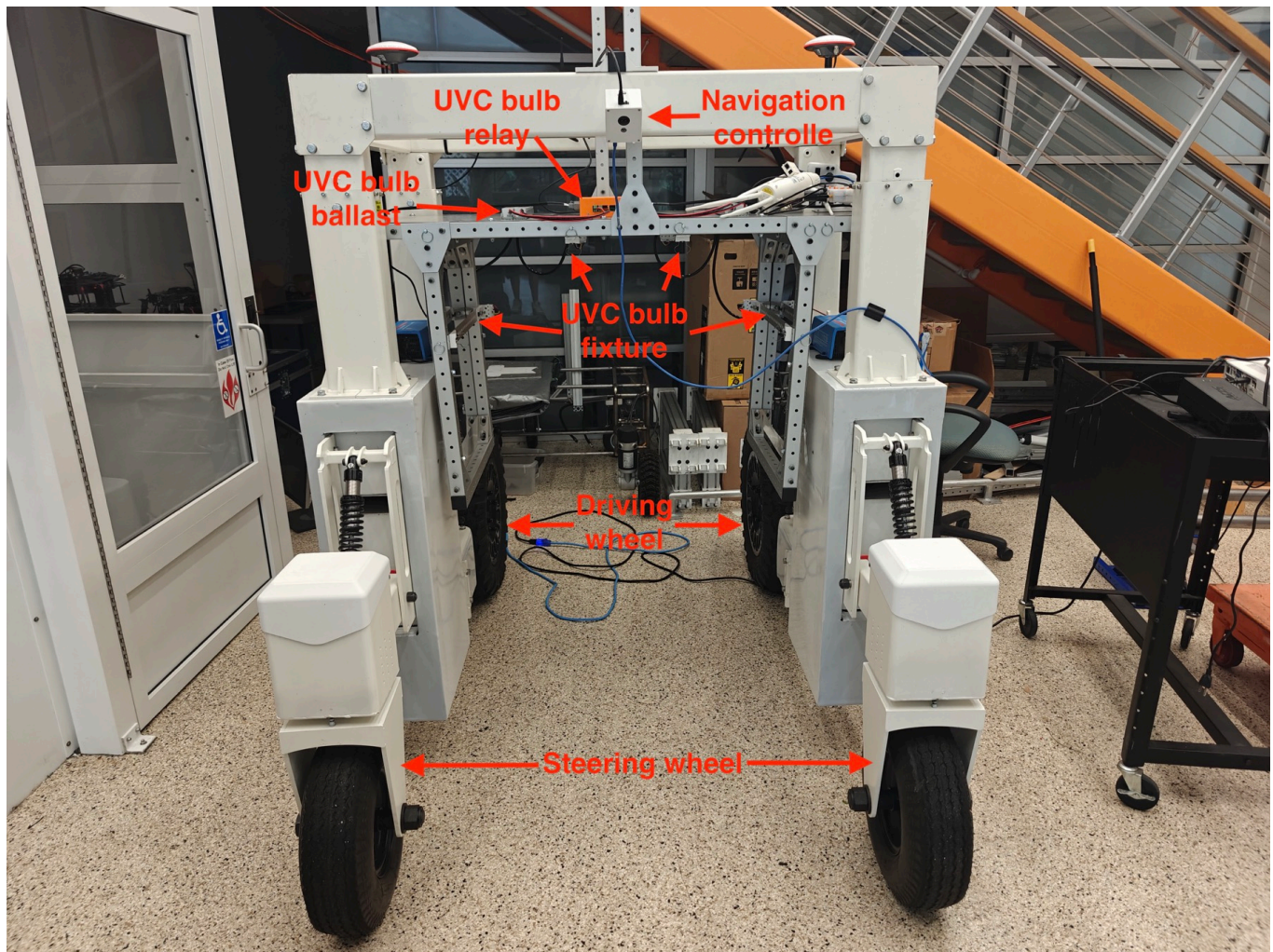


Figure 1.3: Front view: the steering wheels are at the front; the labeled equipment includes the driving wheels, UVC fixtures, UVC ballast and relay, and navigation controller.



Figure 1.4: Side view: the labeled controller and battery compartments are positioned between the steering wheel and the larger driving wheel.

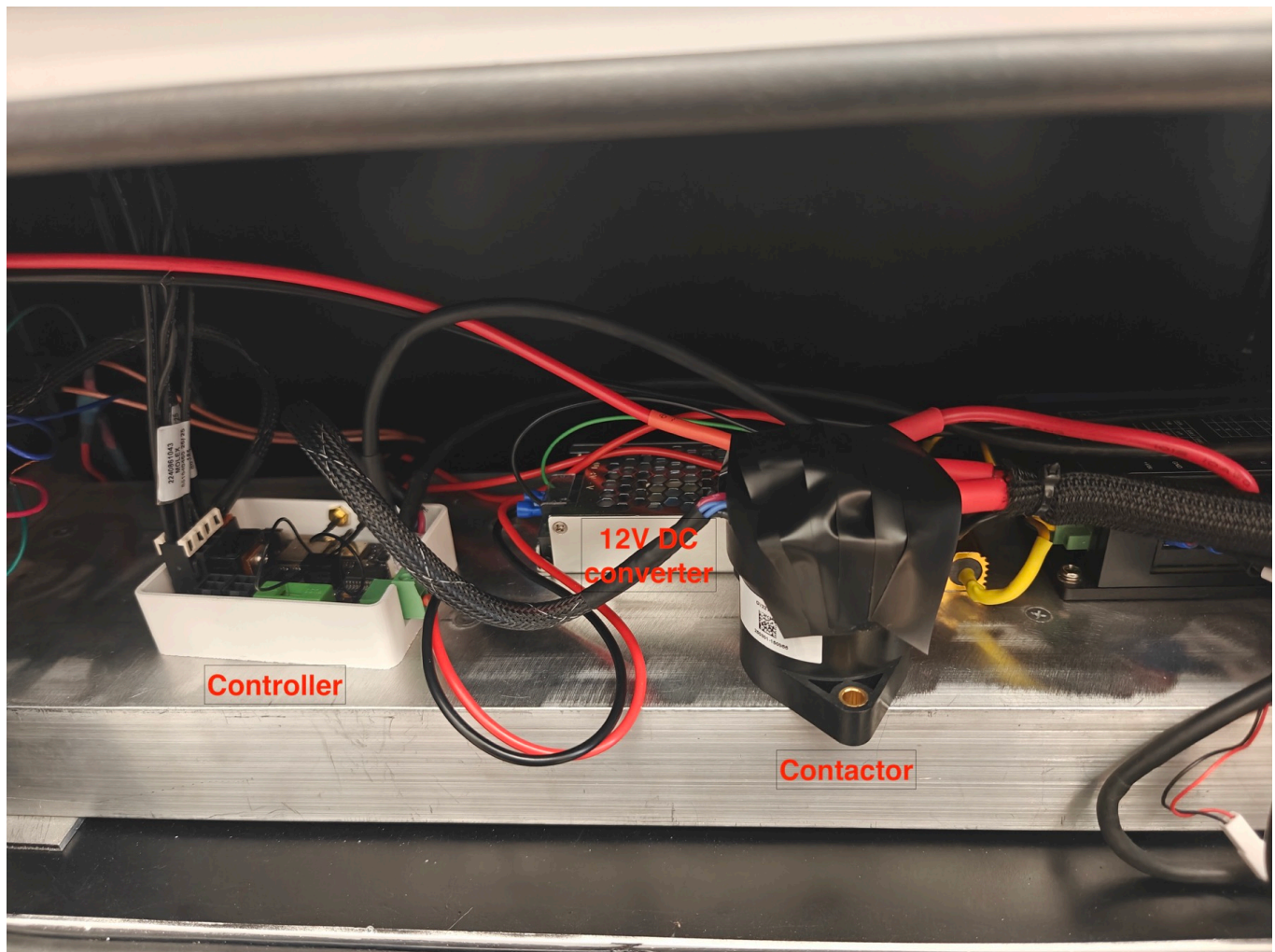


Figure 1.5: Control and switching components inside the compartment: robot controller, 12 V DC converter, main contactor, and associated wiring.

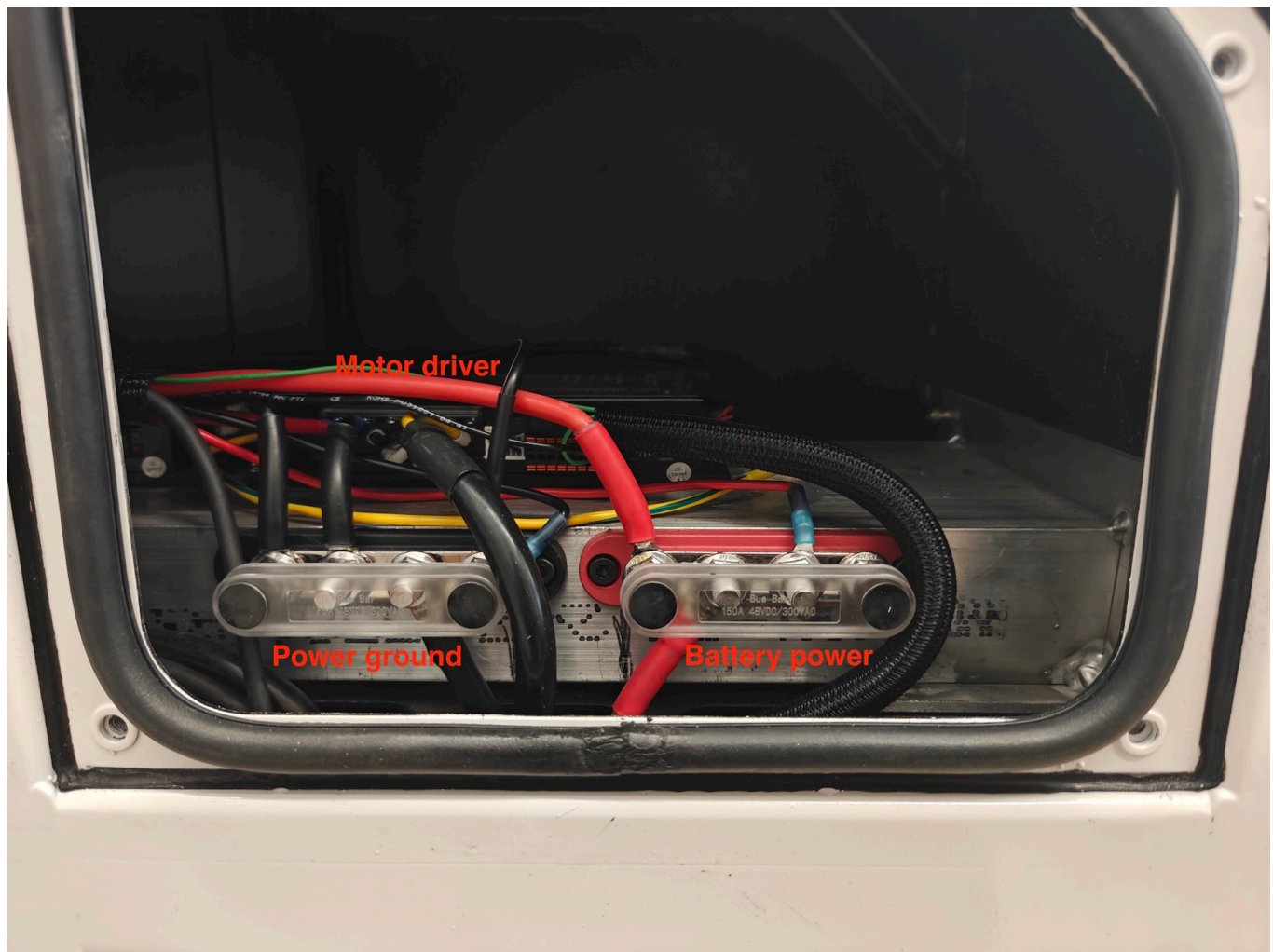


Figure 1.6: Motor-power components inside the compartment: motor driver, battery-power busbar, power-ground busbar, and high-current cabling.

1.3 Robot system diagrams

This page is the visual index for the system. Each caption links to the detailed instructions kept with the related component.

1.3.1 1 · System architecture

Figure 1.7 summarizes the information and power relationships among the major robot-system components.

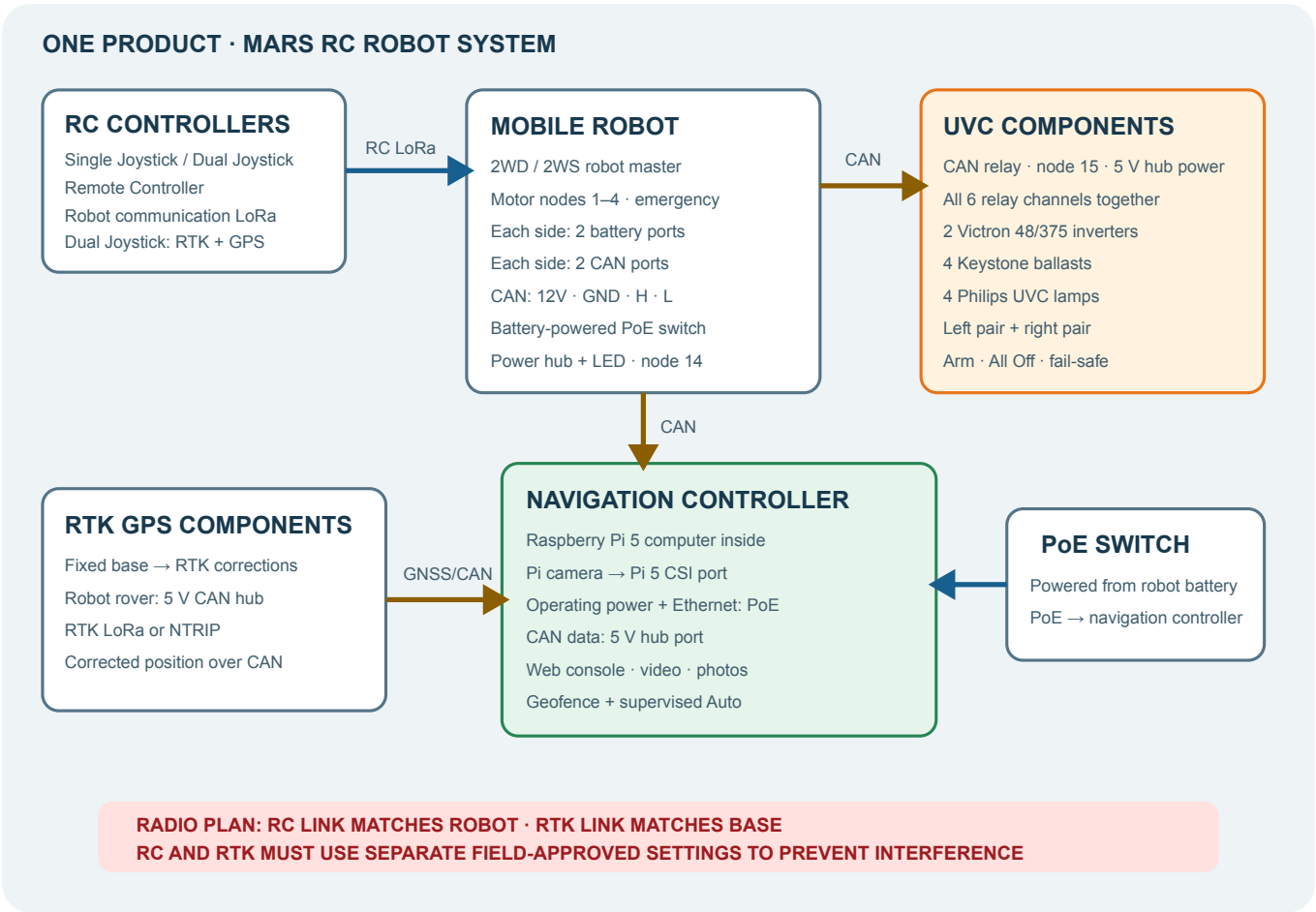


Figure 1.7: RC controllers, mobile robot/UVC, RTK base and rover, and navigation controller. See system overview and complete-system operations.

1.3.2 2 · Physical power, network, radio and CAN connections

Figure 1.8 traces the installed power, network, radio, and CAN connections.

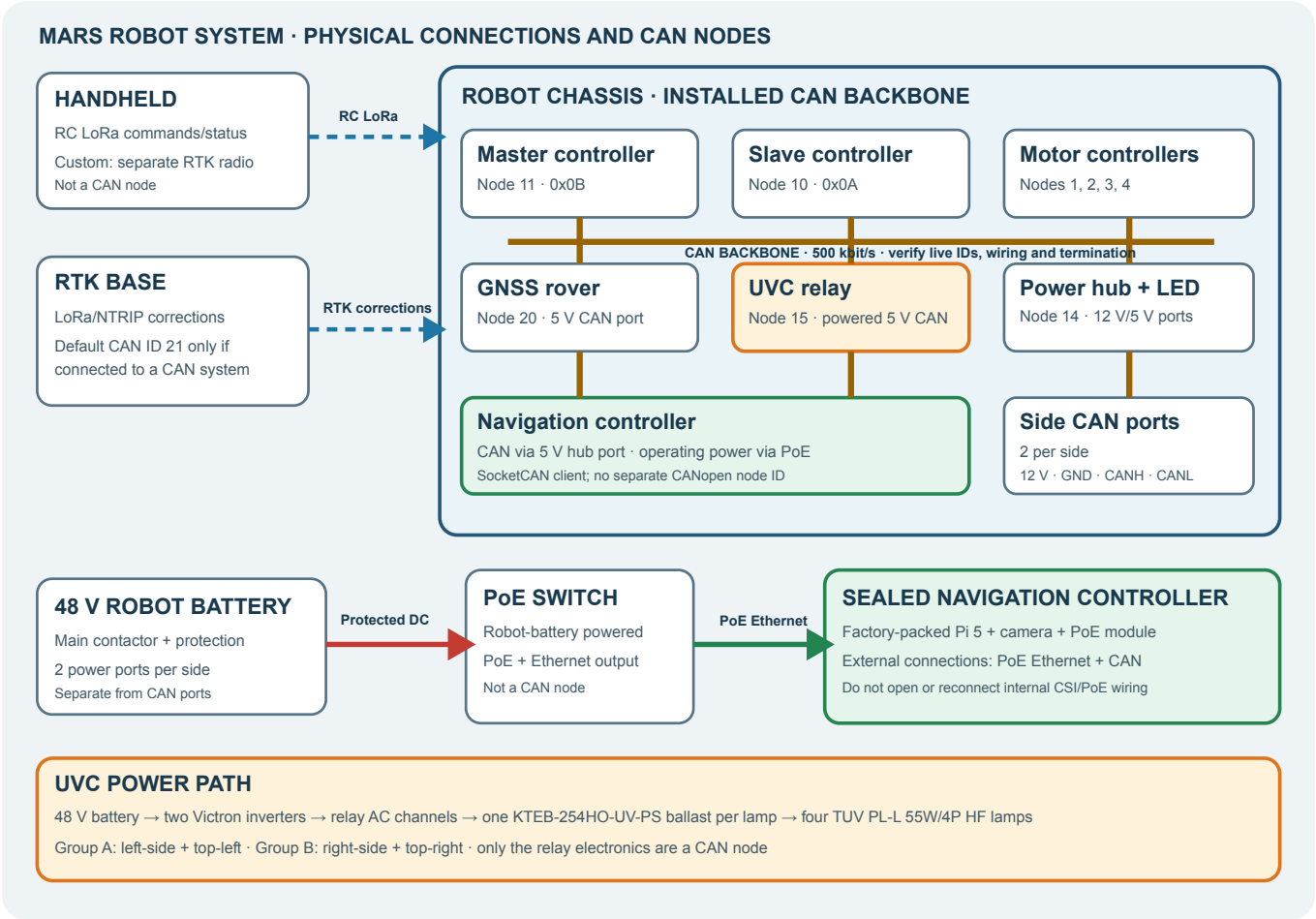


Figure 1.8: Shows the robot chassis, four motors, cross-chassis power hub, PoE path, RTK/RC radios and UVC AC path. See physical connection schedule and robot side ports.

1.3.3 3 · Power-hub ports and device power paths

Figure 1.9 identifies the power-hub ports and the devices powered through each group.

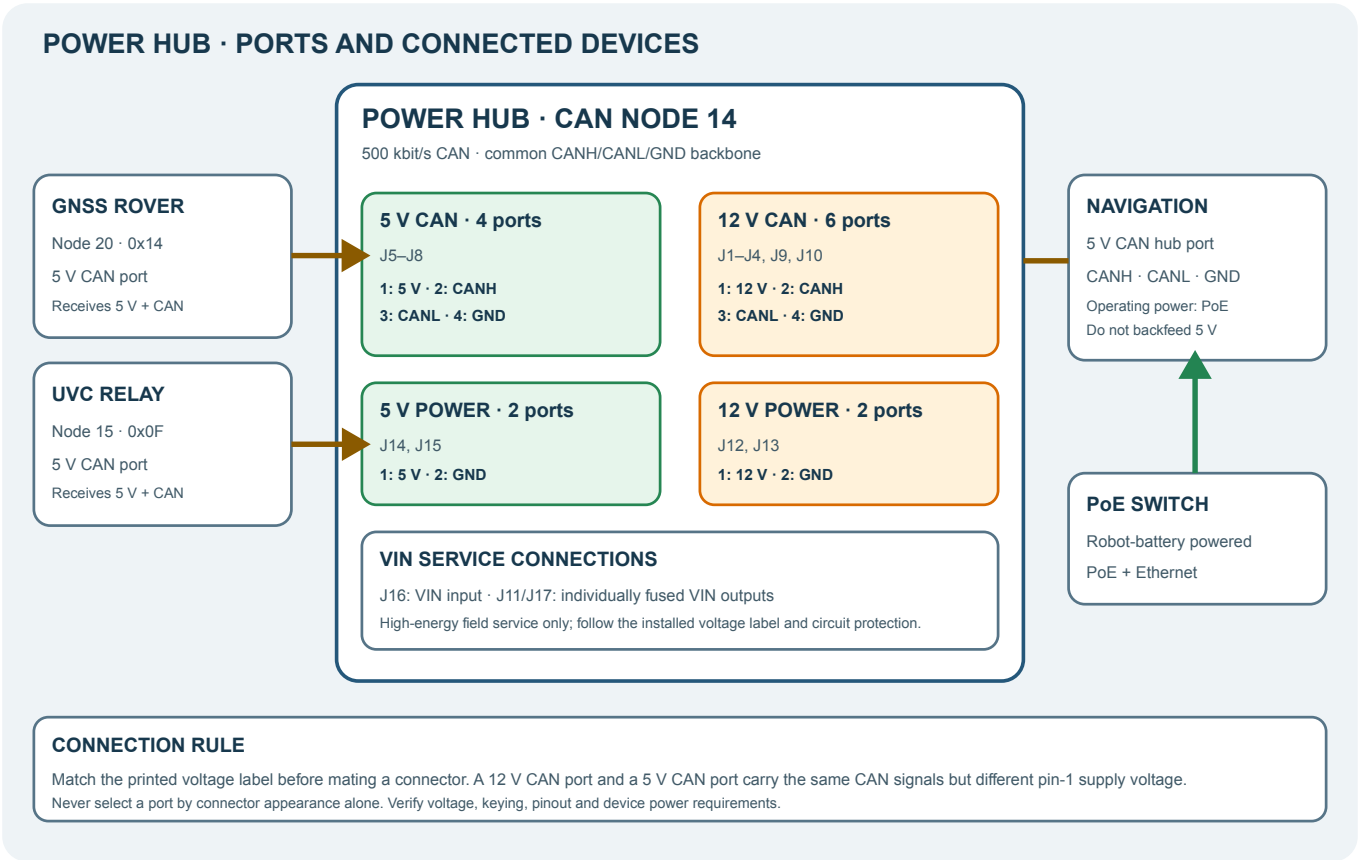
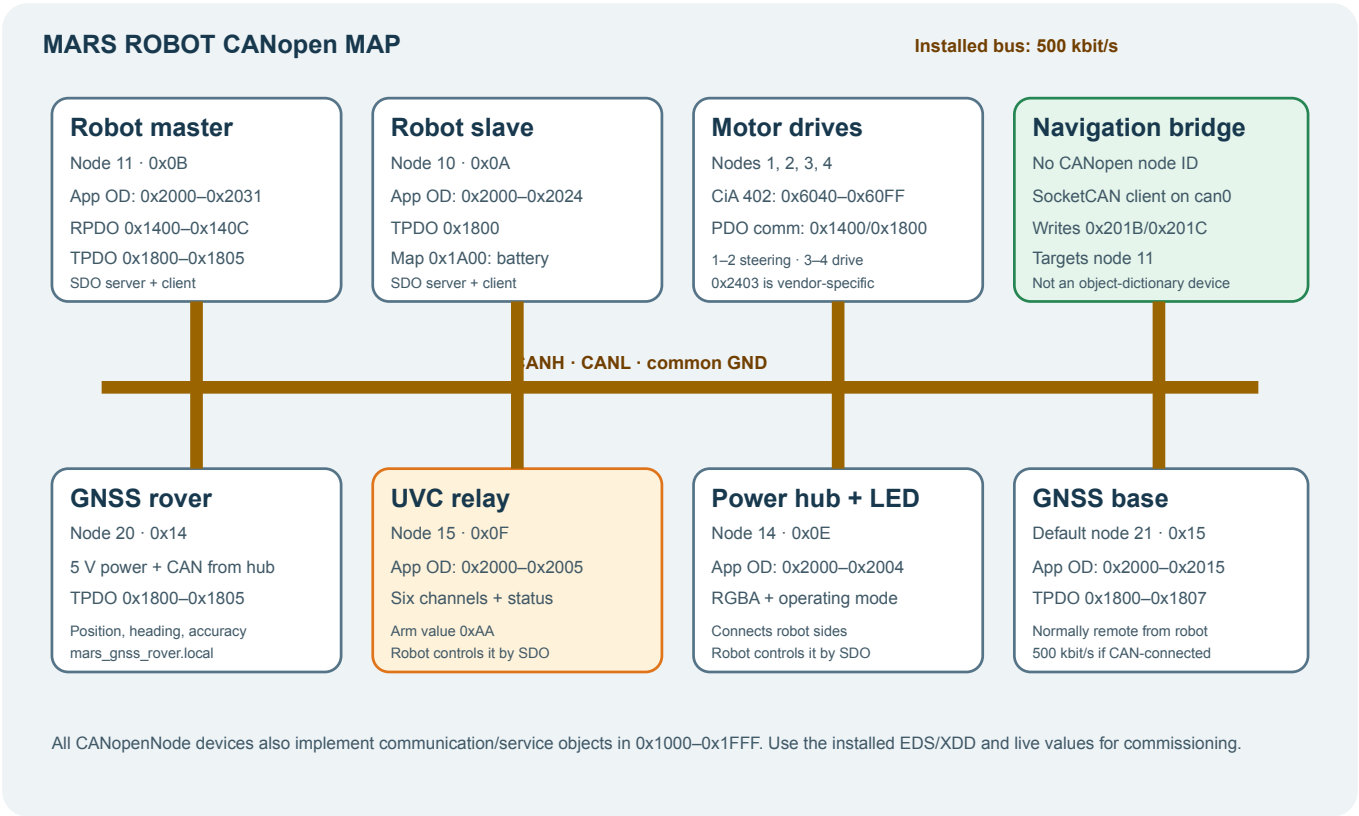


Figure 1.9: The GNSS rover and UVC relay receive 5 V and CAN from the hub. The navigation controller uses a 5 V CAN hub port for communication but receives operating power through PoE. See the power-hub port schedule, GPS connections, and navigation connections.

1.3.4 4 · CANopen topology and object ranges

Figure 1.10 shows the installed CANopen nodes and their principal object ranges.



1.3.5 5 · Power-on state flow

Figure 1.11 shows the expected startup sequence and the corresponding power-hub LED states.

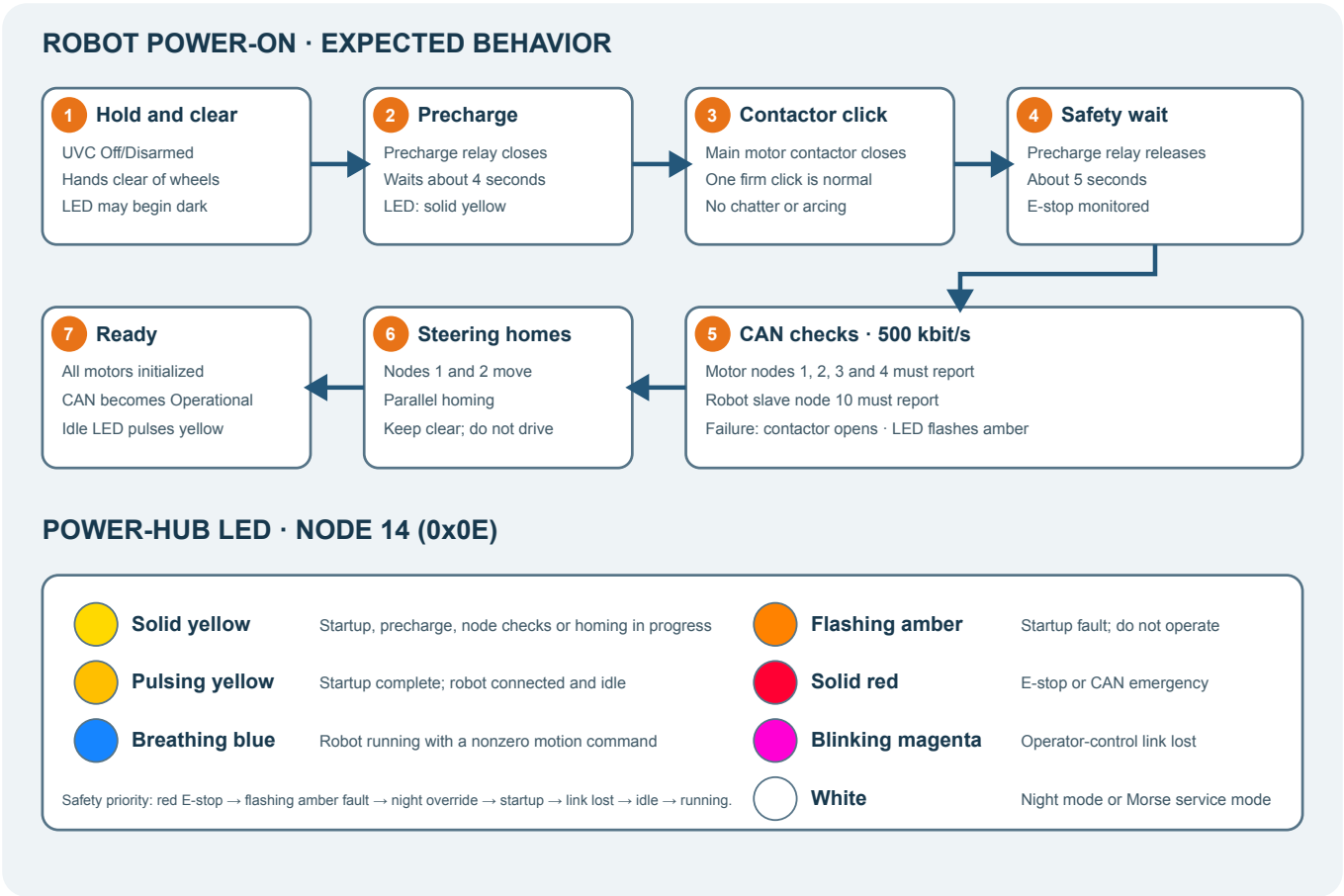


Figure 1.11: Expected precharge, contactor sound, CAN checks, steering homing, ready state and LED changes. See power-on behavior and LED meanings.

1.4 How the complete robot system is connected

This section is the connection overview for the whole product. The blue dashed paths are radios, green is Ethernet/PoE, red is switched battery power, orange is UVC AC power, and brown is CAN. A CAN node ID identifies electronics on the bus; it is not a connector number.

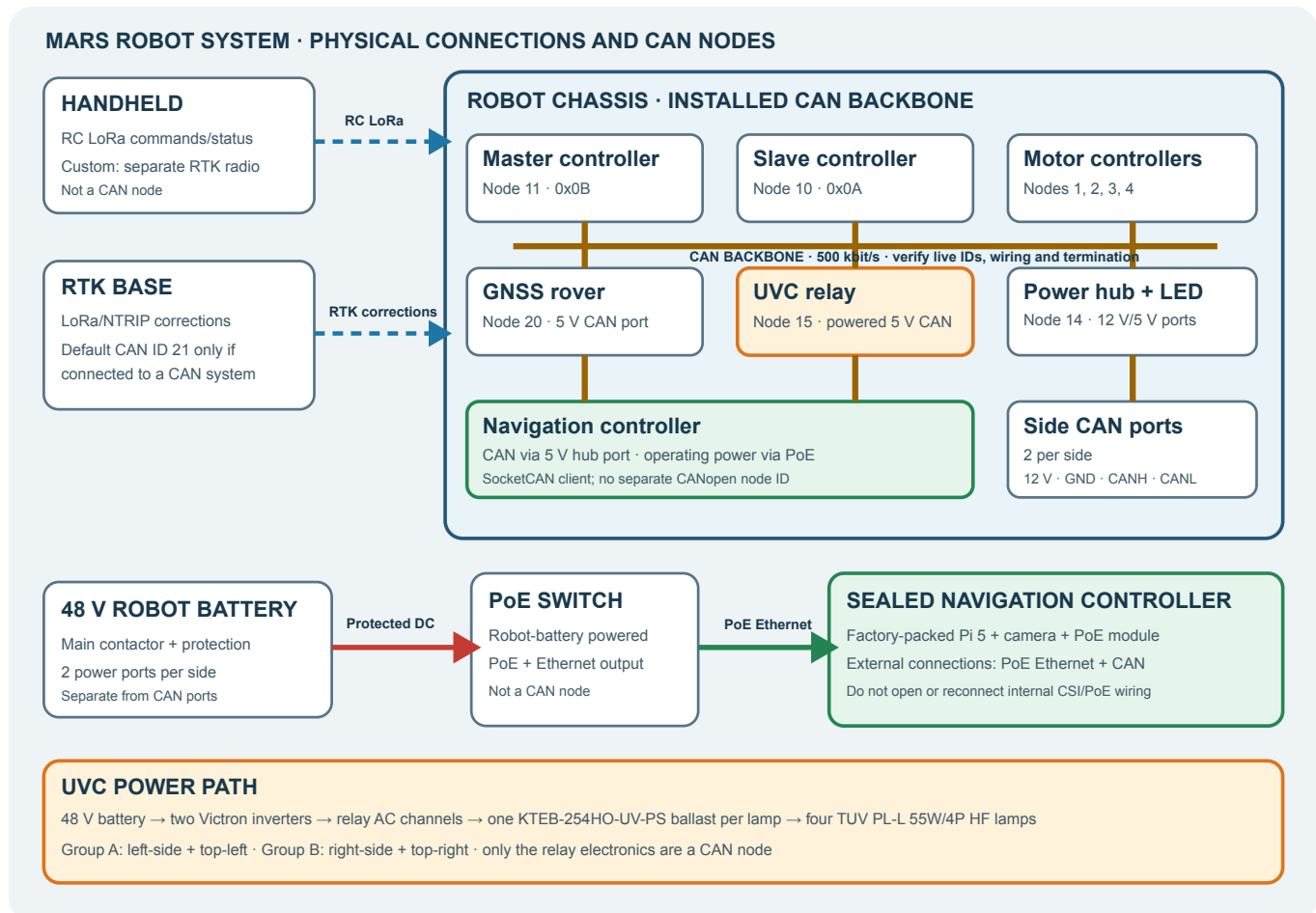


Figure 1.12: The installed motor nodes are 1–4, the power hub/status LED is node 14, the GNSS rover uses a powered 5 V CAN port, and the navigation controller has separate CAN and PoE connections.

Connection rule: keep power Off and the contactor open while making connections. Match the printed port label and voltage. Power-hub CAN ports may place either **12 V or 5 V on pin 1**; both use pin 2 CANH, pin 3 CANL and pin 4 GND. They are not interchangeable.

1.4.1 Physical connection schedule

From	To	Physical link	Purpose and check
Single Joystick Remote Controller or Dual Joystick Remote Controller	Robot master receiver	RC LoRa radio	Manual commands and robot telemetry use MAVLink 2, with one complete MAVLink frame in each RC LoRa packet. The RC settings must match the robot.
RTK base	Robot rover; Dual Joystick Remote Controller RTK receiver when used	RTK LoRa radio	Correction data. It must match the base and must not reuse the RC LoRa settings.

From	To	Physical link	Purpose and check
Robot CAN backbone	Master controller, slave controller, motor nodes 1–4, GNSS rover, UVC relay, power hub/LED, and navigation CAN interface	CANH, CANL and common GND; hub ports provide labelled 12 V or 5 V	Commands, status, position, relay control, heartbeats, and faults. See the power-hub port schedule.
Each robot side	External CAN accessories	Two CAN ports per side: 12 V, GND, CANH, CANL	The two ports are bus connection points, not four separate CAN node IDs. Use short stubs and the approved port assignment.
Power-hub 5 V CAN port	GNSS rover	5 V, CANH, CANL and GND	The rover receives both operating power and CAN communication through this connection.
Power-hub 5 V CAN port	UVC relay, node 15	5 V, CANH, CANL and GND	The relay electronics receive 5 V operating power and CAN communication through this connection. Its isolated relay contacts switch the ballast AC inputs; the 5 V supply never powers a lamp or ballast.
Power-hub 5 V CAN port	Navigation controller	Installed 5 V CAN-port connection: pin 1 = 5 V, pin 2 = CANH, pin 3 = CANL, pin 4 = GND	The navigation controller connects to the assigned 5 V CAN port for its robot CAN connection. Its main operating power and Ethernet connection come from the robot PoE switch. Do not modify the installed harness.
Robot battery through protected branch	Robot PoE switch	DC power	Powers the network switch; verify its rated input and branch protection.
Robot PoE switch	Navigation-controller external PoE port	PoE Ethernet cable	Provides the navigation controller's operating power and Ethernet network connection.
Robot battery through contactor	Two battery-power ports on each side	Switched DC	Accessory power. Verify voltage and polarity before attaching equipment.
48 V battery	Two Victron inverters	Protected DC	One inverter supplies each robot side.
Inverters	Relay AC inputs -> four ballasts -> four lamps	120 V AC	Relay node 15 switches each ballast input. One ballast serves one lamp.

1.4.2 CAN node schedule

Component	Decimal	Hex	Segment / role
Steering motor 1	1	0x01	Steering axis; homed during startup.
Steering motor 2	2	0x02	Steering axis; homed during startup.
Drive motor 1	3	0x03	Velocity-controlled drive axis.
Drive motor 2	4	0x04	Velocity-controlled drive axis.
Right/slave robot controller	10	0x0A	Right/slave CAN controller.

Component	Decimal	Hex	Segment / role
Left/master robot controller	11	0x0B	Robot master and target of navigation velocity/status bridge.
Power hub and status LED	14	0x0E	Connects both robot sides and displays robot startup, running, idle, link and safety states.
UVC relay	15	0x0F	Powered from a power-hub 5 V CAN port; controls all six relay channels.
Robot GNSS rover	20	0x14	Corrected position; web-configurable, firmware default shown.
RTK base CAN interface	21	0x15	Base firmware default; normally the field base is remote and is not physically attached to the robot CAN bus.
Navigation controller	No separate CANopen node ID		SocketCAN client/bridge on can0; targets robot node 11 and observes node 10.
Handheld controllers, PoE switch, inverters, ballasts, lamps	Not CAN nodes		Radio, Ethernet/power, or switched AC equipment.

Installed CAN setting: every device on the robot CAN segment must use **500 kbit/s**. Read each configurable device's live node ID, compare it with the as-built record, and verify that every active node is unique. The table documents firmware defaults and the current robot configuration; an authorized node-ID change can make a live value different.

1.4.3 CAN cable and termination checks

1. Open the contactor and isolate UVC AC power.
2. Verify the connector label and pinout before mating it; never probe by wire color alone.
3. Maintain the approved CAN trunk with short stubs. Keep CAN wiring separated from inverter, ballast, motor-power, and lamp wiring.
4. With all CAN power Off, measure approximately 60 Ω between CANH and CANL on a correctly terminated bus. A substantially different reading requires wiring/termination inspection before energizing.
5. Check common ground continuity, CANH/CANL polarity, shield/ground practice, cable strain relief, the power-hub cross-chassis connection, and the two end terminators.
6. Set every attached CAN device to **500 kbit/s**. Energize in Hold. Confirm motor nodes 1–4, slave node 10, master node 11, power hub node 14, relay node 15 and rover node 20 are unique and producing expected heartbeats without bus errors before enabling motion, Auto navigation, or UVC.

1.5 CANopenNode object dictionaries

This is the field-engineer reference for the objects implemented by the installed software release. The robot bus operates at **500 kbit/s**. Index and subindex are hexadecimal; for example, 0x2013:02 means object 0x2013, subindex 2.

Engineering access only: an SDO or PDO write can move the robot, release steering, alter safety timing, or energize all six UVC relay channels. Put the robot in Hold, restrain the wheels, disconnect UVC AC power, preserve the original EDS/XDD and parameter backup, and obtain authorization before writing. Operators should use the normal controller and webpages, not raw CANopen writes.

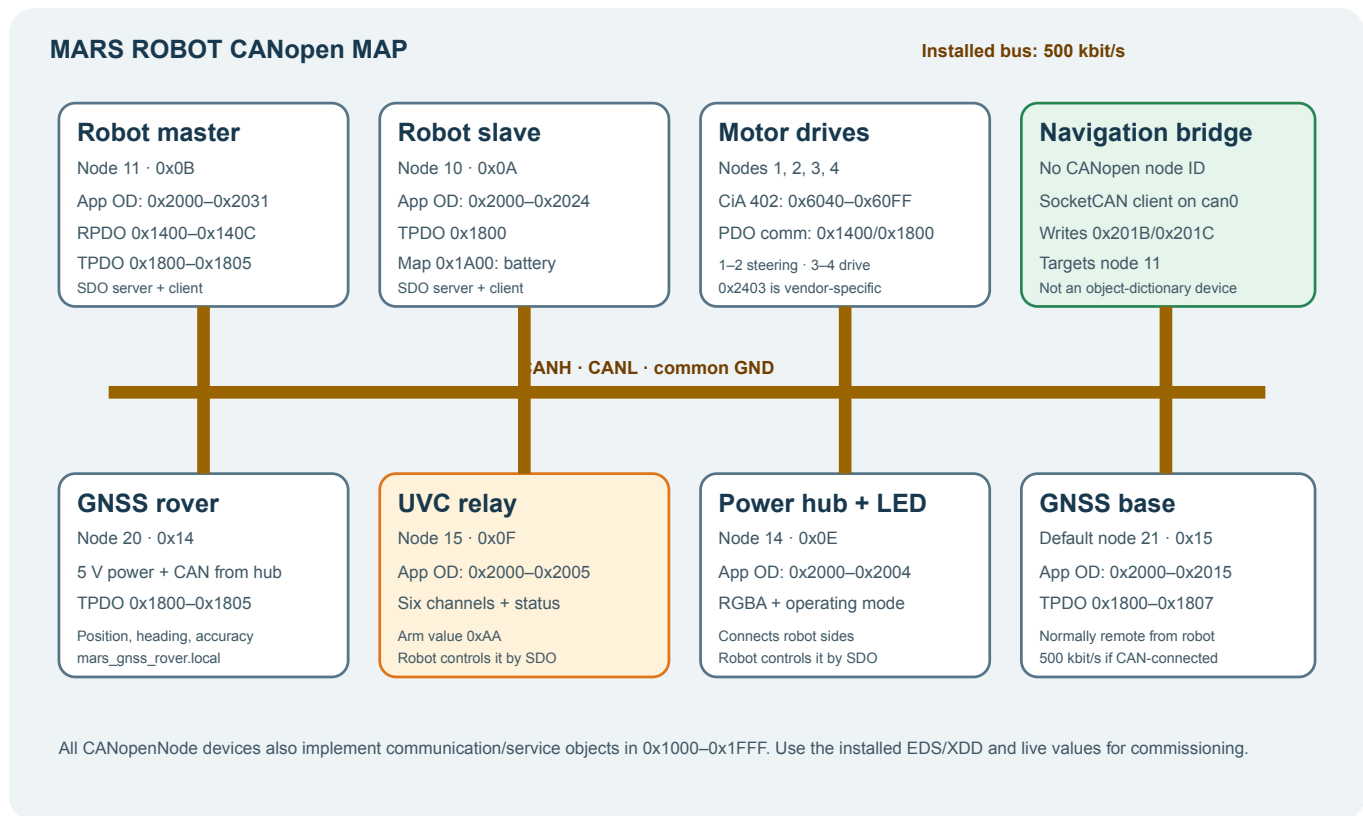


Figure 1.13: Application objects are in the manufacturer-specific 0x2000 range. Motor drives additionally use the CiA 402 drive profile. The navigation controller is a SocketCAN client/bridge and has no object dictionary or CANopen node ID.

1.5.1 Notation and common communication objects

RO = read-only; **RW** = read/write; **WO** = write-only; **TPDO** = transmitted process data; **RPDO** = received process data. For arrays, subindex 00 is the RO element count and data begins at subindex 01.

Index or range	CANopenNode object	Field use
0x1000, 0x1001, 0x1003	Device type, error register, pre-defined error field	Identify the node and diagnose active/history errors. Read before clearing faults.
0x1005–0x1007, 0x1019	SYNC COB-ID, cycle period, synchronous window, counter overflow	Timing configuration. Do not change on one node alone.
0x1008–0x100A	Device name, hardware version, software version	Present in the GNSS dictionaries; use for asset/firmware records.
0x1010, 0x1011	Store parameters, restore defaults	Controlled service commands. Back up first; restart may be required.
0x1012, 0x1014, 0x1015	Time-stamp COB-ID, EMCY COB-ID, EMCY inhibit time	Emergency and time-stamp communication.
0x1016, 0x1017	Consumer and producer heartbeat times	Node-loss supervision. Never lengthen a safety heartbeat without revalidating the system.
0x1018	Identity record	Vendor ID, product code, revision and serial number where populated.
0x1200, 0x1280	SDO server and SDO client parameters	Configuration/diagnostic channel. Normal default SDO COB-IDs are 0x600 + node request and 0x580 + node response.
0x1400..., 0x1600...	RPDO communication and mapping	Defines received real-time data. The robot master implements 0x1400–0x140C/0x1600–0x160C.

Index or range	CANopenNode object	Field use
0x1800..., 0x1A00...	TPDO communication and mapping	Defines transmitted real-time data. Master: six TPDOs; slave: one; rover: six; base: eight.

1.5.2 Robot master node 11 and robot slave node 10

The following application objects are common to both robot-controller dictionaries unless marked otherwise. Arrays 0x2008, 0x2009, 0x200F, 0x2012, 0x2014, and 0x201F–0x2024 use subindices 01–04 for four axes/wheels.

Index	Type / access	Name and engineering meaning
0x2000	uint8 RO	CAN node ID.
0x2001	uint8 RO	CANopenNode bit-rate index. The commissioned physical bus rate is 500 kbit/s.
0x2002, 0x2003	uint16 RW	Maximum linear speed and maximum turning speed.
0x2004, 0x2005	int16 RO · TPDO	Actual linear speed in 0.001 m/s and actual turning speed in 0.001 rad/s.
0x2006, 0x2007	int16 RO · RPDO	Desired linear and turning speed; scales are 0.001 m/s and 0.001 rad/s.
0x2008:01–04	int32 RW · TPDO	Desired motor speed/position for each of four axes.
0x2009:01–04	int32 RW · RPDO	Actual motor speed/position feedback.
0x200A:01–03	Master int32; slave float64 · RW/RPDO	Actual location: latitude, longitude, altitude. Master represents latitude/longitude as degrees $\times 10^7$ and altitude as millimetres. The current slave dictionary uses float64, so never copy raw payloads between the two EDS variants.
0x200B	float32 RW · RPDO	Actual heading.
0x200C	uint8 RO · RPDO	Control mode: 0 Manual, 1 Hold, 2 Auto, 3 Auto Pause.
0x200D, 0x200E	uint16 RW; control word TPDO	Robot status word and control word.
0x200F:01–04	int16 RW · RPDO	Actual motor torque for four axes.
0x2011	uint8 RO	Robot type.
0x2012:01–04	float32 RO · TPDO	Wheel radius for each wheel.
0x2014:01–04	uint16 RW · RPDO	Motor status words.
0x2015, 0x2016	uint16 RO · TPDO	Linear-speed and turning-speed accuracy.
0x2017–0x201A	uint16, int16, uint8, uint16 · RO/TPDO	Battery voltage, current, capacity and temperature.
0x201B, 0x201C	int16 RW · PDO	Automatic-navigation linear and turning commands, 0.001 m/s and 0.001 rad/s. Both must update within the firmware's 500 ms Auto watchdog.
0x201D, 0x201E	uint16 RO	Battery design capacity and controller update rate in milliseconds.
0x201F:01–04	int32 RW · RPDO	Actual motor positions.
0x2020:01–04	float32 RO	Motor-to-wheel ratio.
0x2021:01–04, 0x2022:01–04	float32 RW	Wheel X and Y positions used by robot kinematics.
0x2023:01–04	float32 RW	Gear ratio.

Index	Type / access	Name and engineering meaning
0x2024:01-04	uint8 RW	Installed motor node IDs: 01, 02, 03, 04. Subindices 1–2 are steering motors; 3–4 are drive motors. Treat this commissioned mapping as part of the safety-controlled configuration.

Master-only GNSS and slave-battery objects

Index	Type / access	Meaning
0x2025-0x2028	uint16, int16, uint8, uint16 · RO/RPDO	Slave battery voltage, current, capacity and temperature.
0x2029	int32 RO · RPDO	GNSS heading.
0x202A-0x202C	uint32 RO · RPDO	GNSS horizontal, vertical and heading accuracy.
0x202D	uint16 RO · RPDO	GNSS fix state.
0x202E	uint64 RO · RPDO	GNSS Unix epoch.
0x202F	uint32 RO · RPDO	GNSS time accuracy.
0x2030	uint8 RO · RPDO	GNSS time-valid flags.
0x2031	uint8 RO	Bit mask showing which GNSS TPDO groups have been received.

The slave's TPDO 0x1800, mapped by 0x1A00, carries 0x2017 voltage, 0x2018 current, 0x2019 capacity and 0x201A temperature.

1.5.3 GNSS rover node 20 and GNSS base default node 21

Rover and base firmware use the same application-object layout. The rover is on the robot at `mars_gnss_rover.local`. The base is normally remote and therefore not on the robot CAN trunk; if connected to a CAN test/system segment, configure it for 500 kbit/s.

Index	Type / access	Name, units and use
0x2000	uint16 RW · TPDO	GNSS fix state.
0x2001, 0x2002	int32 RO · TPDO	Latitude and longitude, degrees $\times 10^{-7}$.
0x2003, 0x2004	int32 RO · TPDO	Height above ellipsoid and height above mean sea level, millimetres.
0x2005-0x2008	int16/int32 RO · TPDO	North, east and down velocity and ground speed, millimetres/second.
0x2009	int32 RO · TPDO	Heading of motion. Use the matching XDD/firmware scaling; the current XDD labels it radians $\times 10^{-5}$.
0x200A	int16 RO · TPDO	PDOP $\times 0.01$.
0x200B	int32 RO · TPDO	Vehicle/dual-antenna heading, degrees $\times 10^{-5}$.
0x200C, 0x200D	uint32 RO · TPDO	Horizontal and vertical accuracy, millimetres.
0x200E	uint32 RO · TPDO	Heading accuracy; current XDD scale is radians $\times 10^{-5}$.
0x200F	uint64 RO · TPDO	Unix epoch, milliseconds.
0x2010	uint8 RO · TPDO	GNSS time-valid flags.
0x2011	uint32 RO · TPDO	Time accuracy, microseconds; rover firmware stores receiver <code>tAcc / 1000</code> .
0x2012	uint8 RW	Navigation update rate.

Index	Type / access	Name, units and use
0x2013:01-04	int32 RW	Antenna positions in centimetres: base X, base Y, rover X, rover Y.
0x2014:01-02	uint8 RW	Satellite counts for the two receiver roles.
0x2015	uint8 RO	Rover mode: 0 Dual GNSS, 1 Moving Baseline. Treat the base copy as reserved unless its current firmware explicitly uses it.

GNSS TPDO payload groups

TPDO	Rover default payload	Default period
1	Latitude + longitude	200 ms
2	Ellipsoid height + vehicle heading	200 ms
3	Horizontal + vertical accuracy	200 ms
4	Heading accuracy + fix	200 ms
5	Unix epoch	5 s
6	Time accuracy + time-valid flags	5 s

The base exposes eight TPDO communication/mapping pairs (0x1800-0x1807/0x1A00-0x1A07); use its current XDD to confirm the extra base payload mapping before commissioning.

1.5.4 UVC relay node 15

Index	Type / access	Meaning
0x2000	uint8 RO	Node ID.
0x2001:01-03	boolean RW	Relay Group 1 commands: physical channels 1-3.
0x2002:01-03	boolean RW	Relay Group 2 commands: physical channels 4-6.
0x2003	uint8 RO	Applied relay status bit mask; bit 0 is channel 1 through bit 5 channel 6.
0x2004	uint8 WO	All Off command. Any nonzero write drops all six channels, then self-resets.
0x2005	uint8 RW	Arm interlock: 0xAA = armed; any other value = disarmed.

Installed control path: robot firmware controls relay node 0x0F by SDO. It arms explicitly, commands all six channels together for UVC, reads applied status, and relies on heartbeat/emergency fail-safe behavior. Never bypass the warning confirmation or any installed UVC safety device with an engineering write.

1.5.5 Power-hub and status-LED node 14

To keep all information for this component together, see the power-hub component manual. Its full CANopen table is at Power-hub CANopenNode objects; LED meanings are at Power-hub LED indications.

1.5.6 Installed motor drives: objects used by robot firmware

The installed CANopen motor IDs are **1 and 2 for steering** and **3 and 4 for driving**; the same ordered values are stored at robot OD 0x2024:01-04. The table lists only objects the current master/slave firmware actually accesses. The motor manufacturer's matching EDS and drive manual remain authoritative for the full dictionary, data scaling and limits.

Index	CiA 402 / vendor function	Current firmware use
0x1000:00	Device type	Presence/identity read.
0x1400:01, 0x1800:01	PDO COB-IDs	Configures the command RPDO and feedback TPDO identifiers.
0x6040:00, 0x6041:00	Controlword, statusword	Fault reset, shutdown, switch-on, enable, quick stop and homing-state checks.
0x6060:00	Modes of operation	1 position, 3 velocity, 4 torque; homing mode is selected during steering homing.
0x6071:00, 0x6087:00	Target torque, torque slope	Torque-mode setup.
0x607A:00, 0x6081:00	Target position, profile velocity	Steering position command and profile speed.
0x6080:00, 0x6083:00, 0x6084:00	Maximum speed, acceleration, deceleration	Drive/steering motion limits.
0x60FF:00	Target velocity	Drive-wheel command and explicit zero-speed stop.
0x607C:00, 0x6098:00	Home offset, homing method	Steering homing configuration.
0x6099:01-02, 0x609A:00	Homing seek/zero speeds and acceleration	Steering homing profile.
0x2403:00	Vendor-specific digital-input configuration	Configures the homing input. Do not interpret or change it without the exact drive vendor manual.

1.5.7 Commissioning record

1. Record the firmware folder/version, EDS or XDD revision, physical connector, node ID and 500 kbit/s setting for every installed device.
2. With power isolated, confirm two end terminators and approximately 60 Ω between CANH and CANL.
3. Power in Hold with UVC AC physically disconnected. Confirm NMT state, heartbeat, EMCY state and SDO identity for each node.
4. Read 0x2024:01-04 and confirm the ordered values are 1, 2, 3, 4: steering, steering, drive, drive. Confirm each physical axis label before motion.
5. Compare live PDO communication/mapping objects with the approved EDS/XDD. Do not remap a live system ad hoc.
6. Test read-only feedback first. Then validate controlled writes with wheels restrained and record the result and restored values.

1.6 Robot-mounted UVC lights: PPE and access warning

Normal UVC operation requires an empty treatment/exclusion zone. No operator, bystander, or animal may remain where direct or reflected UV-C can reach them. Wearing PPE does not make normal occupancy acceptable.

1.6.1 Minimum PPE decision

Situation	Required protection
Normal lamp operation	No personnel or animals in the established treatment/exclusion zone. Use signs, cones, flagging, gates or a spotter as the site assessment requires; confirm warning indicators and remote supervision.
Inspection after verified All Off, Disarm, and physical disconnect	Normal mechanical/electrical task PPE plus lockout/tagout according to site procedure.
Authorized energized diagnostic where UV exposure is possible	Only under a written task-specific risk assessment: UV-C-rated goggles with side protection selected for the lamp wavelength; compatible face shield when specified; clothing that fully covers skin; UV-resistant gloves; closed footwear; controlled access and minimum exposure time/distance.

Situation	Required protection
Lamp may generate ozone	Follow the lamp manufacturer's ozone instructions and the site exposure assessment. Do not assume outdoor air movement is sufficient in still air, dense crop canopies, sheds or other partially confined locations.

Not acceptable as the sole protection: ordinary prescription glasses, contact lenses, sunglasses, a clear face shield without rated goggles, sunscreen, thin/unverified clothing, or watching through an unverified window. Confirm every PPE item and viewing material is rated for the installed lamp's wavelength and intensity.

1.6.2 Before anyone approaches the UVC robot

1. Command **All Off** and verify Relay Groups A and B and all four physical lamps are Off.
2. Command **Disarm** and verify Disarmed.
3. Verify the external UVC indicator is Off.
4. Operate the physical UVC disconnect and apply the site lockout/tagout procedure.
5. Wait the manufacturer/site-required time, including ventilation time specified for the installed fixture.
6. Only then enter while wearing the task-required mechanical/electrical PPE.

1.6.3 Possible exposure

Leave the area, shut down from a safe location, and follow the site medical/emergency procedure. UV-C can injure skin and eyes after a short exposure. Eye pain, tearing, a gritty "sand" sensation, light sensitivity, or skin redness/burning requires prompt medical evaluation. Do not return the robot to service until the event is investigated.

Safety basis: FDA ultraviolet radiation guidance and CDC/NIOSH eye-protection guidance. The installed lamp manufacturer's instructions and the site hazard assessment control the exact PPE selection.

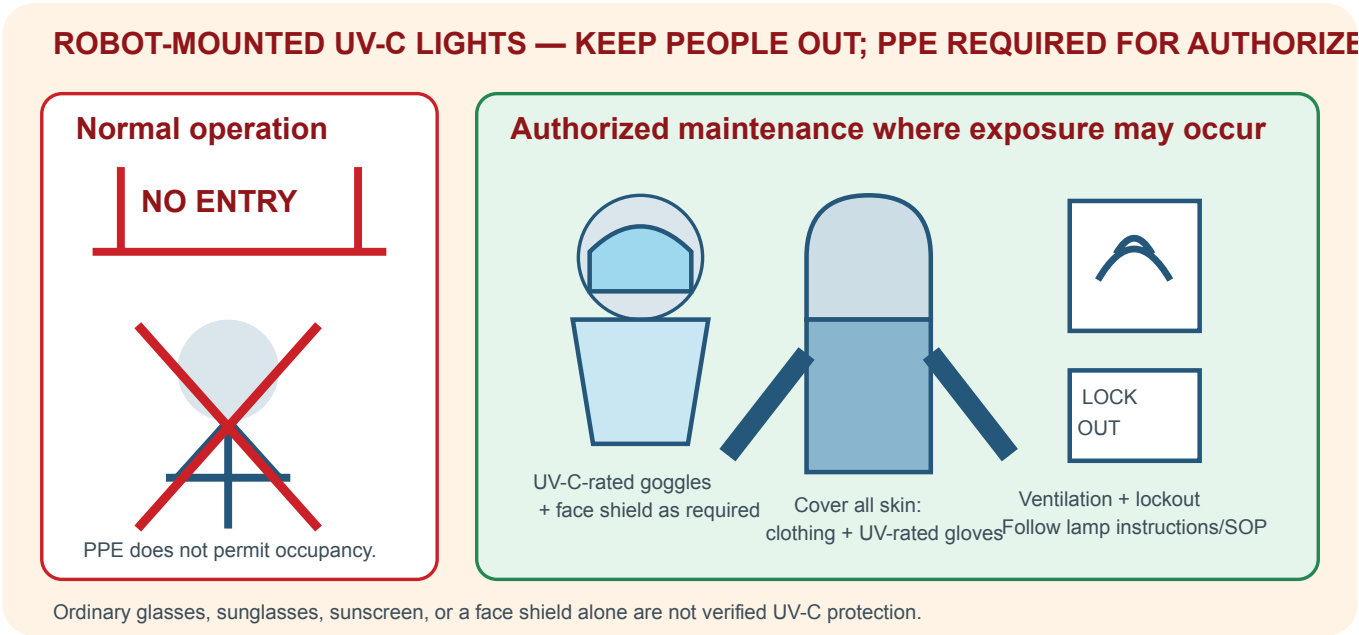


Figure 1.14: Four UVC lamps are mounted on the robot: left side, right side, top-left, and top-right. Establish the field exclusion boundary from a site UV-C exposure assessment that considers measured irradiance, lamp height and orientation, reflections, travel speed, crop rows, slopes and nearby access points. Use remote operation, access control, warnings, distance and any installed guards or automatic safety devices before relying on PPE.

Component · RC controllers

2.1 RC controllers: component overview

Use one handheld controller at a time. Both models send robot commands and receive robot status as MAVLink 2 messages carried by the RC LoRa link. Single Joystick Remote Controller is the compact version; Dual Joystick Remote Controller adds two joysticks, a larger screen, voice alerts, a separate RTK radio, local GNSS, and UVC controls.

2.1.1 Features

Feature	Single Joystick Remote Controller	Dual Joystick Remote Controller
Display	128 × 64 OLED	400 × 300 RLCD
Controls	One joystick, Buttons 1–3, joystick Select	J1/J2 and SW1–SW10
Robot link	One RC LoRa radio	One RC LoRa radio
RTK link	None	Separate RTK LoRa receiver
UVC controls	Not provided	Arm, On/Off, and confirmation warning
Audio	No	Voice warnings and adjustable volume
Setup	OLED menus and Wi-Fi webpage	RLCD menus and Wi-Fi webpage

2.2 Controller specifications

Item	Single Joystick Remote Controller	Dual Joystick Remote Controller
Wi-Fi name	<code>mars_rc_XXXXXX</code>	RC-Controller
Wi-Fi password	<code>alphaagtech</code>	<code>alphaagtech</code>
Web address	<code>http://mars_rc.local</code>	<code>http://rc-controller.local</code>
Fallback address	<code>192.168.10.1</code>	<code>192.168.4.1</code>
Application protocol	MAVLink 2; one complete unsigned frame per RC LoRa packet	
Addressing	Controller system 255 / MAV_COMP_ID_MISSIONPLANNER; robot system 1 / MAV_COMP_ID_AUTOPILOT1	
Default RC radio	915 MHz, 500 kHz bandwidth, SF7, coding rate 4/6, private sync word, 8-symbol preamble and 22 dBm	
Radio choices	903–927 MHz in 2 MHz steps; 125/250/500 kHz; SF7–SF12	
Dual Joystick Remote Controller RTK radio	Not provided	Matches the RTK base and must use field-approved settings that do not overlap the RC link
Dual Joystick Remote Controller antennas	One robot-communication LoRa antenna	Left: RTK LoRa; middle: GNSS; right: robot-communication LoRa
Default low limits	0.5 m/s linear; 0.5 rad/s turning	
Default high limits	1.5 m/s linear; 1.5 rad/s turning	
Startup mode	Hold	

XXXXXX is a unique part of the device identity, so the exact Wi-Fi name varies. The listed AP password is the general robot-device default; the navigation-controller AP has the specific exception documented in its technician-access section. The listed RC radio default is not an instruction to configure RTK the same way.

2.3 RC controller MAVLink communication

MAVLink 2 is the message format used between either remote controller and the robot controller. The controller creates a MAVLink frame, places that complete frame inside one SX1262 RC LoRa packet, transmits it, and briefly listens for one robot reply. MAVLink is therefore the **application protocol carried by the RC LoRa radio**; it is not another radio and it is not the robot’s CANopen bus.

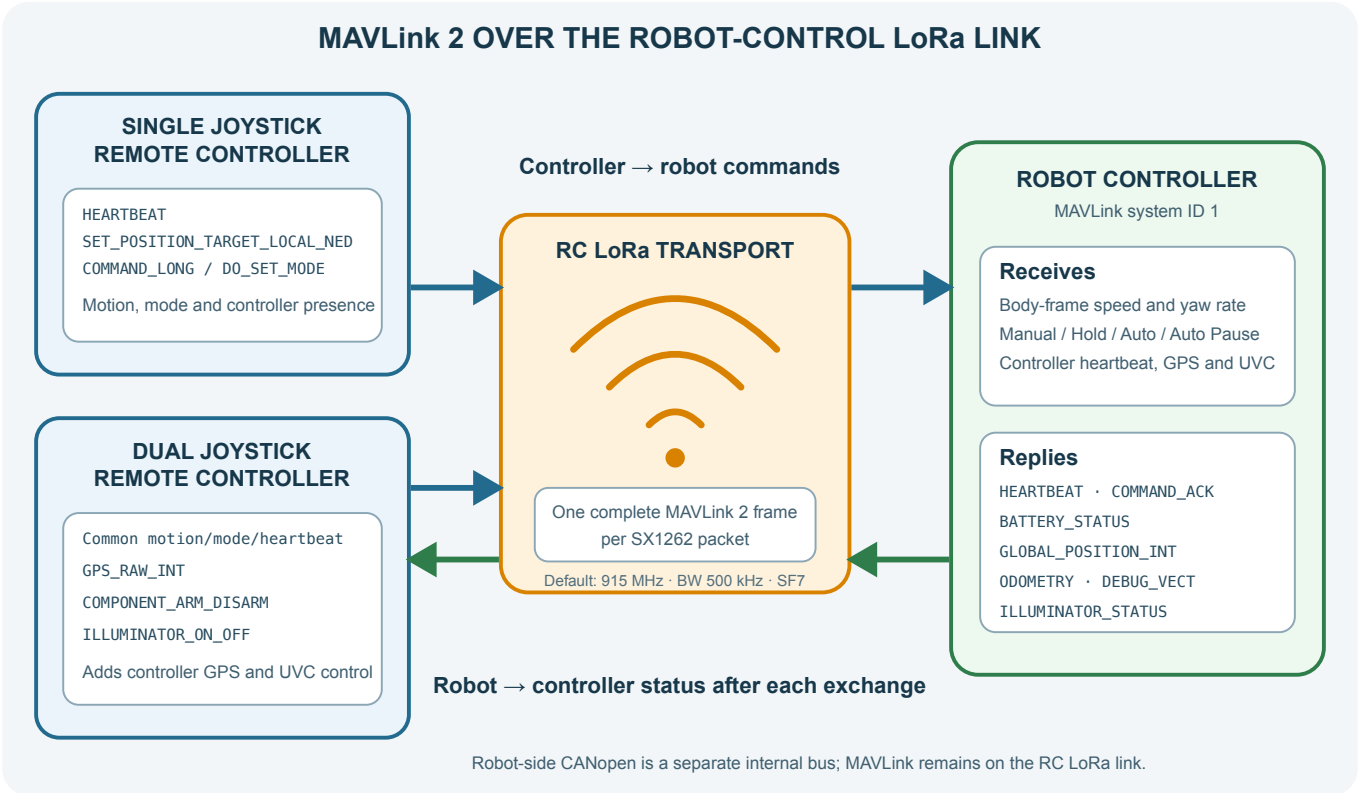


Figure 2.1: Both remote controllers use the common motion, mode and heartbeat messages. The Dual Joystick Remote Controller also sends its valid GPS solution and UVC commands.

Operator meaning: joystick positions, requested mode and UVC selections are encoded as MAVLink commands. Robot mode, motion, batteries, position, emergency state and UVC feedback displayed on the controller come from MAVLink replies. A control press is only a request; the displayed robot feedback is the confirmation.

2.3.1 Protocol and addressing

Item	Installed behavior
Protocol	MAVLink 2 over the robot-control SX1262 LoRa link
Packet rule	One complete MAVLink 2 frame per LoRa packet; maximum radio payload 255 bytes
Remote-controller source	System ID 255 ; component MAV_COMP_ID_MISSIONPLANNER; heartbeat type MAV_TYPE_GCS
Robot target/source	System ID 1 ; component MAV_COMP_ID_AUTOPILOT1
UVC target	Robot system ID 1 ; component MAV_COMP_ID_USER1; used by the Dual Joystick Remote Controller
Frame security	Current frames are unsigned. MAVLink signing, encryption and peer authentication are not provided by this radio profile.

Item	Installed behavior
Protocol separation	RC LoRa carries MAVLink. RTK LoRa carries correction data. The robot's wired 500 kbit/s CAN network uses CANopen. Do not mix their settings or troubleshooting paths.

2.3.2 Controller-to-robot messages

MAVLink message/command	Purpose and fields used	Controller models
SET_POSITION_TARGET_LOCAL_NED	Uses MAV_FRAME_BODY_NED. Only forward velocity v_x in m/s and yaw rate in rad/s are enabled; position, other velocity/acceleration axes and yaw angle are ignored.	Both
COMMAND_LONG with MAV_CMD_DO_SET_MODE	Requests the custom robot mode: Manual = 0, Hold = 1, Auto = 2, Auto Pause = 3. The custom-mode-enabled flag is set.	Both
HEARTBEAT	Identifies an active ground-control-station controller and maintains controller presence.	Both
GPS_RAW_INT	Sends controller latitude/longitude, MSL and ellipsoid altitude, fix class, satellites and available accuracy data only when the onboard GPS fix is valid and no older than 1.5 seconds.	Dual Joystick Remote Controller
MAV_CMD_COMPONENT_ARM_DISARM	Arms or disarms the UVC component; Disarm also forces the requested UVC output Off in the controller.	Dual Joystick Remote Controller
MAV_CMD_ILLUMINATOR_ON_OFF	Requests all installed UVC relay channels On or Off after the controller's UVC safety-confirmation workflow.	Dual Joystick Remote Controller

2.3.3 Robot-to-controller messages

MAVLink message	Controller use
HEARTBEAT	Confirms robot system/component, active mode and emergency state. A matching mode can complete a pending mode request.
COMMAND_ACK	Reports whether a mode or UVC command was accepted, rejected, unsupported, failed or temporarily rejected. Temporarily rejected commands remain pending for retry.
BATTERY_STATUS	Updates installed robot battery state of charge by battery ID; invalid or absent values are not presented as valid capacity.
GLOBAL_POSITION_INT	Updates robot latitude, longitude, altitude, heading-derived forward speed and GPS-valid indication.
ODOMETRY	Updates actual forward velocity and yaw rate.
DEBUG_VECT named ROVER_VEL	Compatibility telemetry for actual linear and turning velocity.
ILLUMINATOR_STATUS	Updates applied UVC output and clears a matching pending UVC On/Off request on the Dual Joystick Remote Controller.
UVC-component HEARTBEAT	Updates the UVC armed state on the Dual Joystick Remote Controller.

2.3.4 Timing and acknowledgement behavior

Behavior	Single Joystick Remote Controller	Dual Joystick Remote Controller
Radio exchange cycle	10 Hz (approximately every 100 ms)	10 Hz (approximately every 100 ms)
Reply window after transmit	Up to 80 ms	Up to 60 ms
Command retry interval	300 ms for a pending mode command	300 ms for pending mode or UVC commands
Controller status schedule	Heartbeat every 1 second	One GPS/heartbeat status slot every 1 second; heartbeat at least every 5 seconds
Robot-link timeout	Link Lost after approximately 1 second without a valid robot reply	Link Lost after approximately 1 second without a valid robot reply

Communication security: a private LoRa sync word and matching frequency/settings separate compatible radio traffic but do not encrypt or authenticate commands. Treat the RC radio channel as a safety-relevant operational link, keep physical emergency-stop coverage, use the approved radio plan, and do not expose protocol settings publicly.

2.3.5 Field-engineer checks

1. Keep the robot restrained in Hold with UVC electrically isolated before protocol testing.
2. Verify both RC endpoints use the same approved LoRa frequency, bandwidth, spreading factor, coding rate, sync word and preamble. Keep RTK LoRa on its separately approved settings.
3. Capture the controller serial log. Confirm transmitted MAVLink 2 frames begin with the MAVLink 2 start byte 0xFD, fit in one 255-byte LoRa payload, and parse without CRC errors.
4. Verify controller source system ID 255 and robot source/target system ID 1. Confirm robot and UVC component IDs match the installed firmware.
5. For motion, verify body-frame vx and yaw-rate units are m/s and rad/s. A 1000-times scaling error indicates confusion with the internal mm/s or mrad/s values.
6. For a mode or UVC problem, inspect the outgoing COMMAND_LONG, the returned COMMAND_ACK result, and subsequent matching HEARTBEAT or ILLUMINATOR_STATUS.
7. For Dual Joystick Remote Controller GPS, require a valid fix newer than 1.5 seconds. No GPS_RAW_INT is expected for an invalid or stale fix.
8. After service, test Hold, low-speed motion, every mode transition, link-loss timeout, emergency indication, both battery reports, robot position, and—on the Dual Joystick Remote Controller—GPS and UVC acknowledgements.

2.4 Single Joystick Remote Controller operation manual

Never power the controller without its LoRa antenna. Hand-tighten the correct robot-communication LoRa antenna before switching on.

1. **Inspect and connect.** Check the case, joystick, buttons, battery, and antenna socket. Fit the LoRa antenna and let the joystick rest in the center.
2. **Power on.** Keep the robot restrained. Switch on the handheld. Confirm the OLED says **Hold**; Robot Lost is normal until the robot starts.
3. **Wait for the robot.** Switch on the robot. Do not continue until the OLED says Robot OK, RSSI is visible, and telemetry changes.
4. **Select Manual.** Short-press Button 1. Confirm the display changes to Manual. A button press is only a request; the displayed robot mode is the confirmation.
5. **Test direction.** Move the joystick slightly forward, reverse, left, and right. Stop immediately if the robot moves in the wrong direction.
6. **Stop.** Release the joystick and short-press Button 1 for Hold. Confirm CMD is zero and wheels stop.
7. **Use setup webpage when required.** Join mars_rc_XXXXXX, enter alphaagtech, and open 192.168.10.1. Change speed/radio settings only while the robot is in Hold.



Figure 2.2: RobotController physical controls: LoRa antenna, joystick, Button 1, and Button 2.

```

[1/4] HOME
ROBOT OK
ROBOT PWR 82%
RC 4.08V RSSI -67dBm
MODE HOLD
CMD L+0.00 T+0.00
ACT L+0.00 T+0.00

```

Figure 2.3: Single Joystick Remote Controller connections and controls: robot-communication LoRa antenna, joystick, Buttons 1–3, Select, charging/power connection, and OLED.

Single Joystick Remote Controller — match each picture to the numbered instruction

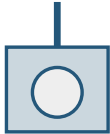
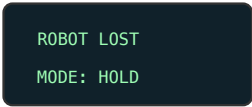
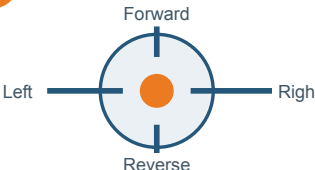
1 Inspect and connect  Antenna fitted Stick centered No damage	2 Power on in Hold 	3 Wait for connection 
4 Select Manual 	5 Test directions slowly 	6 Stop and select Hold 
7 Open Wi-Fi settings when needed  Network: mars_rc_XXXXXX Password: alphaagtech Open: 192.168.10.1		8 Emergency: use physical E-stop  Release the joystick Press robot E-stop Do not approach moving robot

Figure 2.4: Each picture number matches the instruction below.

8. **Emergency.** Release the joystick and use the robot’s physical emergency stop. Do not approach until the robot is confirmed stationary.

2.4.1 Buttons and screens

Control	Result
Button 1 short	Manual/Hold; changing from Auto to Manual or Hold automatically stops active autonomous navigation
Button 1 long	Sleep; Button 1 wakes
Button 2 short	Auto/Auto Pause; during an active mission, selecting Auto Pause pauses autonomous navigation while retaining the mission for Resume; also confirms calibration when prompted
Button 2 held	Enter Auto
Button 3 held	High speed while held
Joystick Select	Home -> Telemetry -> Settings -> Calibration

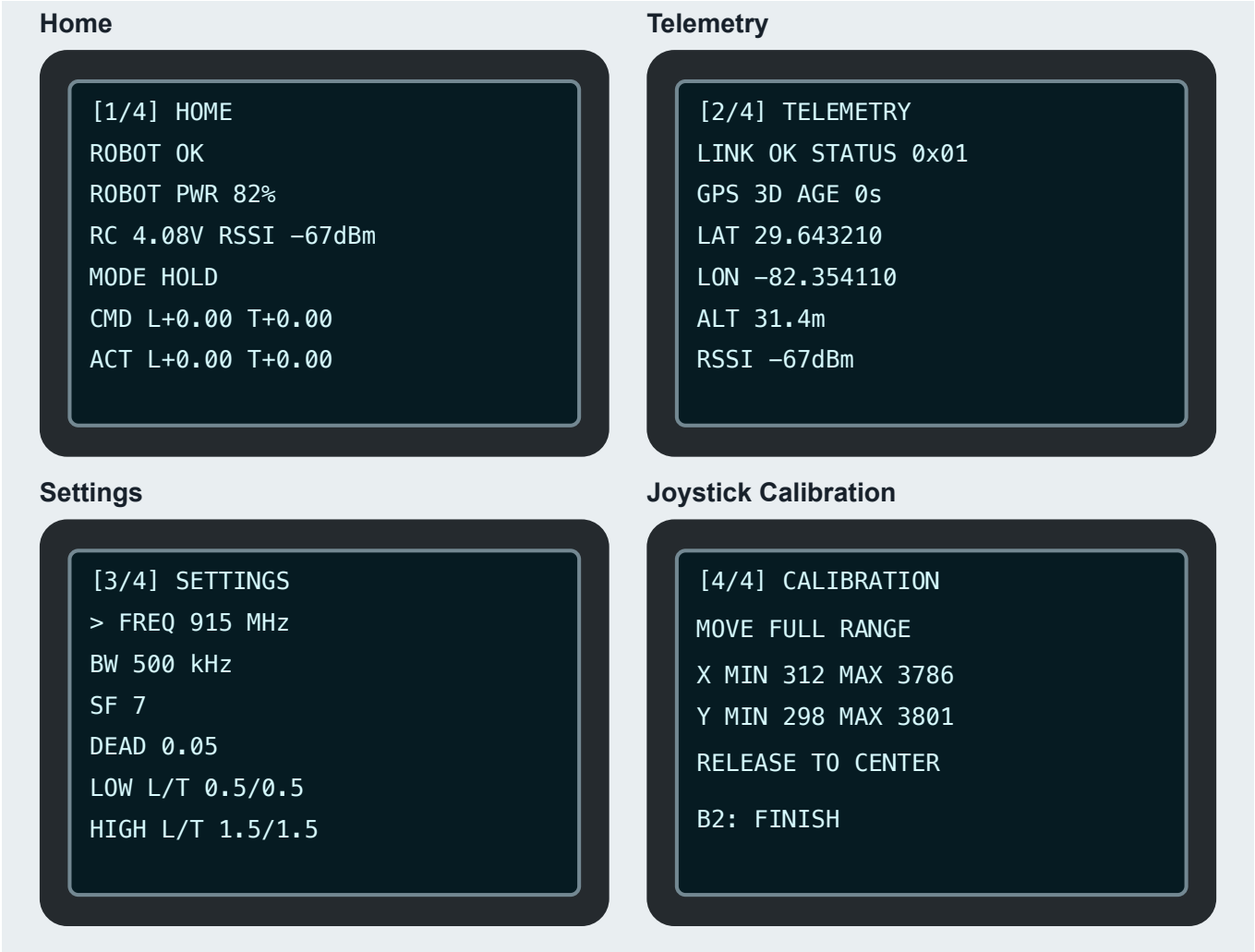


Figure 2.5: Home shows robot/mode/CMD/ACT; Telemetry shows GPS and age; Settings changes radio/speed; Calibration shows joystick progress.

2.4.2 Joystick calibration

1. Restrain the robot and open Calibration with Select; Hold is forced.

2. Let the stick center and short-press Button 2.
3. Move slowly to every endpoint and circle the outside edge several times.
4. Release to natural center and press Button 2 again.
5. Do not touch the stick during center sampling. Wait for Success and verify zero CMD at center.

2.4.3 Wi-Fi settings

1. Join the Single Joystick Remote Controller network and remain connected when the phone reports No Internet.
2. Open the local address. Review battery, GPS, RSSI, and telemetry age.
3. Set slow/high speed values, save, reload, and test while restrained.
4. Record old LoRa values before changing them. Set the robot to the identical frequency, bandwidth, and SF.

2.5 Dual Joystick Remote Controller operation manual



Figure 2.6: Custom RC Controller physical controls: left and right joysticks, Buttons 1–8, center GNSS antenna, and left/right LoRa antennas. Facing the controller, the left antenna is RTK LoRa, the middle antenna is GNSS, and the right antenna is robot-communication LoRa.

Attach all three correct antennas before power-on. Do not swap the RTK LoRa, GNSS, and robot-communication LoRa antennas.

Keep RC LoRa and RTK LoRa on different radio settings. The RTK radio must match the RTK base, while the RC radio must match the robot. Use the approved channel plan and verify both links independently.

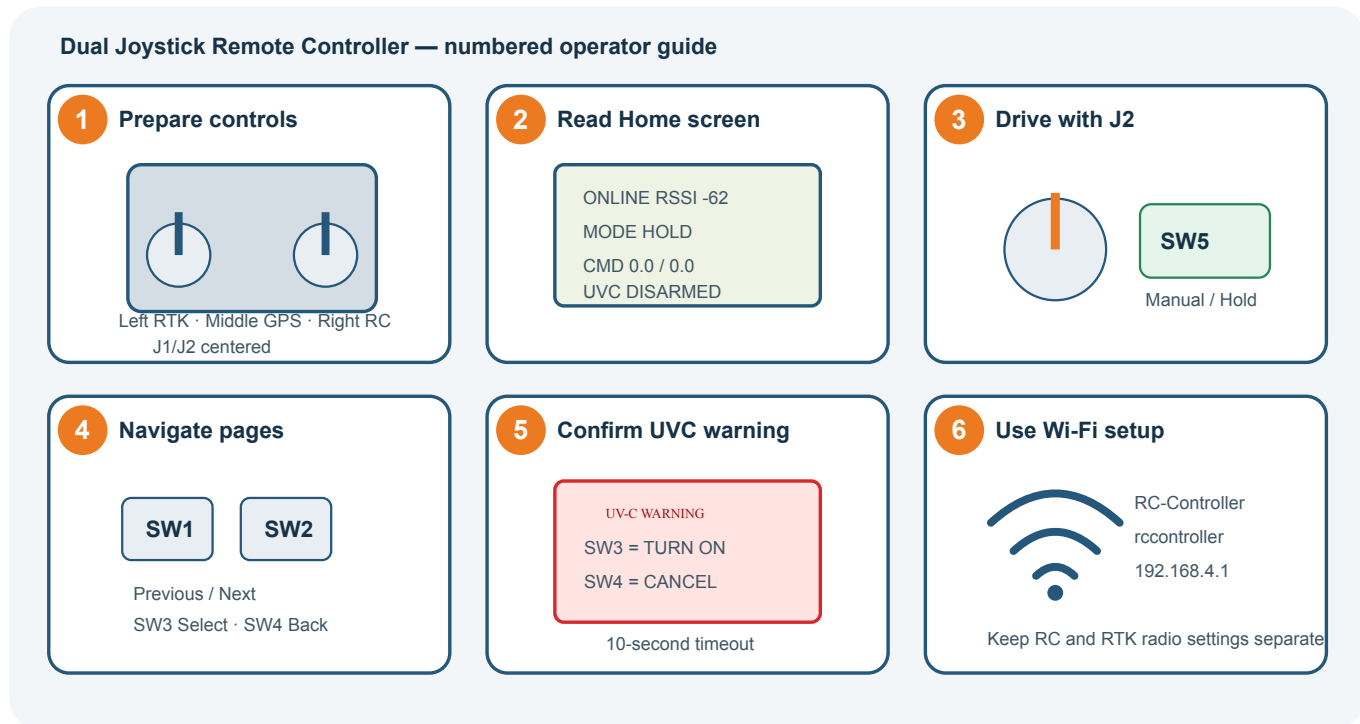


Figure 2.7: Each picture number matches the instruction below.

1. **Prepare.** Inspect the unit. Facing the controller, fit the RTK LoRa antenna on the left, GNSS antenna in the middle, and robot-communication LoRa antenna on the right. Center J1/J2 and make sure no switch is held.
2. **Read Home.** Power on and verify Hold, zero CMD, UVC Disarmed, controller battery, robot Online, and fresh RSSI/telemetry.
3. **Drive.** Press SW5 for Manual and use J2 by default. Test at low speed; SW9 gives high speed only while held.
4. **Use pages.** SW1/SW2 move through pages or menu items, SW3 selects/saves, and SW4 goes back/cancels.
5. **UVC warning.** After arming, an On request opens the warning. Check the secured area; SW3 confirms, SW4 cancels, and 10 seconds without action cancels.
6. **Use Wi-Fi.** Join RC-Controller with `alphaagtech`, open `192.168.4.1`, and configure the RC and RTK radios separately. Never copy the RC settings into the RTK radio; each radio matches its own endpoint and the two links must use non-overlapping field-approved settings.

2.5.1 Calibration and mapping

1. Restrain the robot and open Settings -> Joystick Calibration.
2. Move J1/J2 through every endpoint while watching all four displayed axes.
3. Release both sticks; SW3 saves or SW4 cancels.
4. In Joystick Mapping choose J1 or J2 for driving, reverse axes if necessary, and use the smallest trim/deadband that produces stable zero.
5. In Button Mapping assign SW5–SW10. Each function may be used once; SW1–SW4 are fixed.

2.5.2 Radio, clock, voice, and firmware webpage

- **RC radio:** must match the robot's communication radio.
- **RTK radio:** must match the RTK correction base and must not use the same channel/settings as RC. Separate the links according to the approved radio plan to prevent LoRa interference.
- **Applied:** saved radio settings are active. Pending means wait. Failed means restore known-good values.

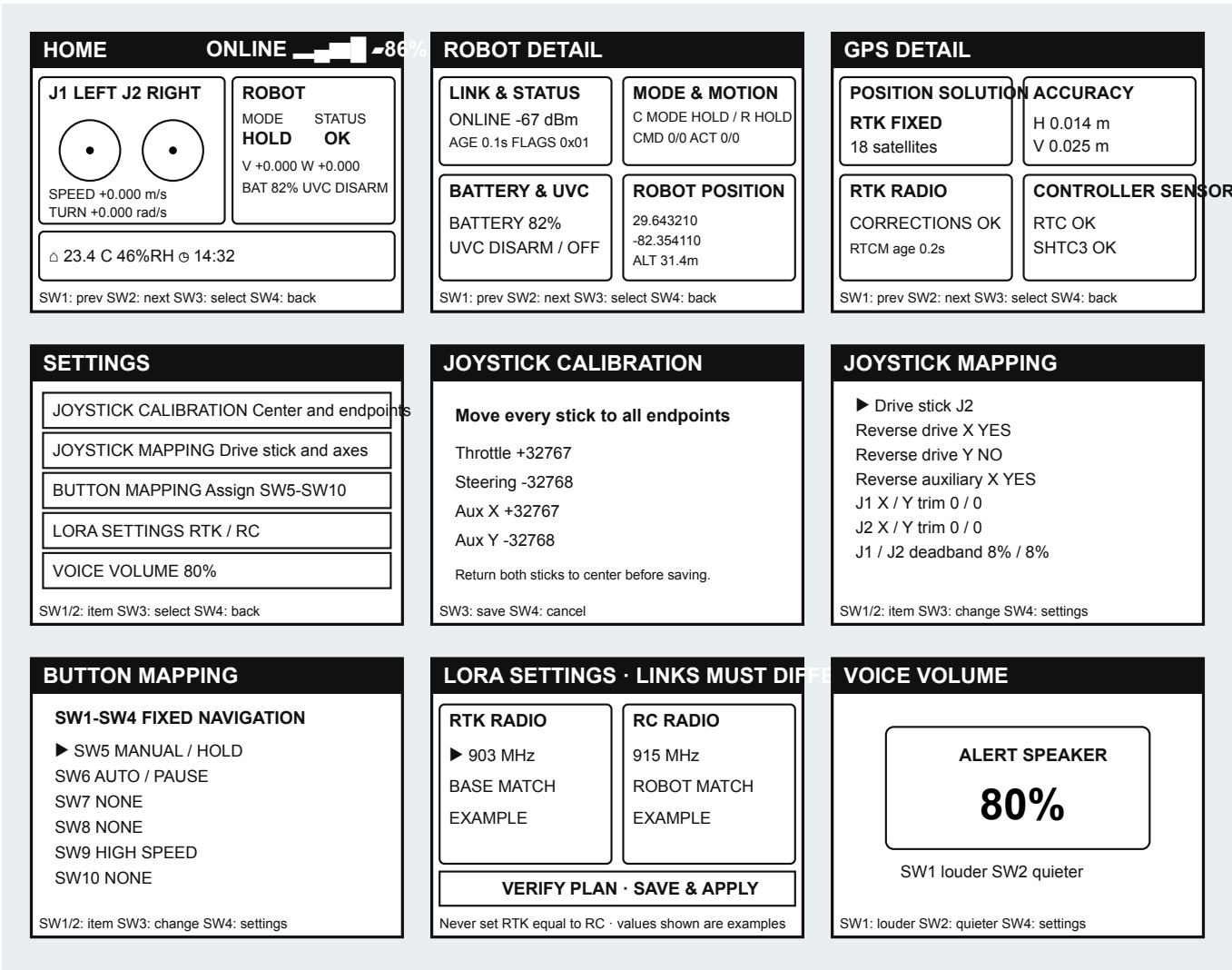


Figure 2.8: Home, Robot Detail, GPS Detail, Settings, Joystick Calibration, Joystick Mapping, Button Mapping, LoRa Settings, and Voice Volume.

- **Clock:** verify the phone time zone, then Sync Browser Time.
- **Voice:** 0% mutes alerts; higher values change in 10% steps.
- **Firmware:** select only the custom ESP32-S3 application image. Username is admin; password is rccontroller. Verify all functions after reboot.

2.6 Controller troubleshooting

Problem	What it means	What to do
Robot Lost / Offline	No valid MAVLink reply from robot system ID 1 for approximately one second.	Select Hold; verify robot power and antennas; move closer; match RC LoRa settings; then inspect MAVLink version, source ID and parse/CRC errors.
Good RSSI but no valid telemetry	RF packets are present but MAVLink frames have the wrong version/ID, fail parsing, or contain unsupported messages.	Keep Hold. Confirm MAVLink 2, controller system 255, robot system 1, component IDs, one frame per packet and matching firmware.
Mode request repeats	No accepted COMMAND_ACK or matching robot heartbeat has completed the request.	Read the ACK result; verify MAV_CMD_DO_SET_MODE, target component and custom-mode value; correct the rejection before retrying.
Dual Joystick Remote Controller GPS is not sent	The onboard GPS fix is invalid or older than 1.5 seconds.	Improve sky view/corrections and verify fresh GPS status; stale fixes are intentionally not transmitted.
Robot creeps at center	Joystick zero is incorrect.	Check free return to center; recalibrate; increase deadband slightly; use only small trim.
Direction is reversed	Axis mapping or wiring does not agree.	Stop; select Hold; correct Dual Joystick Remote Controller axis reversal or service Single Joystick Remote Controller/robot mapping; retest restrained.
Webpage will not open	Phone left the local network or hostname failed.	Stay connected despite No Internet; disable mobile-data switching/VPN; use the numeric address.
Dual Joystick Remote Controller UVC On ignored	Not armed, warning canceled/timed out, or no robot acknowledgment.	Verify UVC Armed, robot link, relay status, then repeat deliberate confirmation.
Emergency overlay remains	Last robot heartbeat reported emergency.	Make the robot safe and reset the physical system; wait for a new non-emergency heartbeat.

Component · Mobile robot and UVC system

3.1 Mobile robot and UVC: component overview

The robot uses two driven wheels and two steering axes. Robot master node 0x0B and slave node 0x0A coordinate motor nodes 1 and 2 for steering and motor nodes 3 and 4 for driving. The installed power hub connects the two robot sides and provides the CAN-controlled status LED at node 0x0E. The master receives handheld or navigation commands, reads system status/GPS, and controls relay node 0x0F. The UVC system has four Philips lamps on left and right physical AC power branches.

3.1.1 Features

- 2WD propulsion and 2WS steering.
- Four installed CANopen motor nodes: steering 1–2 and drive 3–4.
- Automatic precharge, motor-contactor control, steering homing and startup fault detection.
- Power hub joining the two robot sides, with RGBA LED indications for startup, idle, running, link loss and safety states.
- Hold, Manual, Auto, and Auto Pause modes.
- LoRa handheld control and telemetry.
- 500 kbit/s CAN connection to motor controllers, robot master/slave, GPS rover, navigation interface, power hub, and UVC relay.
- Robot Wi-Fi dashboard and configuration webpage.
- Four external ports on each robot side: two contactor-switched battery-power ports and two CAN-bus ports.
- Battery-powered PoE switch supplying the navigation controller network/power path.
- Four robot-mounted UVC lamps: left side, right side, top-left, and top-right.
- Routine UVC control commands all six relay channels together, with Arm, All Off, Disarm, heartbeat fail-safe, and CAN emergency shutdown.

3.2 Robot and UVC specifications

3.2.1 Installed component list

Component	Installed model or rating	Manufacturer / notes
Motor	ELVM80100V48FH-M17-HD	Leadshine
Motor driver	ELD2-CAN7030B	Leadshine
Steering-wheel gearbox	FAD-110D-50-P2-19-70-90-M6-42	OC Drive Machinery Co.
Drive-wheel gearbox	FAD-110D-70-P2-19-70-90-M6-42	OC Drive Machinery Co.
Battery pack	48 V, 80 Ah	Two packs installed, one on each robot side
Inverter	Victron Inverter 48/375 VE	Victron; two installed, one per robot side
UVC ballast	KTEB-254H0-UV-PS	One ballast per installed UVC bulb
UVC bulb	TUV PL-L 55W/4P/HF	Philips; four installed

3.2.2 System specifications

Item	Specification
Controller firmware	1.0.1
Motion	Two drive axes and two steering axes

Item	Specification
Motor nodes	1–2 steering; 3–4 drive
CAN bus	500 kbit/s
Startup timing	Approximately 4 s precharge, main-contactor closure, 0.5 s precharge overlap, then 5 s safety wait before CAN/node checks
Power hub	Cross-chassis connection between robot sides; CAN node 14 (0x0E); RGBA status LED
Robot Wi-Fi	mars_h_XXXXXX; local dashboard
Relay	CAN node decimal 15 (0x0F); electronics powered by 5 V from a power-hub 5 V CAN port; 500 kbit/s CANH/CANL
Relay groups	Group A: left-side + top-left; Group B: right-side + top-right; all six channels are commanded together by robot firmware
Relay Wi-Fi	can_relay_XXXXXX; 192.168.10.1
Fail-safe	All six channels Off/Disarm on CAN emergency or missing robot-master heartbeat
Heartbeat timeout	Approximately 3 seconds at relay

3.3 Robot connections and side ports

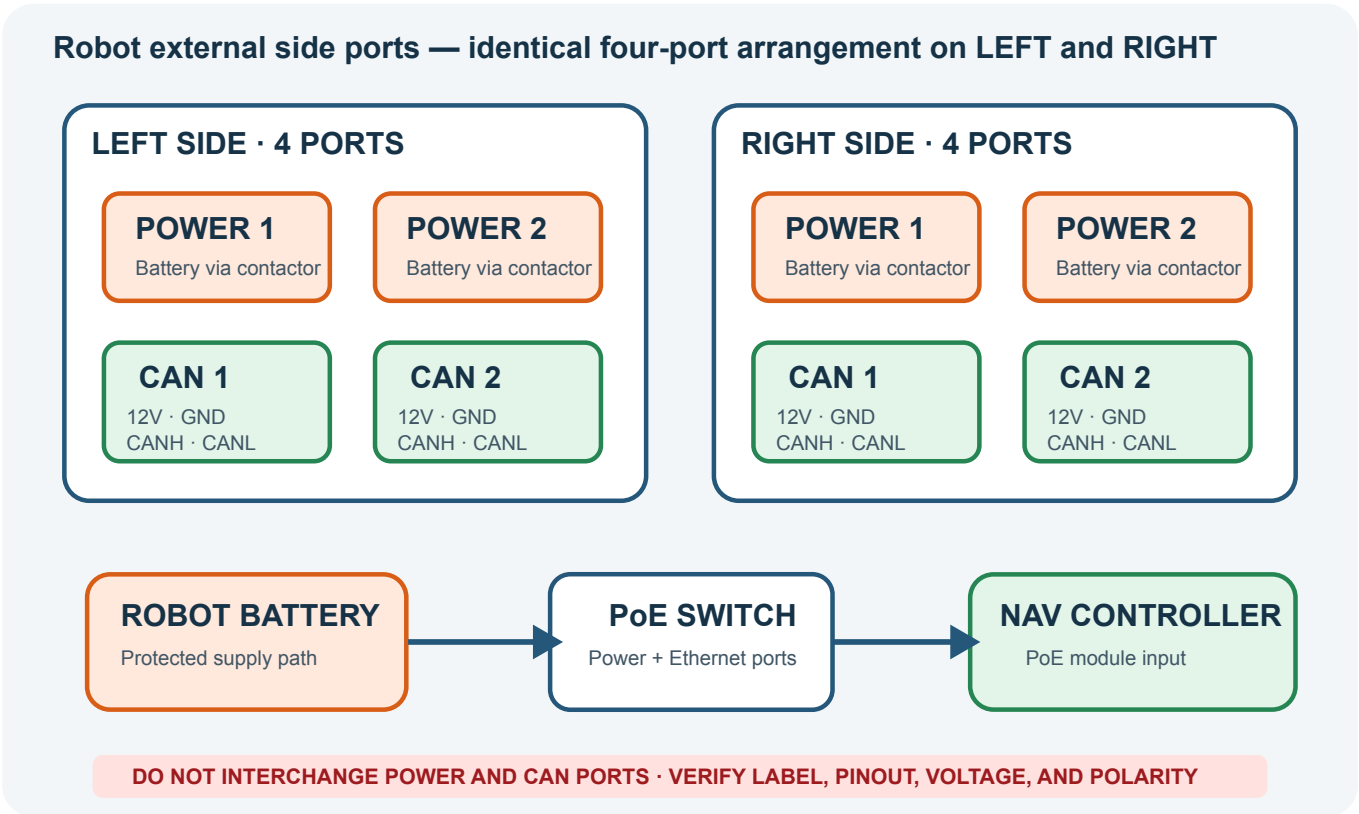


Figure 3.1: Each robot side has four ports: two battery-power ports switched by the main contactor and two CAN-bus ports carrying 12 V, GND, CANH, and CANL. The installed power hub connects the left and right sides and carries the system status LED.

Main contactor: the two power ports on each side are connected to robot battery power through the electrically controlled contactor. They can become energized when the contactor closes. Do not assume a connector is safe because the robot is in Hold.

Connection	Purpose	Operator/engineer check
RC LoRa antenna	Handheld commands and telemetry	Tight, undamaged, RC settings match only the controller
Battery power ports—2 per side	Contactor-switched robot battery supply for approved loads	Correct voltage/polarity, connector label, fuse/protection, contactor Off before mating
CAN-bus ports—2 per side	Combined auxiliary power and CAN communications	Pinout is 12 V, GND, CANH, CANL; never confuse with battery-power port
Power hub + LED, node 14	Cross-chassis connection joining the two robot sides and displaying system status	Hub secure; inter-side connections tight; LED visible; CAN heartbeat and SDO response present
PoE switch	Network connection and PoE supply for the navigation controller	Powered from robot battery power through the installed protected supply path; link and PoE indicators normal
Drive motors, nodes 3–4	Forward and reverse	Correct direction and feedback
Steering motors, nodes 1–2	Wheel angle	Homed, no fault, correct actual angle
Relay Group A	Left-side and top-left lamps	Two correct physical labels and matching applied feedback
Relay Group B	Right-side and top-right lamps	Two correct physical labels and matching applied feedback
Emergency stop	Independent safe stop	Accessible and tested before operation

3.3.1 Side-port connection procedure

1. Select Hold, switch UVC Off/Disarm, open the main contactor, and follow the site lockout procedure.
2. Read the port label. Identify **BATTERY POWER** or **CAN BUS**; do not choose by connector appearance alone.
3. Inspect pins, seals, keying, strain relief, cable rating, and damage. Do not connect a wet, bent, or unlabeled plug.
4. For a CAN port, verify the cable pinout end-to-end: 12 V, GND, CANH, CANL. Check bus termination and avoid adding an unintended terminator.
5. For a battery port, verify the approved load, operating voltage, polarity, branch protection, and current rating before mating.
6. Mate without forcing, secure the connector, route the cable away from steering/wheels/UVC fixtures, then restore power.
7. Confirm the intended device only: supply voltage, CAN online state, or PoE/link indication. Stop and isolate immediately for heat, odor, arcing, wrong voltage, or CAN faults.

3.4 Battery compartments and charging

The robot has **two separate battery packs**: one in the locked compartment on the left side and one in the locked compartment on the right side. Open one compartment, support and move its battery out far enough to reach the battery charge port, charge it with the approved charger, then return and secure it before repeating the procedure on the other side.

THE BATTERY CHARGE PORT IS NOT AN AC PORT. Never connect wall-outlet/mains AC, an AC power cord, or any other direct AC source to the battery charge port—even if a connector appears to fit. Only the battery-output connector from the charger approved for this battery may connect to that port. Connect the charger's AC input only to a correctly rated wall outlet and follow the charger label.

Battery charging can involve hazardous voltage, high fault current, heat and fire risk. Never charge a wet, damaged, swollen, leaking, unusually hot, or unlabeled battery. Keep ignition sources away and follow the battery, charger, and site emergency instructions.

3.4.1 Prepare the robot and battery

1. Park the robot on stable, dry ground in the designated charging area with ventilation and unobstructed emergency access.



Figure 3.2: The charge port is on the battery itself. Its connector shape does not make it an AC mains inlet.



Figure 3.3: The charger has two different connections: its AC input plug connects to the correctly rated wall outlet, while its battery-output connector connects to the battery charge port.

2. Stop any mission, center the joystick, select Hold, and verify zero motion. Command UVC All Off, verify all four lamps Off, Disarm, and open the physical UVC disconnect.
3. Switch the robot off using the approved shutdown procedure. Do not begin while the contactor is closed or the robot is capable of motion.
4. Identify the battery side to charge. Confirm the compartment key and the charger approved for that exact battery pack. Charge one battery at a time unless the approved equipment and site procedure explicitly allow otherwise.
5. Unlock and open the selected battery compartment. Support the cover, disconnect or release the battery only as specified by its manual, and carefully slide or lift the battery out far enough to access the charge port. Support its weight; do not pull on cables or allow the battery to fall.
6. Inspect the battery housing, display, cables, terminals, charge port, protective cover, QR label, and charger. Stop and tag the equipment for service if anything is loose, wet, contaminated, cracked, corroded, swollen, leaking, burnt, or unusually hot.

3.4.2 Connect and charge

7. Keep the charger disconnected from AC power while making the battery-side connection unless the charger's own instructions require a different sequence.
8. Identify the charger's **battery-output connector** and connect it to the battery charge port without forcing it. Do not connect the wall-outlet AC plug or any direct AC source to the battery.
9. Check that the connector is fully seated and the cable is protected from the compartment edge, wheels, steering, and foot traffic. Then connect the charger's AC input plug to a correctly rated wall outlet and energize it according to the charger instructions.
10. Confirm normal charging on the charger and battery indicators. Monitor the battery and charger for faults, unusual heat, odor, smoke, noise, swelling, leakage, or cable damage. Stop and follow the battery emergency procedure for any abnormal condition.

3.4.3 Finish and secure

11. When charging is complete, switch off or unplug the charger from AC power before disconnecting its battery-output connector, unless the charger manual specifies another order. Never pull on the cable.
12. Inspect the charge port and refit its protective cover. Carefully return the battery fully into its compartment, reconnect and secure it according to the battery manual, and confirm no cable is pinched or strained.
13. Close and lock the compartment. Verify it is latched, then repeat the complete procedure for the battery on the other side.
14. Before returning the robot to service, confirm both batteries are secured, both compartment locks are closed, no charger remains attached, both battery states are acceptable, and the robot passes the normal power-on sequence.

3.4.4 Battery Bluetooth app and manual

Each battery has built-in Bluetooth communication. Scan the QR code printed on that battery to obtain the manufacturer's app and battery manual. Before installing, verify that the QR destination and app publisher match the battery manufacturer. Follow the battery manual for pairing, status interpretation, alarms, and maintenance. Bluetooth/app readings supplement—but do not replace—the battery display, charger indicators, physical inspection, or charging safety checks.

3.4.5 Charging troubleshooting

Symptom	Action
Charger does not start	De-energize it. Verify the approved charger, correctly rated AC input, seated battery-output connector, battery charge port, and charger indication. Never test the battery port with direct AC.
Charger reports a fault	Stop charging, record the code/indicator, isolate the charger, and follow the charger/battery manual. Do not repeatedly reconnect to clear an unexplained fault.
Battery or connector becomes hot, smells, swells, leaks, or smokes	Use the site battery emergency plan immediately. Do not touch, move, open, or energize the robot unless the trained emergency procedure directs it.

Symptom	Action
Battery will not slide out or return securely	Do not force it or operate the robot. Inspect the retainers, cable routing, compartment, battery alignment, and damage; obtain service assistance.
QR code or Bluetooth app does not work	Use the printed battery model and manufacturer support information to obtain the official manual/app. Do not install an app from an unverified publisher.
One side charges but the other does not	Treat each pack independently. Verify the second battery's charger compatibility, charge port, connector, battery condition, and fault status; do not infer its state from the first battery.

3.5 Power hub and status LED: component manual

The power hub distributes regulated 12 V and 5 V power, provides multiple CAN connection points, joins devices on the 500 kbit/s robot CAN backbone, and controls the visible red/green/blue/white status LED.

3.5.1 Features and specifications

Item	Installed specification
Power distribution	Six 12 V CAN ports, four 5 V CAN ports, two 12 V power-only ports and two 5 V power-only ports
CANopen node	14 (0x0E)
CAN rate	500 kbit/s
LED channels	Red, green, blue and dedicated white; commanded/applied values 0–255
PWM	500 Hz, 9-bit timer; output limited to approximately 50% maximum PWM duty for fixture current/thermal control
Automatic control	Robot master writes mode object 0x2004 by SDO, retries failures and reasserts the current mode at least every 2 seconds
Local emergency behavior	A nonzero CAN EMCY selects solid red locally
Wi-Fi AP	rgba_led_XXXXXX; password alphaagtech; channel 1; maximum two clients
Web addresses	http://rgba_led.local or http://192.168.10.1

3.5.2 Port, connector and pinout schedule

Function / board reference	Board connector	Cable-side mating housing	Pinout and use
12 V CAN J1–J4	Molex 46999-0014 Mini-Fit Jr., 4 circuits	Molex 39-01-2040 (5557 series) Female crimp terminals are selected separately for the installed wire gauge/current.	1: 12 V; 2: CANH; 3: CANL; 4: GND. For approved 12 V CAN devices.
12 V CAN J9–J10	Amphenol Sine ATP13-4P-BM03GRY	ATP06-4S plug, plus the specified wedge lock, seals and size-12 socket contacts	1: 12 V; 2: CANH; 3: CANL; 4: GND. Sealed 12 V CAN connection.
5 V CAN J5–J8	Molex 43045-0402 Micro-Fit 3.0, 4 circuits	Molex 43025-0400 Female terminals selected separately for wire gauge/current.	1: 5 V; 2: CANH; 3: CANL; 4: GND. The GNSS rover and UVC relay receive 5 V operating power and CAN here. Navigation CAN also uses this group, but navigation operating power comes from PoE.

Function / board reference	Board connector	Cable-side mating housing	Pinout and use
12 V power only J12–J13	Molex 76825-0002 Mega-Fit, 2 circuits	Molex 171692-0102 Female terminals selected separately for wire gauge/current.	1: 12 V; 2: GND. Approved 12 V loads only; no CAN.
5 V power only J14–J15	Molex 43045-0200 Micro-Fit 3.0, 2 circuits	Molex 43025-0200 Female terminals selected separately for wire gauge/current.	1: 5 V; 2: GND. Approved 5 V loads only; no CAN.
VIN input J16	2-pin generic connector symbol; production part number is not assigned in the board design	Match the connector fitted to the serviced board; confirm against the production BOM before ordering	1: VIN; 2: VIN–. Qualified field service only.
Fused VIN outputs J11, J17	2-pin generic connector symbol; production part number is not assigned in the board design	Match the connector fitted to the serviced board; confirm against the production BOM before ordering	1: fused VIN output; 2: VIN–. Qualified field service only.

Housing part numbers do not include crimp contacts, seals, secondary locks or wire. Use the connector manufacturer's application specification and an approved crimp tool; choose contacts for the actual conductor size, insulation diameter and circuit current. Never substitute a housing solely because it appears to fit.

Wrong-voltage connection can destroy equipment. Read the printed voltage label before mating a connector. Never select a port by connector shape alone, and never connect a 5 V device to a 12 V CAN or power port.

Installed device assignments

- **GNSS rover:** connect to a labelled 5 V CAN port. Pin 1 supplies its 5 V operating power; pins 2/3 carry CANH/CANL and pin 4 is GND. See GPS connections.
- **UVC relay:** connect node 15 to its assigned 5 V CAN port. Pin 1 supplies the relay electronics with 5 V; pins 2/3 carry CANH/CANL and pin 4 is GND. The relay contacts separately switch 120 VAC to the four ballast inputs. Never connect the relay electronics to a 12 V CAN port.
- **Navigation controller:** connect its installed CAN harness to the assigned power-hub 5 V CAN port. The port pinout is 5 V, CANH, CANL and GND; the robot PoE switch provides the navigation controller's main operating power and Ethernet. Do not alter the installed harness. See navigation connections.

3.5.3 LED indications and operator response

Appearance	Meaning	Operator response
Dark briefly after switch-on	Hub is booting or waiting for the robot master's first SDO mode command.	Wait and watch the dashboard. Persistent darkness requires the troubleshooting checks below.
Solid yellow	Startup: precharge, delay, node checks, steering homing or motor initialization.	Remain in Hold and keep clear of all wheels.
Pulsing yellow , about 3-second cycle	Startup complete, operator link present and commanded motion is zero.	Normal ready/idle state; verify the dashboard before changing mode.
Breathing blue , about 3-second cycle	Robot running with a nonzero motion command.	Normal only for intentional motion; press E-stop for unexpected movement.
Flashing amber , about 1.5 Hz	Startup fault latched.	Do not operate. Read the robot dashboard fault and follow startup troubleshooting.
Solid red	Physical E-stop or CAN EMCY.	Keep clear, correct the cause before reset, and never use the LED as proof of electrical isolation.

Appearance	Meaning	Operator response
Blinking magenta , about 1 Hz	Operator-control link lost after startup.	Use physical safety controls and verify actual motion/UVC state from a safe location.
Constant white	Night-mode override.	Not a readiness indication; use the dashboard for the operating decision.
White Morse flashes	Telegraph/service mode.	Read the service message and return to automatic status mode after service.

Priority: E-stop red overrides everything; startup-fault amber is next. Both override Night mode. Otherwise Night white overrides Startup, Link Lost, Idle and Running. The LED is an information display, not a safety-rated interlock.

3.5.4 Power-hub webpage

1. Keep the robot in Hold with UVC Off/Disarmed. On the phone, join `rgba_led_XXXXXX` using `alphaagtech`.
2. Open `http://rgba_led.local`; if it does not resolve, open `http://192.168.10.1`. Confirm the page says the hub is connected.
3. **Robot Status Mode:** use for an authorized indicator test only. The robot master normally selects the mode automatically and can overwrite a manual webpage selection within two seconds.
4. **Telegraph:** type up to 32 supported letters/numbers/spaces. The hub selects mode 6 and repeats the text in Morse on the white channel.
5. **Manual Color:** move R/G/B/W sliders to test a channel. Any slider edit selects mode 0. Use **All Off** to set every channel to zero.
6. **CAN Bus Settings:** the installed values are **500 kbit/s** and node ID **14**. Do not change them during normal operation. Save & Reboot persists a change and immediately restarts the hub.
7. After service, restore the automatic robot-selected mode and verify the physical LED follows robot state.

3.5.5 Power-hub CANopenNode objects

Index	Type / access	Meaning
0x2000	uint8 RO	Node ID; installed value 14.
0x2001:00	uint8 RO	Highest subindex/count = 4.
0x2001:01-04	uint8 RW	Requested red, green, blue and white intensity, 0-255.
0x2002:00	uint8 RO	Highest subindex/count = 4.
0x2002:01-04	uint8 RO	Applied red, green, blue and white intensity.
0x2003	uint8 WO	All LEDs Off; any nonzero write clears all color commands, selects Manual and self-resets.
0x2004	uint8 RW	Mode: 0 Manual, 1 Startup, 2 Running, 3 Error, 4 E-stop, 5 Night, 6 Telegraph, 7 Idle, 8 Link Lost.

Common CANopen communication/service objects are documented in CANopenNode object dictionaries. Raw SDO writes are for authorized field engineers only.

3.5.6 Maintenance

- With power isolated, inspect both cross-chassis connectors, strain relief, seals, pins, mounting and harness clearance.
- Clean the LED lens without scratching it; keep it visible from the operator's normal observation point.
- During scheduled testing, verify each R/G/B/W channel, every automatic status mode, node-14 heartbeat and applied values at 0x2002.

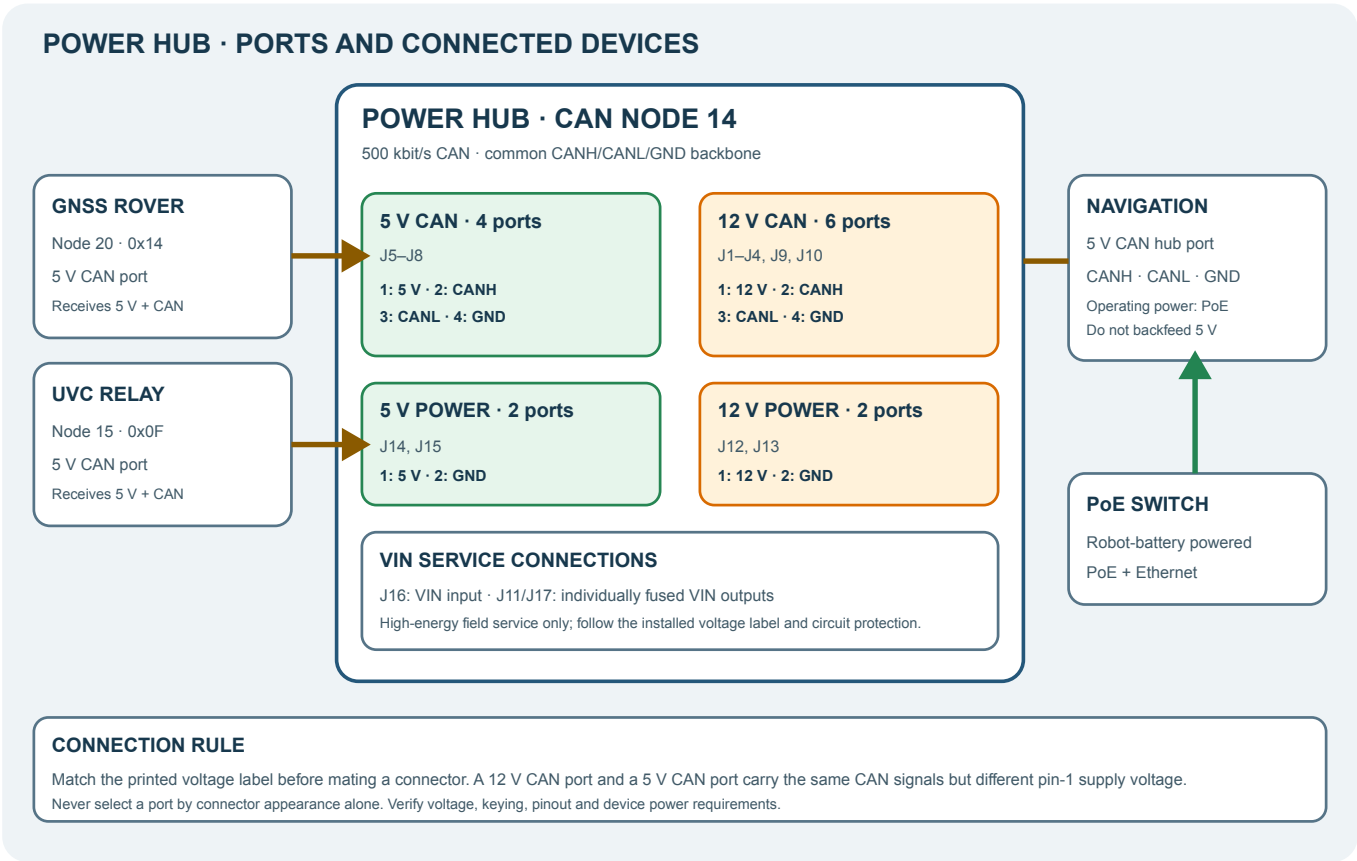


Figure 3.4: The GNSS rover and UVC relay receive 5 V operating power and CAN from 5 V CAN ports. The navigation controller uses another 5 V CAN-group connector for communication, while PoE supplies its operating power.

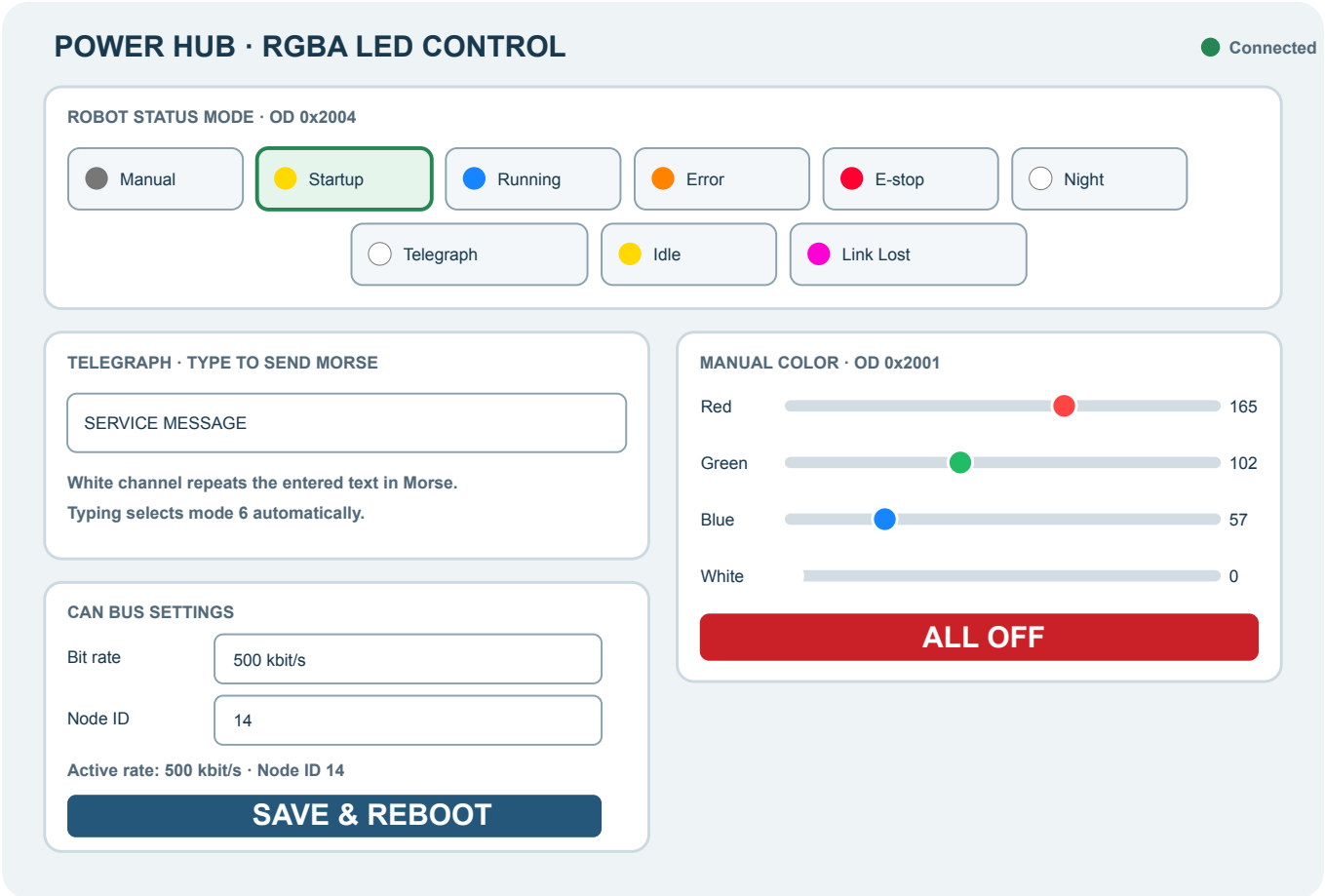


Figure 3.5: Selecting a manual slider switches the hub to Manual mode; Save & Reboot changes the CAN configuration and restarts the hub.

- Record firmware revision, AP credentials under site policy, 500 kbit/s/node-14 configuration and any connector repair.

3.5.7 Power-hub troubleshooting

Symptom	Likely area	Action
LED stays dark	Hub power, cross-chassis connector, LED connector, CAN/SDO command or all channels zero	Hold and isolate power. Inspect both side connections and LED wiring; then verify node 14, 0x2004 and 0x2002.
Wrong color/pattern	Manual/web override, wrong mode, swapped LED channels or stale CAN command	Compare 0x2004 and applied 0x2002; restore robot automatic mode and check R/G/B/W wiring.
Red remains after reset	Physical E-stop, active/repeated EMCY or robot still commanding E-stop	Do not bypass. Find and clear the safety cause, then verify a CAN error-reset and normal robot mode.
Webpage unavailable	Wrong AP, mDNS failure, hub not powered or two-client limit reached	Join <code>rgba_led_XXXXXX</code> , disconnect unused clients and use 192.168.10.1.
Hub offline on CAN	Wrong bit rate/node, CANH/CANL, ground, termination or cross-chassis connection	Hold; verify 500 kbit/s, node 14, wiring and approximately 60 Ω with power off.
One robot side also loses communication/power	Cross-chassis hub or harness fault	Stop operation and isolate power. Inspect the hub and both side harnesses before troubleshooting individual devices.

3.6 Expected robot behavior after power-on

Steering wheels move automatically during startup. Before switching on, select Hold, center the controller, keep UVC Off/Disarmed, restrain the robot, and keep hands, feet, tools and cables clear of every wheel and steering linkage.

3.6.1 Normal power-on sequence

1. **Initial safe state.** Switch on with the robot in Hold and all commands at zero. The main motor contactor and precharge relay begin open. The power-hub LED can remain dark briefly while the CAN stack and LED proxy start.
2. **Motor-driver precharge.** The precharge relay closes and charges the motor-driver DC link through the precharge resistor for approximately **4 seconds**. The robot must not drive during this period. Once node 14 receives the status command, the power-hub LED is **solid yellow**.
3. **Main motor contactor closes.** After precharge, the main motor contactor closes. A single firm **click or clunk** is expected and confirms that motor-driver power has been connected. The precharge path remains on for about 0.5 second, then releases. Repeated clicking, buzzing, visible arcing, smoke or an electrical smell is not normal.
4. **Safety delay.** Firmware waits approximately **5 more seconds** while monitoring the emergency stop. The LED remains solid yellow.
5. **CAN node checks.** At 500 kbit/s, the master waits for motor nodes 1, 2, 3 and 4, then robot-slave node 10. Startup stops and the contactor opens if required heartbeats do not arrive.
6. **Steering homing.** Steering motors 1 and 2 home in parallel. The steering wheels will turn to find their reference and apply their configured offsets. Some motor/gear sound is expected; striking an obstruction, oscillating indefinitely, or continuing beyond the allowed homing sequence is not.
7. **Motor initialization.** The controller initializes steering nodes 1–2 for position control and drive nodes 3–4 for velocity control. All four must succeed.
8. **Ready.** The controller clears the startup fault, marks the robot running, and commands CANopen Operational. With a healthy control link and zero motion command, the LED changes to **pulsing yellow**. A nonzero driving command changes it to **breathing blue**.

During startup, solid yellow means startup in progress, pulsing yellow means ready/idle, breathing blue means commanded motion, flashing amber means startup fault, and solid red means emergency. See the single authoritative power-hub LED

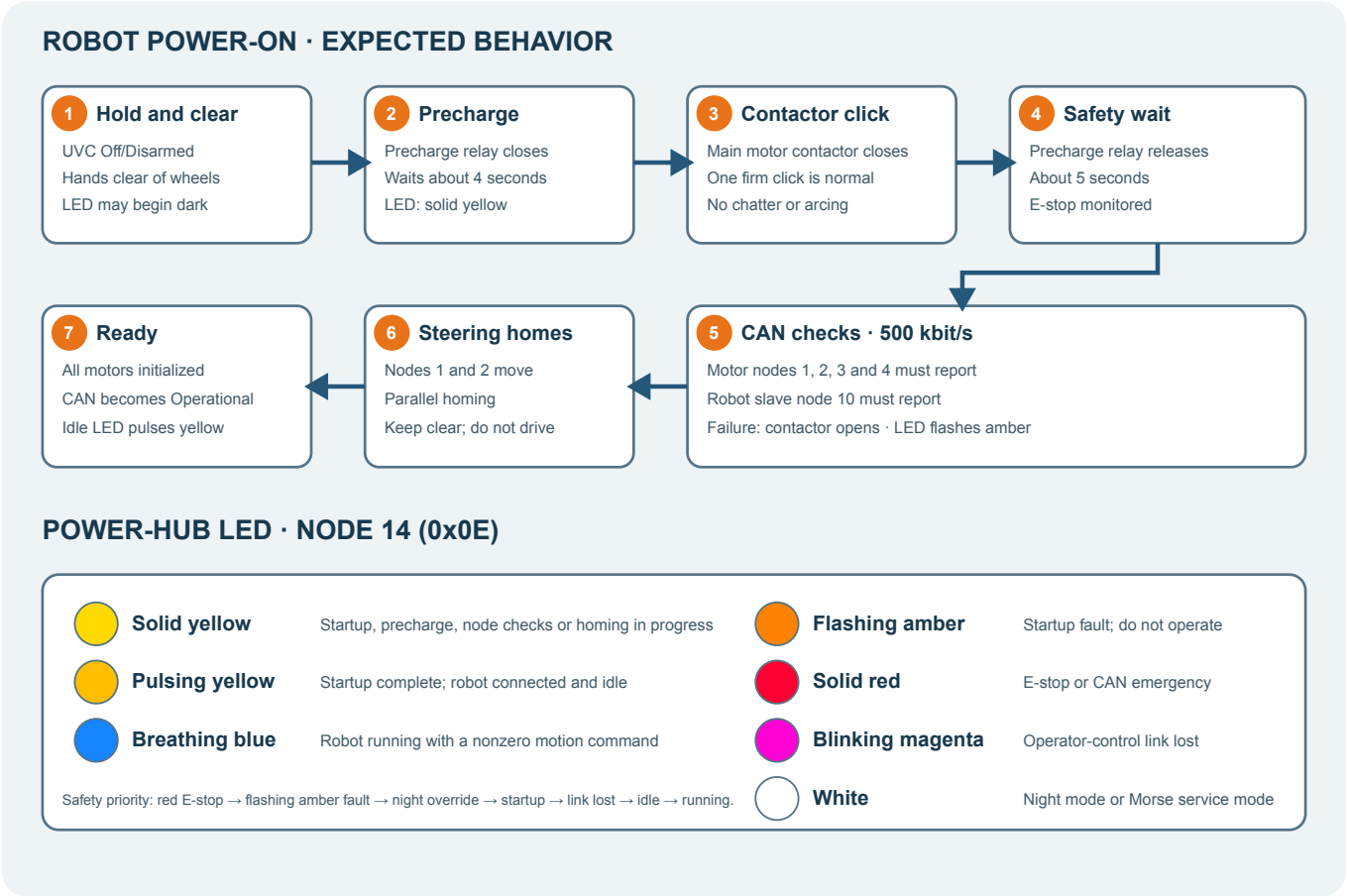


table for every pattern, priority and response.

3.6.2 If startup does not finish

1. Keep the robot in Hold and UVC Off/Disarmed. Do not push or manually force steering while powered.
2. If the expected contactor click never occurs, check the physical E-stop, remote CAN emergency, precharge/contactor power path and dashboard startup state. The firmware intentionally holds the contactor open during an emergency.
3. If the contactor chatters, buzzes, arcs, smells hot or repeatedly opens and closes, press E-stop, isolate power and have a qualified field engineer inspect the precharge circuit, contactor supply and terminals.
4. For flashing amber, record the dashboard fault number before cycling power. Check nodes 1–4, node 10, 500 kbit/s wiring/termination, motor power and homing sensors as appropriate. Cross-reference the power-hub troubleshooting table if the LED or both-side connection is also abnormal.
5. After correction, perform a restrained startup test. Do not release the robot for operation until the sequence reaches Startup OK and normal idle indication.

3.7 Robot operation manual

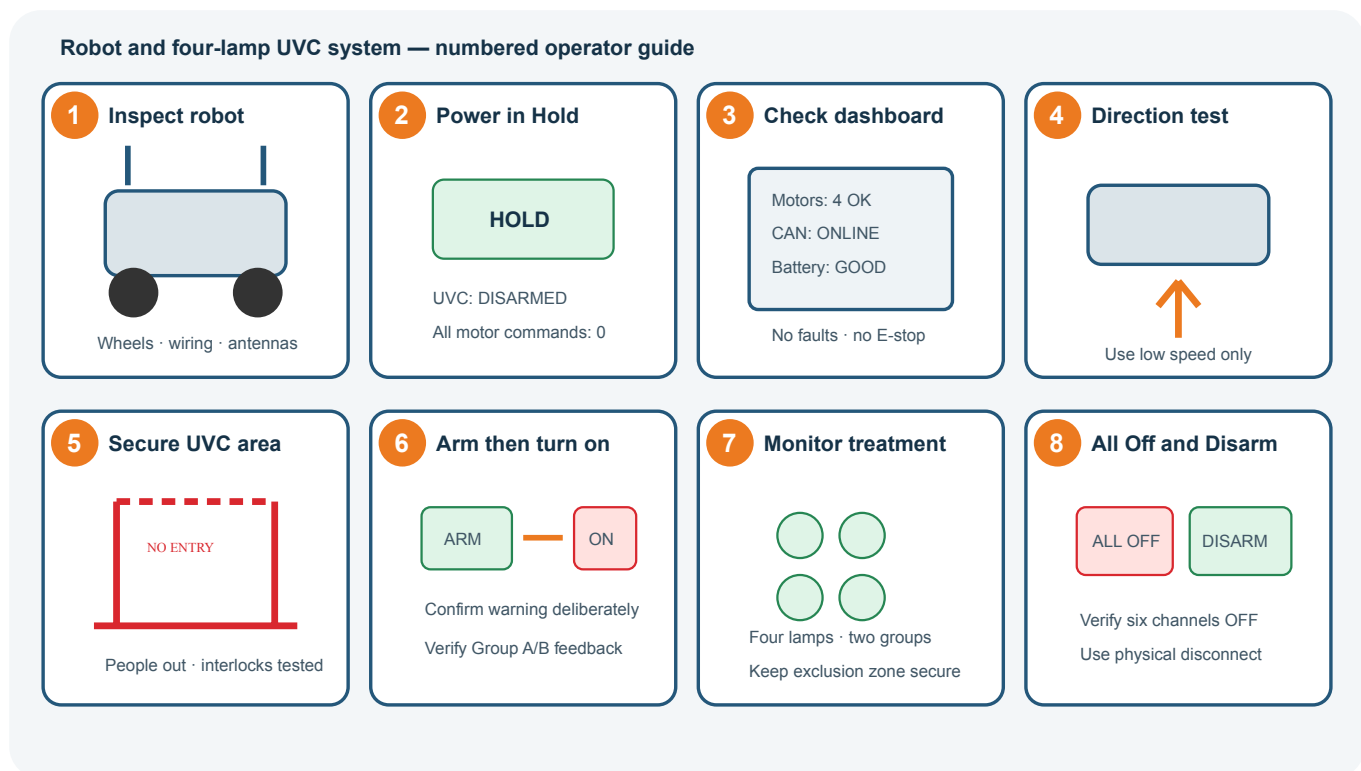


Figure 3.7: For driving, follow pictures 1–4 and 8. For UVC, follow pictures 1–3 and 5–8, applying them to both relay groups and four physical lamps.

1. Inspect wheels, steering, covers, power-hub cross-chassis connections, status LED, wiring, antennas, batteries, emergency stop, four UVC fixtures, and work area.
2. Power up with the robot restrained, mode Hold, UVC Disarmed, and all commands zero. Follow the complete power-on sequence; expect one main-contactor click and automatic steering homing.
3. Wait for Startup OK and the normal pulsing-yellow idle indication. Open the robot dashboard. Confirm motor nodes 1–4 OK, slave node 10 and power hub node 14 online, batteries healthy, CAN online, no emergency, relay online, both UVC groups Off, and all four lamps physically Off.
4. Use the selected controller to enter Manual. Test forward, reverse, left, and right at low speed. Compare command and actual feedback.
5. Drive with smooth inputs. Keep people clear and continuously watch link, LED, motor, battery, and emergency state.

6. To stop normally, release the joystick, select Hold, and verify zero motion and the idle indication.
7. For unsafe motion, use the physical emergency stop immediately; expect the LED to become solid red.
8. At finish, park, Hold, both UVC groups Off/Disarm, and follow the approved power-down order.

3.7.1 Robot webpage

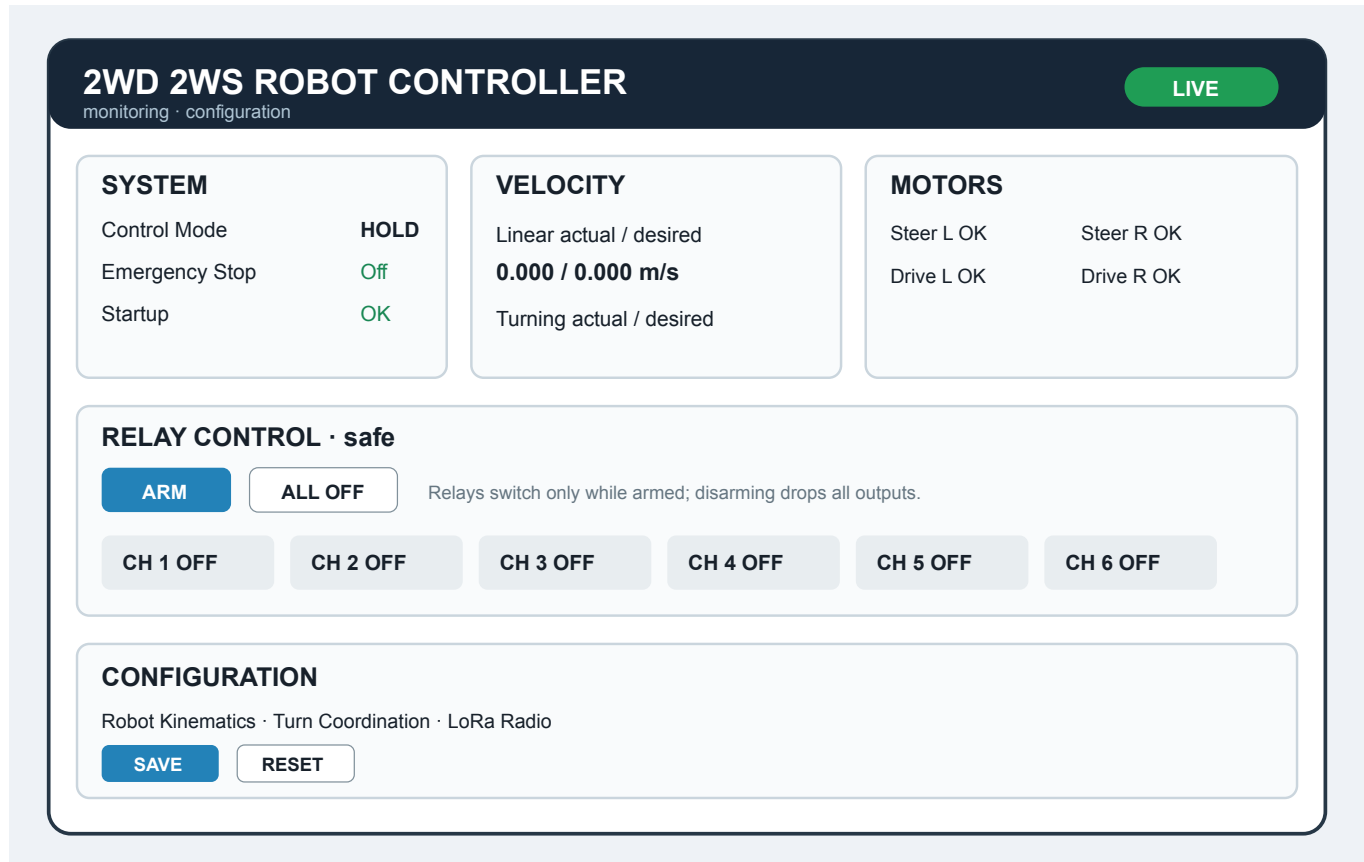


Figure 3.8: The dashboard must show actual feedback, not only requested values. Use the verified physical labels Group A/Left and Group B/Right wherever the service page exposes individual relay outputs.

- **Status cards:** link, mode, emergency, GPS, and battery.
- **Motor cards:** OK/Fault/Offline, actual and commanded drive/steering.
- **Relay card:** online state, armed state, request, and applied outputs mapped to Group A/Left and Group B/Right.
- **Configuration:** radio/CAN/motion settings intended for trained service staff. Record old values first.

3.8 Four-lamp UVC operation manual

UV-C exposure hazard. Never energize lamps while a person or animal is inside the established field treatment/exclusion zone. The installed robot command switches all six relay channels together; four channels feed the four lamp/ballast circuits.

1. Inspect the robot, relay, four lamp housings, Group A/B wiring labels, warning indicators, guards and physical disconnect.
2. Park the robot in Hold. Verify UVC Disarmed, all six relay channels Off, and all four lamps physically Off.
3. Confirm relay node 0x0F is powered from its 5 V CAN port, online at 500 kbit/s, receiving the master heartbeat, and reporting no CAN emergency.
4. Use the approved UV-C field assessment to establish the exclusion boundary for lamp height/orientation, crop rows, terrain, reflections, speed and dose. Remove people and animals; post signs and place cones, flagging, gates or spotters as

Four UVC lamps · two power branches · all six relay channels switch together

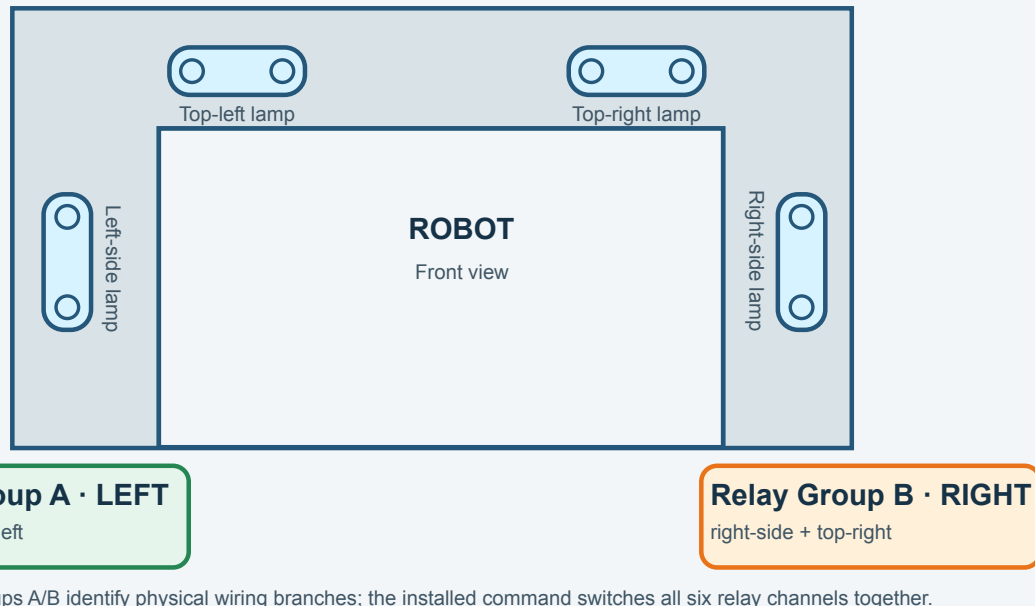


Figure 3.9: Group A identifies the left-side plus top-left power branch. Group B identifies the right-side plus top-right branch. They are physical service groups, not separate selections in the installed robot command.

required.

5. Arm UVC. Verify Armed while all six channels and four physical lamps remain Off.
6. Request On; on Dual Joystick Remote Controller read the warning and press SW3 within 10 seconds or SW4 to cancel. Verify all six requested/applied bits and all four installed lamps from outside the exclusion zone.
7. Monitor the full exclusion boundary, heartbeat, relay connection, applied feedback, travel speed, all four lamps and external indicator throughout treatment. Stop if anyone approaches, the robot deviates, or any required status becomes abnormal.
8. Command All Off; verify all six applied bits, four physical lamps and indicators Off; Disarm; then use the physical disconnect before entry.

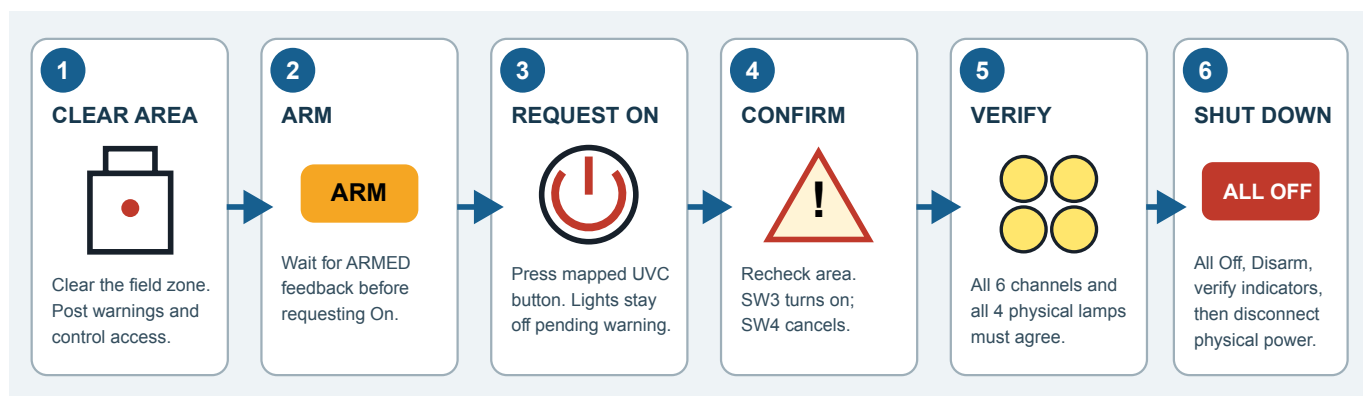


Figure 3.10: Safe sequence: establish field exclusion zone -> Arm -> deliberate On confirmation -> all six channels together -> monitor -> All Off -> physical verification -> Disarm -> disconnect.

For the complete field procedure, including dose/speed, exclusion-zone setup, treatment records and stop conditions, see Operations · UVC treatment.

3.9 Robot and UVC troubleshooting

Problem	Operator response
One motor Offline/Fault	Select Hold; do not drive; inspect power/CAN/mechanical obstruction; have service staff read the status word.
Steering direction or angle wrong	Stop; Hold; verify homing and feedback; do not compensate with aggressive joystick input.
Relay Offline	Keep UVC physically disconnected; verify relay power, the 500 kbit/s CAN setting, node 0x0F, and master heartbeat.
UVC switch snaps Off	The relay is disarmed, CAN emergency is active, heartbeat is missing, or the request was rejected.
One UVC channel differs	All Off, Disarm, physical disconnect, and service that channel. Do not continue with partial feedback.
Robot webpage unavailable	Confirm robot Wi-Fi, remain connected despite No Internet, use the numeric address, and check robot power.

3.10 UVC lamp specifications

The robot uses four **Philips TUV PL-L 55W/4P HF** germicidal lamps. This is a clear, single-ended twin-tube UV-C fluorescent lamp intended for professional disinfection equipment. Use only the exact approved lamp and a compatible high-frequency ballast/fixture.

Specification	Value
Manufacturer / family	Philips (Signify) TUV PL-L
Ordering name	TUV PL-L 55W/4P HF 1CT/25
Bulbs.com SKU / ordering code	294645 / TUV PL-L 55W/4P/HF
Technology / shape / finish	Germicidal compact fluorescent; single twin tube; clear
Base	2G11, four-pin, single-ended
Electrical input	55 W; nominal 105 V lamp voltage; nominal 0.525 A lamp current
UV-C peak	253.7 nm
UV-C output	18.7 W at 100 operating hours
Useful life	9,000 hours nominal; 20% UV depreciation at useful life
Dimensions	Approximately 535 mm maximum overall length; 38 mm maximum overall width; tube diameter 18 mm
Weight	134 g per lamp
Mercury	Contains 4.4 mg Hg; recycle as a mercury-containing lamp
Safety class	Risk Group 3 ultraviolet product; unshielded radiation can severely injure skin and eyes

Do not look at or expose skin to an operating lamp, even briefly. The bare-lamp manufacturer requires protection from its radiation. For this agricultural robot, use the validated field exclusion zone, remote supervision, access control, warnings, physical disconnect, and every guard or automatic safety device fitted to the site. The lamp's visible glow is not a reliable measure of UV-C output.

Sources: Bulbs.com product page and Signify product datasheet.

3.11 Four-lamp locations and relay groups

Position	Relay group	Pair
Left-side lamp	Group A · Left	Top-left lamp
Top-left lamp	Group A · Left	Left-side lamp

Position	Relay group	Pair
Right-side lamp	Group B · Right	Top-right lamp
Top-right lamp	Group B · Right	Right-side lamp

Service identification: every fixture, connector, wire, fuse/ballast and relay output must carry its matching lamp-position and group label. After repair, confirm the mapping with safe indicator loads or continuity testing while de-energized; never identify groups by exposing personnel to UV-C.

3.12 Replace a UVC bulb

Authorized service personnel only. The lamp contains mercury and the fixture may contain hazardous voltage. Follow site lockout/tagout and the fixture/ballast manufacturer's instructions. Never replace a lamp with the robot energized.

1. **Make the UVC system safe.** Park in Hold, command both relay groups Off, verify all four lamps and indicators Off, then Disarm.
2. **Isolate energy.** Switch off and physically disconnect the UVC supply. Apply lockout/tagout. A qualified person must prove the fixture/ballast is de-energized with an appropriate tester.
3. **Cool and put on PPE.** Wait for the lamp and ballast to cool. Wear clean gloves, safety glasses, and the site-required electrical/UV PPE. Keep a replacement-lamp container nearby.
4. **Open the fixture.** Support and remove the guard/cover according to the fixture instructions. Note the lamp base orientation, clips, seals, reflector, wiring and every installed safety device. Do not damage them.
5. **Remove the old lamp by its base.** Release any retaining clip. Grip the plastic 2G11 base—not the glass tubes—and pull straight out evenly. Do not twist, bend, or lever the twin tubes.
6. **Inspect and match.** Inspect socket, pins, clips, guard, seals, reflector, ballast wiring, and heat damage. Confirm the replacement label says TUV PL-L 55W/4P HF, 2G11 four-pin, clear. Reject cracked lamps or bent pins.
7. **Install.** With clean gloves, align all four pins and press the base straight into the socket until fully seated. Refit retaining clips, guard, seals, and covers exactly. Clean fingerprints using only the manufacturer-approved method.
8. **Controlled test.** Establish the field exclusion zone, remove lockout under the site procedure, restore power, and test only the affected relay group remotely. Verify both lamps in that group, no flicker/arc, correct applied feedback, warning indicator, every installed safety device, Off, and Disarm. Record position, group, date, and operating hours.

3.12.1 Broken lamp procedure

1. Do not use a vacuum cleaner.
2. Keep people away and ventilate the area for at least 30 minutes, following the Signify datasheet and site mercury procedure.
3. With the specified gloves/PPE, collect pieces without raising dust. Place debris and cleanup material in a sealed plastic bag or approved mercury-lamp container.
4. Send the lamp to the local approved recycling/hazardous-waste facility; do not place it in ordinary trash.
5. Inspect and decontaminate the fixture before installing another lamp.

3.12.2 Replacement interval

Track hours for each of the four positions. Replace at the site's validated UV-dose limit and no later than the approved useful-life policy. A lamp may still glow after useful UV-C output has declined. Replace early for blackened ends, unstable starting, flicker, damaged glass/base/pins, ballast or socket overheating, or failed dose verification.

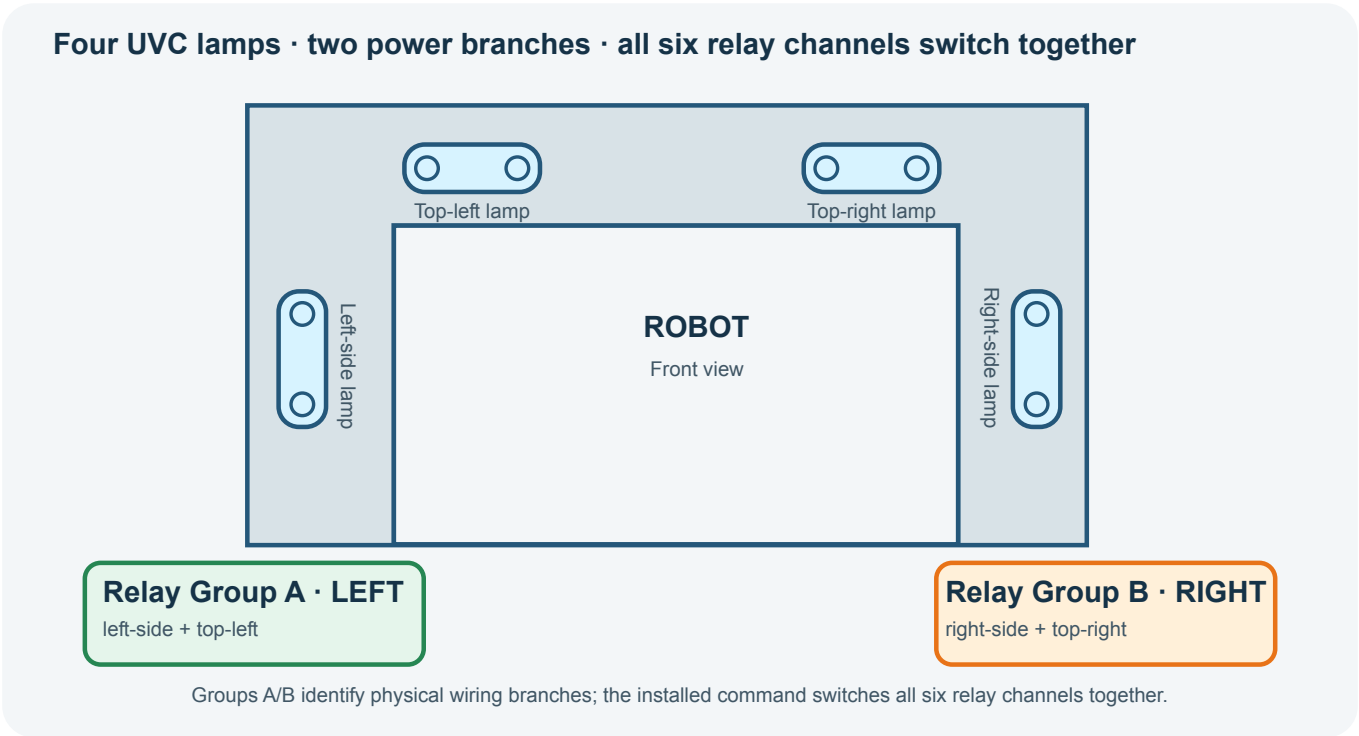


Figure 3.11: Four lamps total: one left-side, one right-side, and two on top. Group A is the left-side pair and Group B is the right-side pair.

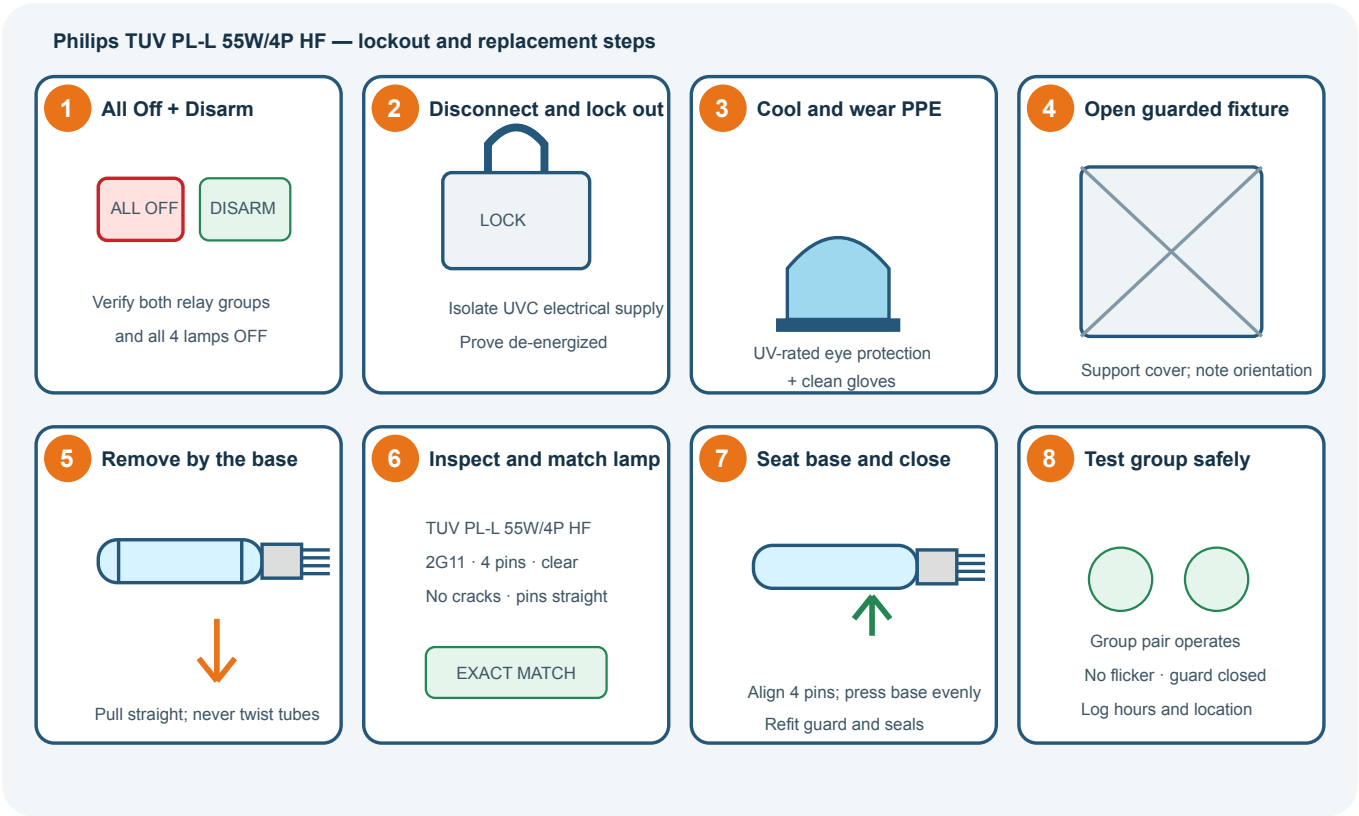


Figure 3.12: Each picture number matches the replacement instruction.

3.13 Recycle used UVC bulbs

The Philips TUV PL-L lamp contains mercury. Do not place it in ordinary trash, incinerate it, intentionally break it, or use a lamp crusher. Follow the site environmental procedure and all applicable federal, state and local requirements.

Used mercury-containing UVC lamp — safe recycling workflow



Figure 3.13: Keep every removed lamp intact, closed in a protective container, labelled and traceable until an approved recycler accepts it.

- 1. Isolate before removal.** Complete the Off, physical verification, Disarm, disconnect and lockout steps in Replace a UVC bulb. Let the lamp cool and wear clean gloves and safety glasses.
- 2. Handle by the base.** Remove the lamp without twisting or stressing the glass. Never tape lamps together or bundle them with rubber bands.
- 3. Package immediately.** Place the intact lamp in its original sleeve/carton or a closed, structurally sound rigid container sized to prevent movement and breakage. Add compatible cushioning without pressing on the twin tubes.
- 4. Label the container.** Mark it “Universal Waste—Lamp(s),” “Waste Lamp(s),” or “Used Lamp(s),” as the site program requires. Add the accumulation start date, lamp model, quantity and responsible work area.
- 5. Store safely.** Keep the closed container dry, protected from traffic, weather and impact, and separate from new lamps. Do not leave removed lamps loose on the robot, in a service vehicle or in the field.
- 6. Use an approved recycler.** The site environmental/EHS contact must select a mercury-lamp recycler, collection agency or compliant mail-back/pickup service and confirm transport and acceptance requirements before shipment.
- 7. Keep records.** Record robot ID, lamp position/group, removal date, operating hours, reason for removal, container ID, quantity, recycler and pickup/shipment record.
- 8. If a lamp breaks.** Stop work, keep people and animals away, prevent fragments from spreading, and follow the site's mercury-lamp spill procedure and current environmental-agency guidance. Do not use a normal vacuum or dry broom. Package and label all fragments and cleanup materials for the approved handler.

Environmental guidance: U.S. EPA—recycling mercury-containing bulbs, U.S. EPA—business lamp-recycling program, and U.S. EPA—broken mercury-containing lamp cleanup. State or local rules may be more stringent.

3.14 UVC robot travel-speed calculator

Engineering estimate only—not a treatment prescription. Plant species, growth stage, wavelength, surface angle, shadowing, reflector geometry, lamp warming, aging, and repeated passes change the delivered dose and can damage plants. A field engineer must measure irradiance at the plant with a calibrated 253.7 nm UV-C radiometer and validate the dose in a controlled plant trial before approving a speed.

Philips TUV PL-L 55W/4P HF — dimensional reference

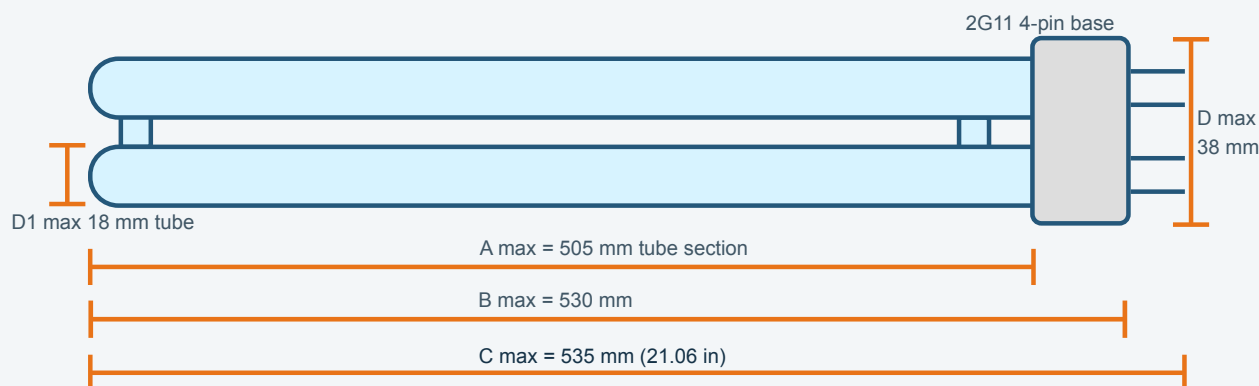


Figure 3.14: Maximum dimensions: C overall 535 mm; B 530 mm; A tube section 505 mm; overall width D 38 mm; individual tube diameter D1 18 mm.

UVC speed calculator — estimate, measure, then validate

1 Enter real geometry

- target plant dose
- active lamps / relay group
- plant distance
- UV output and aging
- exposure length
- measured utilization factor

2 Calculate estimate

$$I = N \times P \times \text{age} \times U \div (4\pi r^2)$$

$$\text{time} = \text{target dose} \div I$$

$$\text{speed} = \text{exposure length} \div \text{time}$$

Estimated speed and time

3 Measure and validate

- Place UV-C radiometer at plant
- Measure each group and position
- Replace estimate with measured I
- Run a controlled plant trial
- Approve speed in site SOP

Figure 3.15: Use calculated irradiance only for preliminary engineering. Enter measured irradiance whenever available.

Target dose at plant

mJ/cm² — must come from the approved treatment protocol Active lamps reaching the same plant

Usually 2 for one left/right relay group; use 4 only if all four illuminate that point UV-C output per lamp

W at 100 h; not the 55 W electrical input Plant-to-lamp distance

cm, shortest perpendicular distance Lamp output remaining

% — useful-life depreciation can reach 20% Geometry/utilization estimate

% of lamp UV-C estimated to reach the target; calibrate with a radiometer Effective exposure length along travel

cm of the measured UV footprint, not automatically the bulb length Measured irradiance at plant (preferred)

mW/cm² from calibrated 253.7 nm radiometer; overrides distance/utilization estimate

Calculate estimated speed Reset

Enter the approved dose and geometry, then calculate.

3.14.1 Calculator method

If measured irradiance is entered, the calculator uses it directly. Otherwise it uses a conservative point-source screening model:

$$I = N \times P_{uvc} \times \text{aging} \times \text{utilization} \div (4\pi r^2)$$

$$\text{exposure time} = \text{target dose} \div I$$

$$\text{robot speed} = \text{effective exposure length} \div \text{exposure time}$$

This model does not represent the twin tubes, reflector, plant canopy, lamp orientation, or left/right group overlap. It is unsuitable as final validation. At 9,000-hour useful life, the datasheet lists 20% depreciation, so an aging value of 80% represents that nominal endpoint before other losses.

3.14.2 Field validation procedure

1. Use the approved target dose for the specific plant, pathogen/pest, growth stage, and number of passes.
2. Place a calibrated 253.7 nm radiometer sensor at the nearest, farthest, shaded, and representative plant surfaces. Keep personnel outside the UV-C area and use remote readings where possible.
3. Operate Group A and Group B separately and record irradiance for left-side/top-left and right-side/top-right pairs. Confirm which lamps actually reach each plant location.
4. Enter the lowest relevant measured irradiance in the calculator. Use the measured longitudinal footprint for exposure length.
5. Run a controlled low-dose trial, inspect plants over the protocol's observation period, and confirm delivered dose with a dosimeter/radiometer.
6. Document approved speed, lamp hours, distance range, group selection, pass spacing, number of passes, and rejection limits in the field SOP.

3.15 UVC AC power: two inverters and four ballasts

The UVC system is split into two complete AC power branches. The robot's nominal 48 VDC system feeds one Victron inverter on the left and one on the right. Each inverter supplies 120 VAC to two relay-controlled ballast inputs. Each Keystone ballast drives exactly one Philips lamp.

120 VAC and hazardous lamp starting voltage are present. Only qualified electrical personnel may open inverter, relay, ballast, or fixture wiring. A 2G11 lamp is never connected directly to the inverter or relay output; its dedicated electronic ballast must remain between the 120 VAC relay port and lamp pins.

3.15.1 Victron inverter specifications — each side

Item	Victron Inverter 48/375 VE.Direct, 120 V
Quantity	2 total: one left, one right
Function	48 VDC battery power to pure sine-wave 120 VAC
AC outlet	NEMA 5-15R
Continuous output	375 W at 25°C; 300 W at 40°C
Time-limited / peak	450 W for 10 s cold start; 600 W for 2 s peak
Output	120 VAC $\pm 3\%$; 60 Hz $\pm 0.1\%$
48 V model DC input range	36.8–62.0 VDC
Default low shutdown / restart	37.2 V shutdown; 43.6 V restart/alarm
Charged detect	56.0 V default
Maximum efficiency	88% for 48/375
Zero-load power	9 W; approximately 2 W in default ECO mode
Environment	–40 to +65°C; derate 1.25%/°C above 40°C; maximum 95% non-condensing humidity
Enclosure	IP21; steel chassis/plastic cover

Item	Victron Inverter 48/375 VE.Direct, 120 V
Size / mass	86 × 165 × 260 mm; 3.0 kg
Protection	Short circuit, overload, high/low battery, overtemperature, AC on output, and excessive DC ripple

3.15.2 Keystone ballast specifications — each lamp

Terminology: the electrical component is a *ballast* (sometimes informally called a “blaster”). The installed model is KTEB-254HO-UV-PS.

Item	KTEB-254HO-UV-PS
Quantity	4 total: one ballast per UVC lamp
Type	T5HO electronic, programmed start, high power factor
Compatible lamp configuration used	One FT55W/2G11 lamp per ballast
Input	120–277 VAC ±10%, 50/60 Hz
One-lamp input at 120 V	Approximately 62 W and 0.52 A; power factor 0.96; maximum THD 15%; crest factor below 1.70
Minimum start temperature	0°F / –18°C
Maximum case temperature	194°F / 90°C
Features	Class P, Type 1 Outdoor listing; lamp end-of-life protection; Sound Rated A; FCC Part 18 Class A non-consumer limits
Standard case	Approximately 16.89 × 1.20 × 1.00 in; steel case; verify the exact suffix/nameplate installed
Wiring	The 120 VAC input of each ballast is connected to one relay output port. Use Keystone one-lamp diagram F1-11 on the lamp side. Unused yellow leads for one-lamp operation must be capped and insulated to 600 V exactly as specified.

3.15.3 Side loads and capacity

Branch	Loads	Nominal 120 VAC input	Engineering note
Group A · Left	2 ballasts × 1 lamp	About 124 W, 1.04 A total	Below the inverter's 375 W/25°C rating; validate start/inrush, temperature, wiring, and protection.
Group B · Right	2 ballasts × 1 lamp	About 124 W, 1.04 A total	Independent from left; one-side failure must not be mistaken for total UVC shutdown.
Both groups	4 ballasts × 1 lamp	About 248 W AC total	Split between two inverters. Approximate nominal DC demand is about 5.9 A total at 48 V and 88% efficiency, excluding cabling and transient losses.

Relay topology: each ballast's AC input is switched through its assigned relay output. Verify that every relay contact, connector, fuse, conductor, enclosure, spacing, and disconnect is rated for 120 VAC, ballast inrush, fault current, environment, and applicable code. Do not switch 120 VAC with a DC-only relay rating.

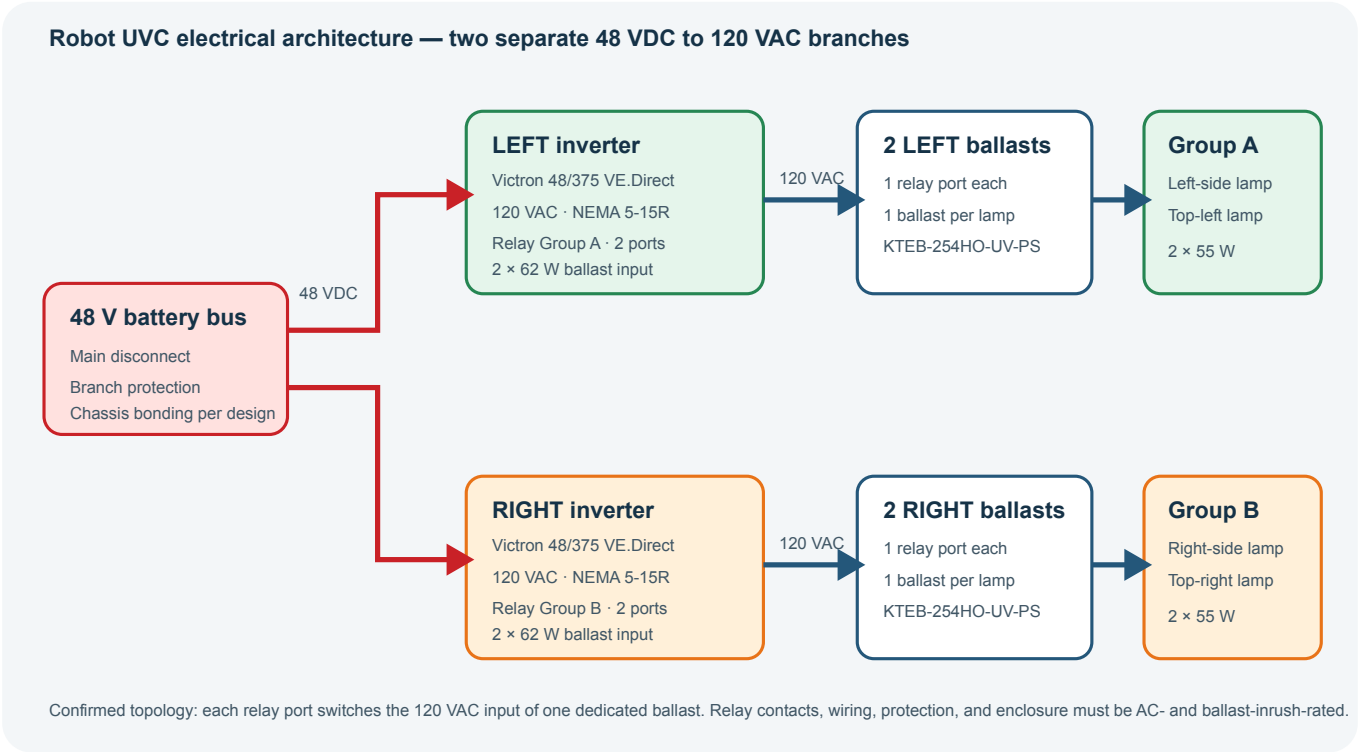


Figure 3.16: Actual chain: 48 VDC -> side inverter -> relay output port -> one ballast -> one lamp. Group A/Left has two branches; Group B/Right has two branches.

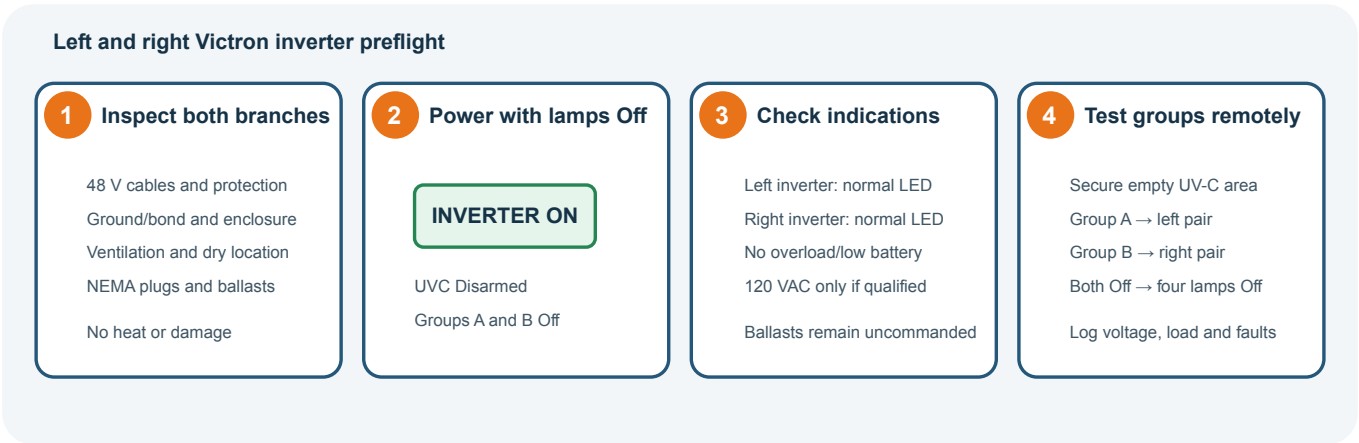


Figure 3.17: Test with the irradiation area empty and secured. Electrical measurements require qualified personnel and properly rated instruments.

3.15.4 Preflight and group test

1. With UVC locked out, inspect both 48 V inverter feeds, branch protection, grounding/bonding, ventilation, mounting, NEMA plugs, relay wiring, four ballasts, sockets, and lamp leads.
2. Restore inverter power while UVC remains Disarmed and all four relay ports Off. Confirm both inverters indicate normal operation and no low-battery, overload, ripple, or temperature alarm.
3. With the area secured, test one relay port at a time: Left-side, Top-left, Right-side, Top-right. Verify exactly one matching ballast/lamp responds to each port.
4. Test Group A: only left-side and top-left operate. Test Group B: only right-side and top-right operate.
5. Command both groups Off, Disarm, and verify four physical lamps Off. Test master-heartbeat loss and CAN emergency shutdown.
6. Record 48 V input, AC output, inverter/ballast temperature, branch current, relay port, lamp position, and alarms under steady operation.

3.15.5 Inverter and ballast troubleshooting

Symptom	Checks
Both lamps on one side fail	Keep area locked out. Check that side's inverter LED/alarm, 48 V branch protection, disconnect, NEMA connection, 120 VAC output, and group control.
One lamp fails; its paired lamp works	The inverter is probably supplying the side. Check that lamp's relay port, ballast AC input, ballast, socket, lamp pins, and one-lamp wiring.
Inverter overloads at start	Check ballast faults, shorted wiring, actual inrush, temperature derating, and branch load. Do not increase protective-device ratings without an engineered review.
Inverter cycles in ECO mode	The electronic ballast/load detection may not remain stable. Qualified staff must validate ECO thresholds or use the approved always-On inverter mode; never defeat UVC safety controls.
Lamp flickers or blackens	Stop the group. Check lamp age/pins/socket and ballast wiring/case temperature; replace damaged components and verify UV output.
Relay feedback Off but lamp remains on	Emergency: use physical UVC disconnect. Suspect welded/miswired relay contact or alternate feed; lock out and service before reuse.

Sources: Victron VE.Direct 120 V technical specifications, Victron manual, Keystone ballast catalog, and KTEB-254HO-UV-PS specification sheet.

Component · RTK GPS base and rover

4.1 RTK GPS: component overview

The fixed **base** knows its own position and sends correction messages. The **rover** is mounted on the robot and combines satellite signals with those corrections. RTK gives the navigation system a much more precise position than ordinary GPS.

4.1.1 Features

- Base survey-in or stored fixed coordinates.
- RTCM correction output by LoRa and optional NTRIP.
- Robot rover correction input by LoRa or NTRIP.
- Local Wi-Fi status/configuration pages.
- Fix, satellite, accuracy, correction, and battery information.
- Rover position/status output over CAN.

RTK and RC are separate LoRa links. Configure the base and RTK receiver to match each other. Configure the controller and robot RC radios to match each other. Do not assign the RTK link the same settings as the RC link; RC transmissions can interfere with RTK corrections.

4.2 GPS base and rover specifications

Item	Base	Robot rover
Local hostname	<code>http://mars_gnss_base.local</code>	<code>http://mars_gnss_rover.local</code>
Default AP name	<code>mars_gnss_base_XXXXXX</code>	<code>mars_gnss_rover_XXXXXX</code>
Factory AP password	<code>alphaagtech</code>	
Direct AP address	<code>http://192.168.10.1</code>	
CANopen firmware default	21 (0x15), configurable	20 (0x14), configurable
CAN bit rate	500 kbit/s for this robot system ; changes take effect after restart	
Factory GNSS mode	Self Survey; 60 s minimum and 2.0 m target in firmware	Dual GNSS; 5 Hz update
Factory RTK LoRa	915 MHz, 500 kHz, SF6; base and rover must match and must be separated from RC LoRa	
Correction rate/path	RTCM 1 Hz; LoRa; optional local/online NTRIP and serial output	LoRa; optional serial RTCM or NTRIP input
NTRIP default	Disabled; port 2101; port 80 reserved for webpage	
Serial default	RTCM output disabled; saved baud 115200	RTCM input disabled; NMEA output disabled; saved baud 115200

Values displayed by the live webpage are the saved device settings and take precedence over factory defaults. The screenshots in this manual are real UI renders without attached GNSS hardware, so their no-data readings and placeholder values are illustrations only.

4.3 GPS physical and data connections

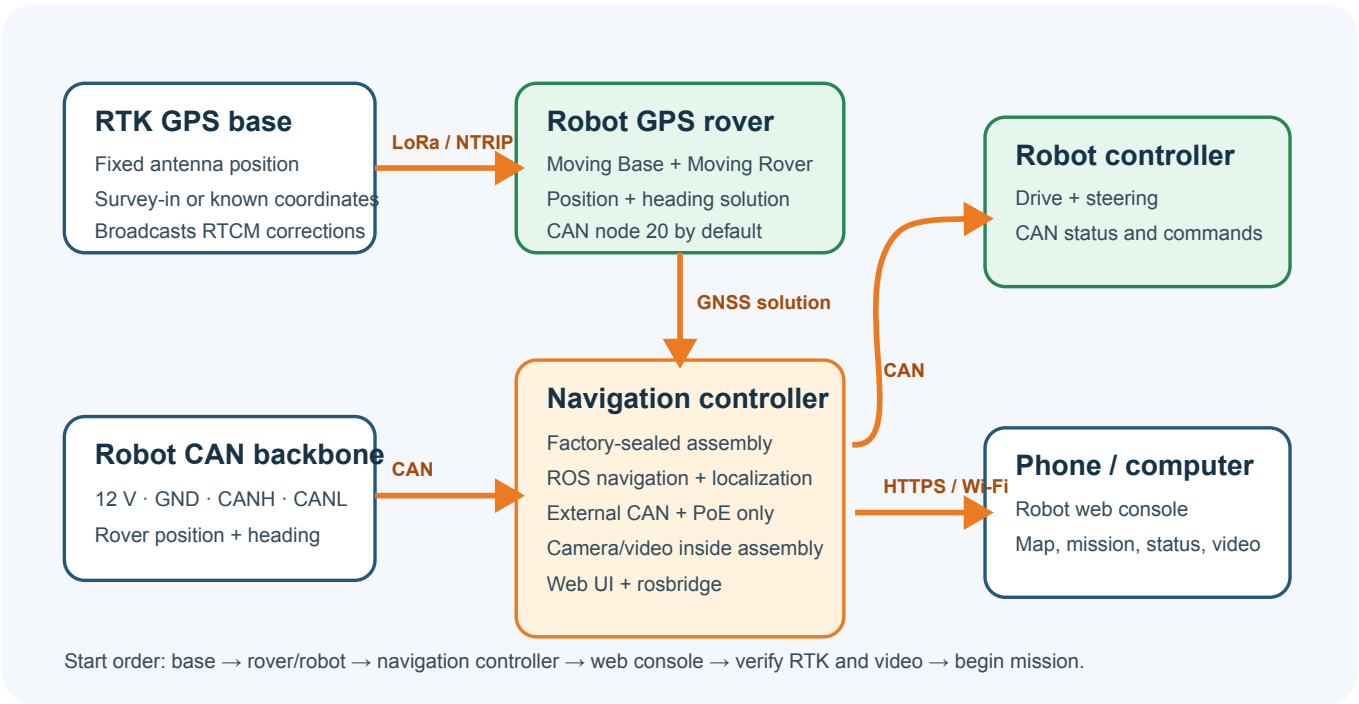


Figure 4.1: The field base distributes RTCM by RTK LoRa or NTRIP. The robot rover contains two GNSS receivers/antenna positions for position and heading, then publishes the computed robot solution on CAN.

Connection	Base	Robot rover
GNSS antenna(s)	One surveyed/stable antenna point with clear sky; never move after survey	Moving Base and Moving Rover antennas rigidly mounted with clear sky and measured offsets
Antenna coordinate frame	Not applicable	Offsets in centimeters: +X forward, +Y left, measured from robot reference point
RTK LoRa antenna	Correction transmitter	Correction receiver; same frequency, bandwidth and SF as base
Configuration Wi-Fi AP	Always on; local page and optional local NTRIP service	Always on; local page only; hostname <code>mars_gnss_rover.local</code>
Internet/correction Wi-Fi	Optional station link for online NTRIP caster output	Separate optional station link to base AP or Internet router for NTRIP input
Serial	Optional RTCM output, 4800–921600 baud	Optional RTCM input or GPGL output; do not enable serial RTCM input and NMEA output together
CAN	Node 21 if connected to a CAN system; normally remote from robot	Power-hub 5 V CAN port: pin 1 = 5 V supply, pin 2 = CANH, pin 3 = CANL, pin 4 = GND; default node 20; 500 kbit/s

Rover supply: connect the rover only to a labelled power-hub 5 V CAN port. A 12 V CAN port has the same signal purpose but the wrong supply voltage and can damage the rover. See the power-hub connector and pinout schedule.

Radio separation: the base/rover RTK LoRa link must match itself, and the controller/robot RC LoRa link must match itself. Do not use the same frequency/settings for both links. Record the approved channel plan.

4.4 GPS base webpage and setup

4.4.1 Open the base page

1. Mount the base antenna on a rigid point with the clearest available sky. Attach GNSS and RTK LoRa antennas before power.
2. Join `mars_gnss_base_XXXXXX` using `alphaagtech`, unless site credentials were changed.

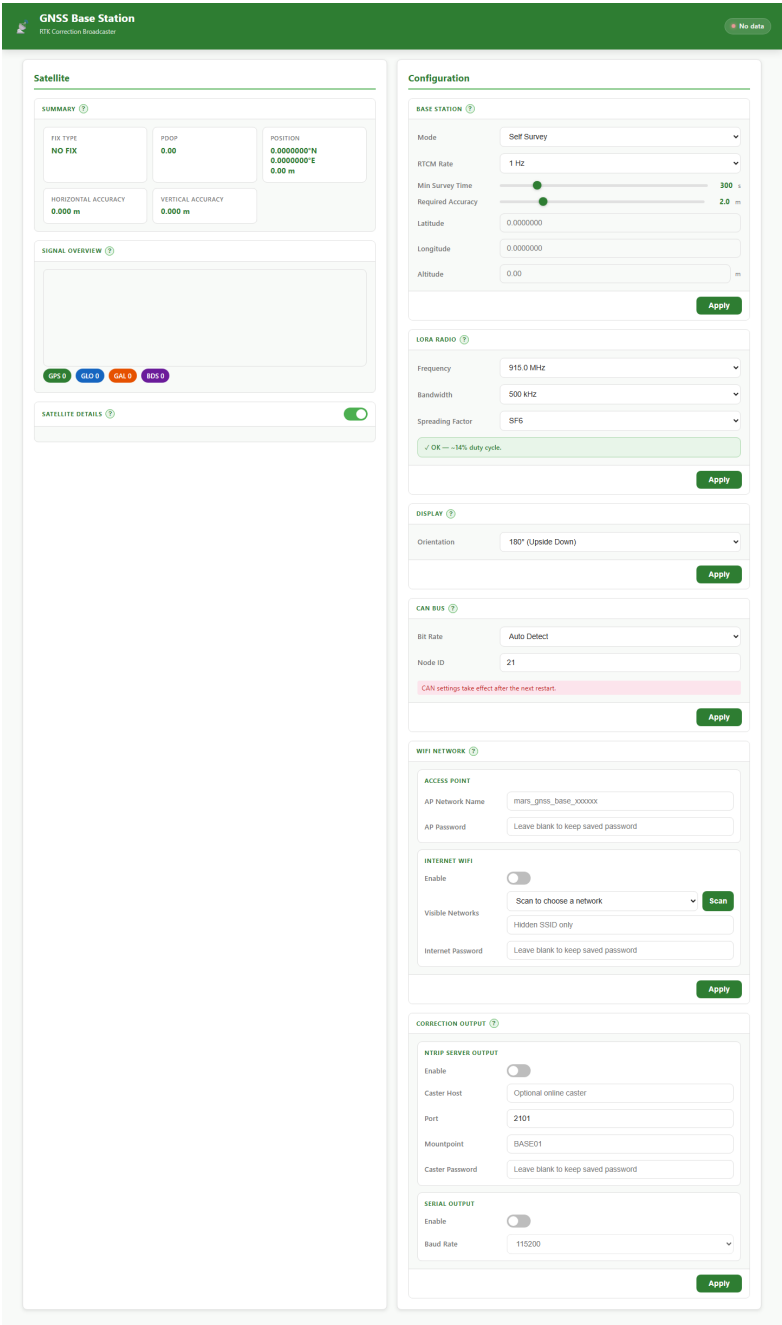


Figure 4.2: GNSS Base Station webpage. On a live base, satellite and survey data replace the no-data placeholders.

4. Select **Apply**, then confirm the displayed position matches the record before allowing correction use.

4.4.4 Configure correction outputs

1. In **LoRa Radio**, set frequency, bandwidth, and spreading factor to the approved RTK plan. The page's bandwidth-status line must say the selected link can carry the RTCM rate. Select **Apply**.
2. For a rover connected to the base AP by Wi-Fi, enable **NTRIP Server Output**, leave Caster Host blank, and set port/mountpoint. The rover uses the base AP address, normally 192.168.10.1.
3. For an online caster, first enable **Internet WiFi**, scan/select the router and apply. Then enable NTRIP output and enter caster host without `http://`, port, mountpoint, and caster password.
4. For wired RTCM, enable **Serial Output** and select the receiving device's matching baud rate.
5. Do not use port 80 for NTRIP. Watch Stream Status and verify corrections at the rover rather than assuming Apply succeeded.

4.4.5 Other base controls

Panel	Instruction
Display	Select the orientation that makes the onboard display upright; Apply.
CAN Bus	Select 500 kbit/s and a unique node ID. Default is 21. Auto Detect is a diagnostic option, not the commissioned robot-system setting. Restart is required before CAN changes take effect.
WiFi Access Point	Change AP name/password only during controlled service. Record the new credentials before Apply; the phone may disconnect when they change.
Satellite Details	Turn on for engineering diagnosis of constellation, SNR, elevation/azimuth, correction and ephemeris data; turn off for a simpler operator view.

4.5 GPS rover webpage and setup

4.5.1 Open and read the rover page

1. Keep the robot in Hold. Confirm both GNSS antennas, RTK LoRa antenna, and CAN cable are secure before power.
2. Join `mars_gnss_rover_XXXXXX` with `alphaagtech`, unless site credentials were changed.
3. Open `http://mars_gnss_rover.local`. Use `http://192.168.10.1` only as the direct-AP fallback.
4. Check the top solution badge and the **Moving Base** and **Moving Rover** summaries: satellites, PDOP, fix type, latitude/longitude, RTCM state, correction age, signal charts, and constellation counts.
5. Do not start Auto navigation unless the solution badge is OK, both receivers are plausible, corrections are fresh, and the site-required fix/accuracy is met.

4.5.2 GNSS mode and antenna geometry

1. Select **Dual GNSS** for the normal two-receiver position/heading calculation, or **Moving Baseline** only when the installed system and field procedure require it.
2. Set update frequency from 1–20 Hz. Factory default is 5 Hz; higher rates increase processing and serial bandwidth requirements.
3. Measure the center of each antenna relative to the robot reference point. Enter Moving Base X/Y and Moving Rover X/Y in centimeters, where +X is forward and +Y is left.
4. Check the diagram direction and signs. The calculation rejects invalid antenna geometry and a baseline under 20 cm; the webpage also warns below 10 cm at save time.
5. Select **Save Configuration**. After a mode/geometry change, keep the robot stationary and verify heading against a known direction before navigation.

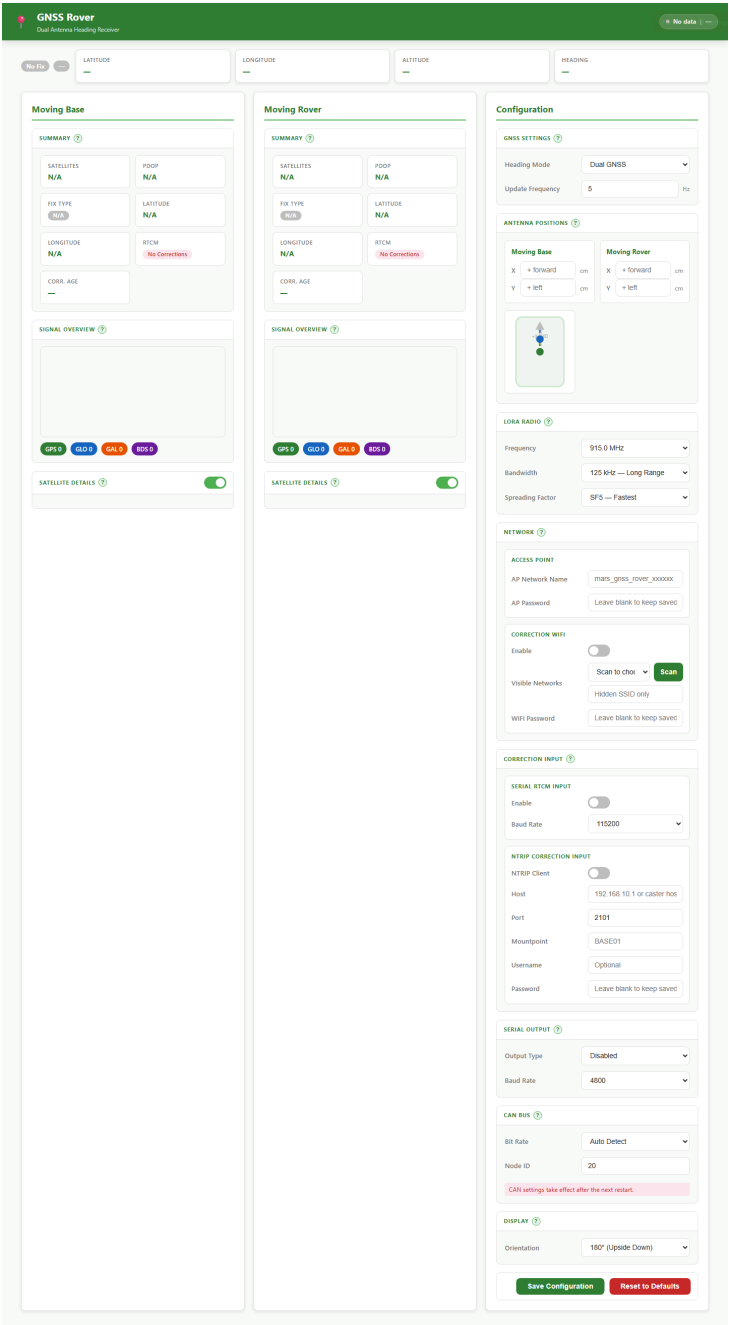


Figure 4.3: GNSS Rover webpage. The two satellite columns correspond to the rover’s Moving Base and Moving Rover receivers.

4.5.3 Select and configure correction input

Method	Setup	Expected status
RTK LoRa	Match base frequency, bandwidth and SF exactly; leave the approved RC/RTK separation intact.	LoRa corrections; correction age remains current.
Base local NTRIP	Enable Correction WiFi; select the base AP; use host 192.168.10.1, base port/mountpoint and credentials; enable NTRIP Client.	NTRIP corrections.
Online NTRIP caster	Enable Correction WiFi and join an Internet router; enter caster host, port, mountpoint, username and password; enable NTRIP Client.	NTRIP connects and corrections become fresh.
Serial RTCM	Disable NMEA Serial Output, enable Serial RTCM Input, and match 4800–921600 baud to the sender.	Serial corrections.

Use one documented correction method for the job. The firmware can receive through several paths; avoid an ambiguous setup where unrelated streams compete. The status field identifies the most recently active path.

4.5.4 CAN, serial output, display and network

1. Leave NMEA **Serial Output** disabled unless another device explicitly requires GPBGA. If enabled, choose a baud rate high enough for the GNSS update frequency; the page warns when it is too low.
2. For **CAN Bus**, select **500 kbit/s** and a unique node ID. Default rover ID is 20. Auto Detect is for diagnosis only. Save, then restart before expecting CAN changes.
3. Set Display Orientation so the onboard display is upright.
4. The rover configuration AP is always available. Correction WiFi is a separate client connection; changing one does not automatically change the other.
5. Use **Reset to Defaults** only under an authorized recovery procedure. It erases saved radio, antenna, correction, network, CAN, and display settings and requires complete recommissioning.

4.5.5 Meaning of rover solution messages

Message	Meaning / response
OK	Geometry calculation succeeded; still verify RTK fix, accuracy, correction age and CAN delivery.
No fix	A receiver lacks a usable GNSS fix. Improve sky view and wait.
Invalid coords	Input position is invalid; inspect GNSS data and configuration.
UTM zone split	The two receiver coordinates fall in different UTM zones; inspect coordinates and placement.
Baseline <20cm	Measured antenna separation is too short for the solution.
Antenna cfg	Saved offsets are unset, too close, or inconsistent.
RelPos invalid / missing	Moving-baseline relative-position data is invalid or absent; check mode, receivers, corrections and wiring.

4.6 GPS base and rover field workflow

1. **Place and identify the base.** Use the recorded base point and antenna height, or establish a new rigid point with clear sky.

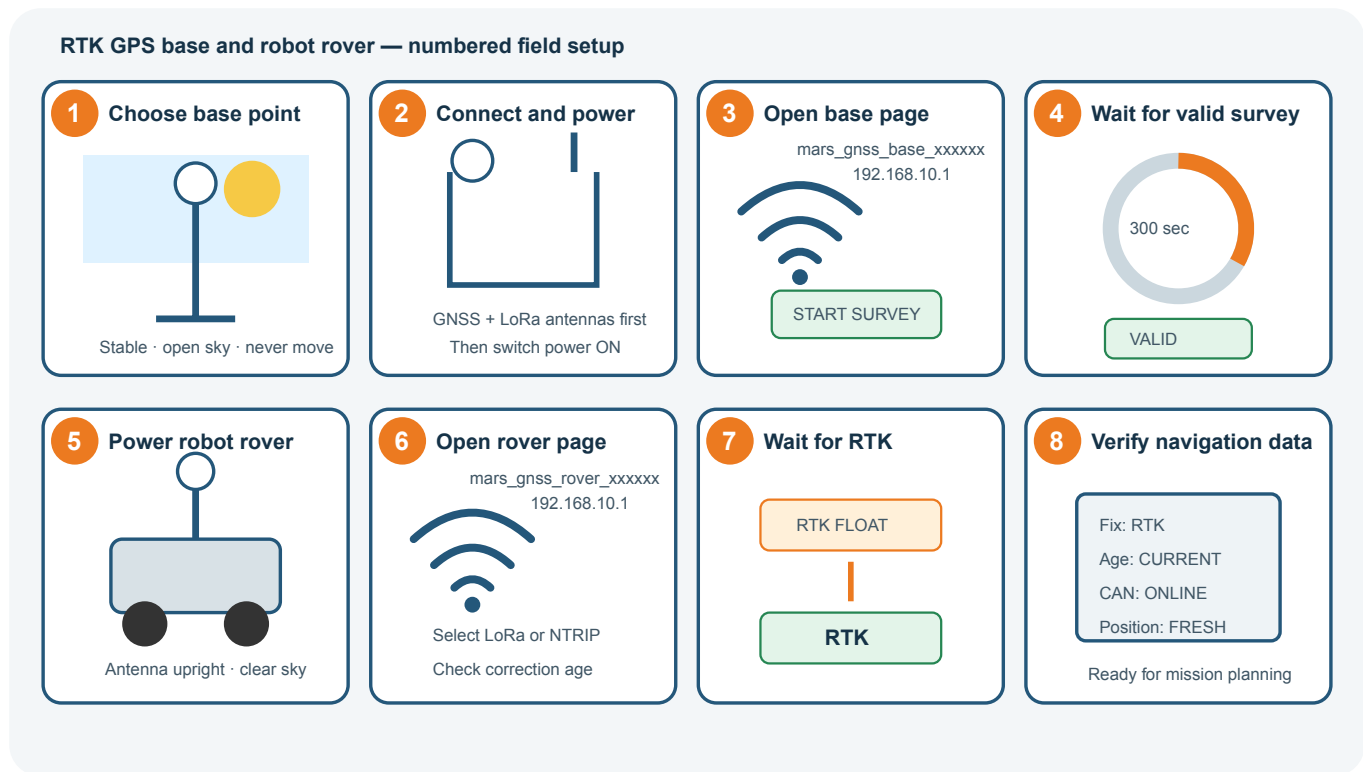


Figure 4.4: Use this field sequence after the detailed base and rover webpage setup above.

2. **Power the base.** Attach antennas first. Open `mars_gnss_base.local` and verify the saved mode, position, corrections, and RTK radio plan.
3. **Complete position setup.** Finish Self Survey or verify Fixed Location. Do not move the antenna after this step.
4. **Verify correction output.** Confirm LoRa, NTRIP, or serial output is active using the selected field method.
5. **Power the robot rover.** Check the two GNSS antenna mounts/offsets, LoRa antenna, and CAN connection.
6. **Open `mars_gnss_rover.local`.** Verify mode, both receiver summaries, antenna geometry, chosen correction input, and fresh correction age.
7. **Verify the robot solution.** Require the site-approved RTK fix and accuracy, stable heading, OK solution status, and correct position/heading while the robot is stationary.
8. **Verify delivery.** Confirm rover node 20 and fresh GPS data at the robot and navigation console. Only then allow Auto; continue monitoring correction age and solution quality.

4.7 GPS troubleshooting

Problem/status	What to inspect	Operator response
<code>mars_gnss_base.local</code> or <code>mars_gnss_rover.local</code> does not open	Wrong Wi-Fi, mDNS unsupported, changed AP credentials	Stay joined to the device AP and use 192.168.10.1. Confirm the page title before editing.
Base survey never completes	Poor sky, antenna movement, multipath, damaged cable, target too strict	Keep antenna rigid; improve sky view; inspect hardware; restart the approved survey.
Base restarts at an unexpected position	Wrong fixed record or survey point moved	Stop Auto. Compare displayed coordinates/ECEF-derived position with site record; recommission the base.
Rover says No fix	One/both receivers lack usable GNSS	Move to open sky, inspect both antennas/power, and wait.

Problem/status	What to inspect	Operator response
Rover stays RTK Float or corrections age	Base invalid, LoRa mismatch/interference, NTRIP/Internet failure, multipath	Check base stream first; verify chosen correction path and radio separation; improve sky view.
NTRIP waiting for WiFi	Correction WiFi disabled/not connected	Enable it, scan/select the correct network, apply credentials, and recheck status.
NTRIP missing host or mountpoint	Incomplete client settings	Enter the exact host, port and mountpoint; do not include <code>http://</code> .
Baseline <20cm / Antenna cfg	Offsets or physical separation wrong	Hold the robot; measure antennas; correct X/Y signs/units; save and verify heading.
RelPos invalid or missing	Moving Baseline mode lacks valid relative position	Verify mode, receiver operation, corrections and antenna geometry.
Good rover page, no robot/navigation position	Wrong 500 kbit/s setting, node ID, wiring or navigation path	Keep Hold; verify CAN port pinout, live rover node, 500 kbit/s , restart after CAN changes, heartbeats and navigation topics.
CAN setting changed but no effect	Restart not completed	Record the change, safely restart the device, then verify the live node and 500 kbit/s setting.

Component · Automatic navigation controller

5.1 Navigation controller: component overview

The navigation controller uses corrected GPS position and an operator-planned route to command the robot. It is made of a **Raspberry Pi 5 computer**, a **Pi camera connected to a Raspberry Pi 5 CSI port**, and a **PoE module that provides its power**. The robot's battery-powered PoE switch supplies the controller's Ethernet/PoE connection. The operator uses a browser to draw the safe boundary, create waypoints, upload the mission, start, pause, resume, and monitor.

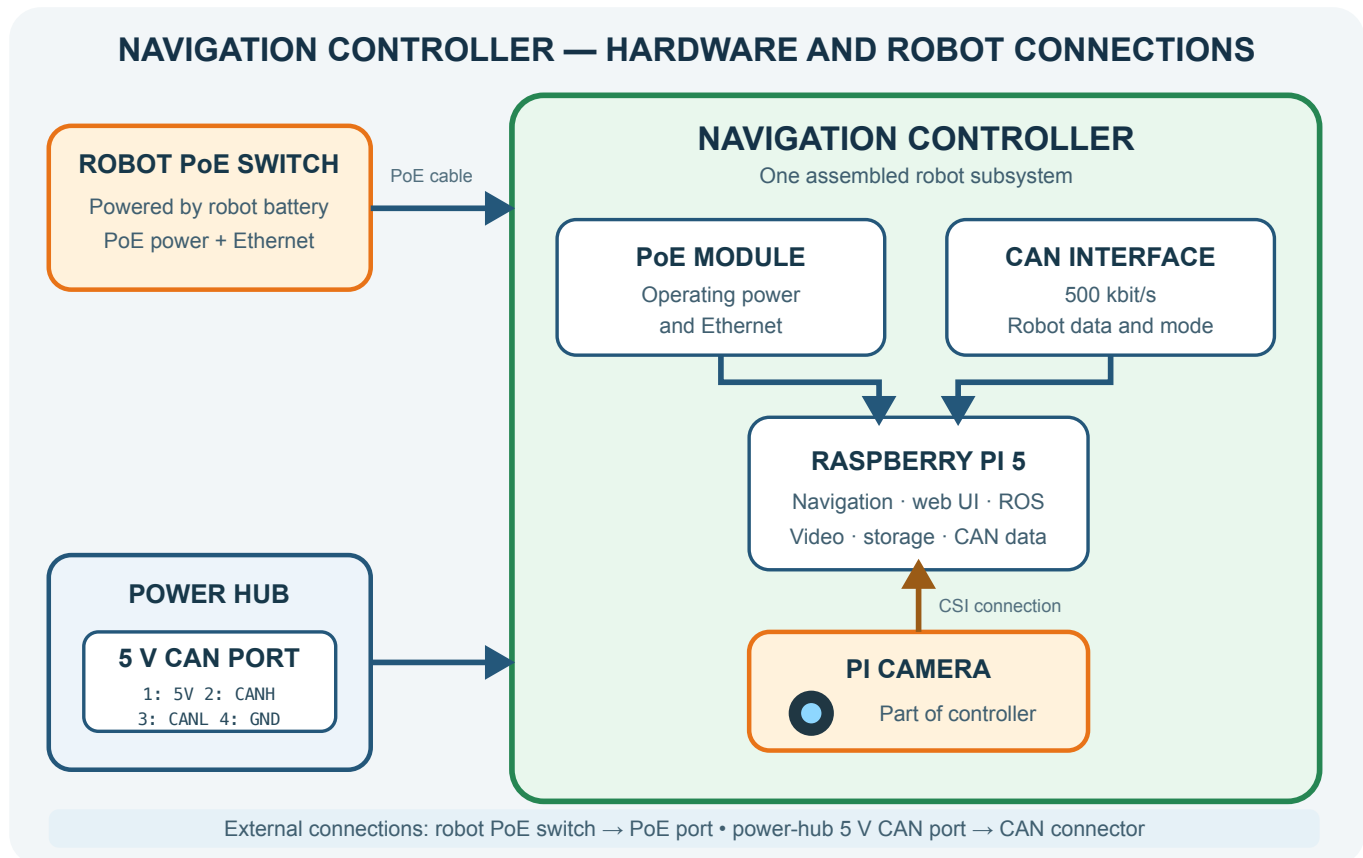


Figure 5.1: The navigation controller includes its Pi camera and connects to the robot through PoE Ethernet and the power hub's 5 V CAN port.

5.1.1 Features

- ROS Noetic navigation and dual-GNSS localization.
- CAN communication with robot controller.
- Map, waypoint, mission, geofence, status, and photo controls.
- CSI-connected Pi camera H.264 live video through MediaMTX WebRTC.
- PoE network and power connection from the robot PoE switch.
- Mission pause on excessive localization uncertainty.
- Geofence start refusal and automatic pause outside boundary.
- Automatic navigation-service startup after the navigation controller boots; operators do not manually launch the normal service stack.

- System monitoring, rosbridge, nginx web access, and managed service recovery.

5.2 Navigation-controller specifications

Item	Specification
Computer	Raspberry Pi 5
Camera	Pi camera connected to a Raspberry Pi 5 CSI port
Power/network module	PoE module supplied by the robot PoE switch
PoE switch source	Robot battery power through the installed protected supply path
Software	camera_catkin_ws, video-stream branch
Production service	camera.service
Main launch file	launch/camera.launch
Expected Ethernet	eth0 with IPv4 address
Operator-device network	Wi-Fi connection to mars_XXXXXX for the local robot webpage plus simultaneous cellular Internet for Google map tiles
Robot bus	can0
ROS bridge	Local port 9090; browser secure proxy through nginx
Camera source	RTSP 127.0.0.1:8554/cam
Browser video	WebRTC/WHEP at /cam/whep; media UDP 8189
Default geofence buffer	2 m outward

5.3 Navigation-controller external connections

Navigation-controller connections: connect the labeled PoE Ethernet cable to the robot PoE switch and connect the installed CAN harness to the assigned power-hub 5 V CAN port. The Pi camera is already part of the assembled controller.

External connection	Purpose	Operator check
Robot PoE switch -> navigation-controller PoE port	Operating power and Ethernet	Correct labeled PoE port; cable latched and undamaged; switch PoE/link indication present
Power-hub 5 V CAN port -> navigation-controller CAN connector	Robot mode/status and Auto commands	Installed harness connected to the labeled 5 V CAN port; pinout 5 V, CANH, CANL and GND; 500 kbit/s; correct termination; harness unmodified
Camera window/lens	Live view and recordings	Clean, unobstructed, secure, and aimed correctly
Enclosure/cooling openings	Protects the assembled electronics	Closed, dry, securely mounted, airflow clear

5.3.1 External connection and startup procedure

1. Keep the robot in Hold, UVC Off/Disarmed, contactor open, and power isolated.
2. Inspect the sealed navigation-controller enclosure. Do not open it or disturb its internal Pi camera, CSI ribbon, Raspberry Pi 5, or PoE module.
3. Connect the labeled PoE Ethernet cable from the robot PoE switch to the navigation-controller external PoE port. This is the controller's operating-power path. Route it away from wheels, steering joints, lamp fixtures, ballast leads, and sharp edges.
4. Connect the installed CAN harness to its assigned power-hub 5 V CAN port. Verify the port pinout—1: 5 V, 2: CANH, 3: CANL, 4: GND—and confirm the connector is fully seated and the harness has not been altered. See the power-hub port schedule.

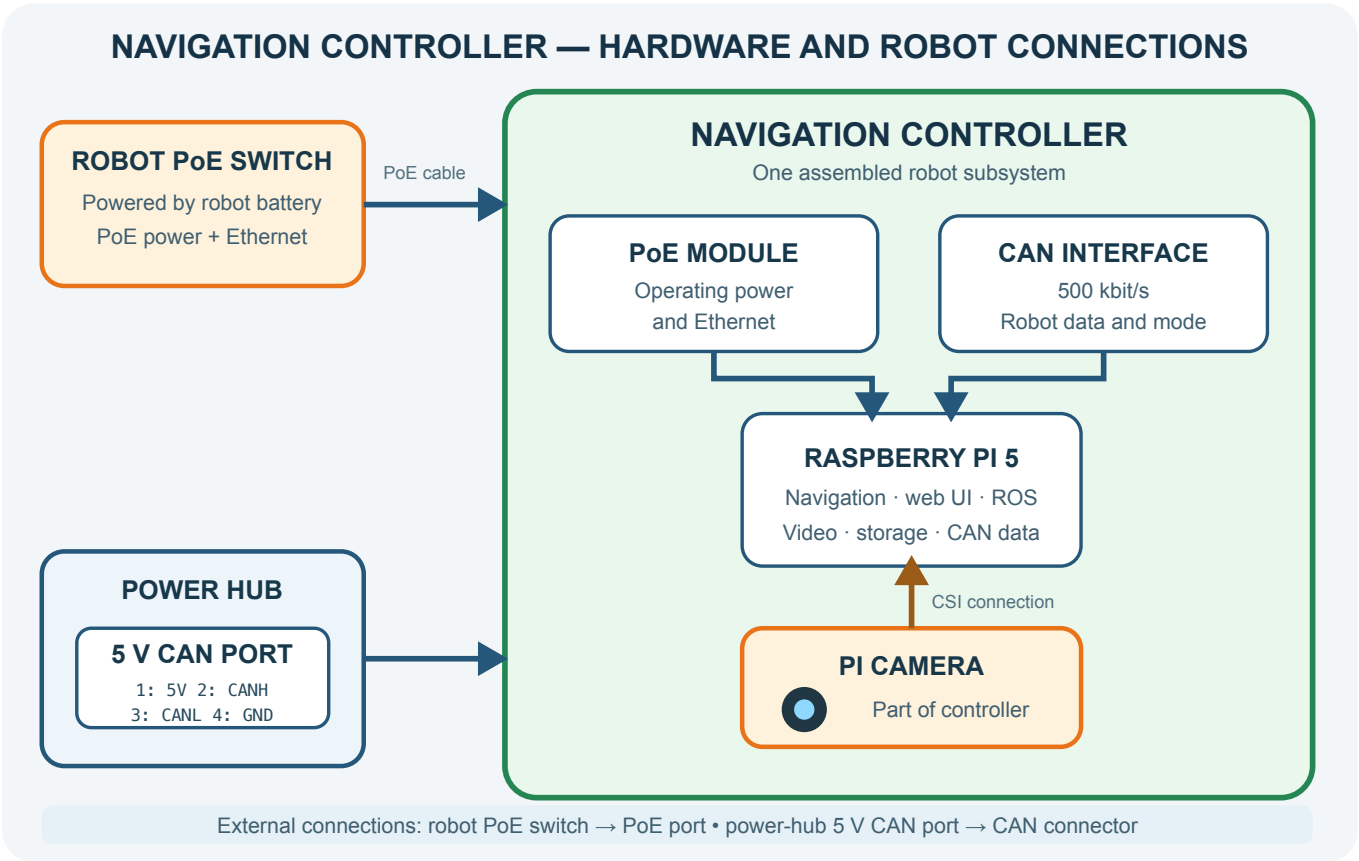


Figure 5.2: The Pi camera is part of the assembled navigation controller. The controller’s external robot connections are PoE Ethernet and the power hub’s 5 V CAN port.

5. Secure both cables with strain relief, then restore robot power according to the startup procedure.
6. Allow the navigation controller to finish booting. Its navigation, web, ROS, camera/video, and supporting services start automatically; do not manually start the normal service stack.
7. Confirm the webpage becomes available, then verify PoE/link indicators, ROS Connected, live video, CAN status, and current RTK position. If the page or a service does not become ready, keep Hold and use the troubleshooting procedure.
8. If any check fails, return to Hold, isolate power, and correct the external connection. Internal enclosure service is for authorized technicians only.

5.4 Automatic navigation operation manual

Do not navigate from video alone. Confirm RTK, map position, robot health, the physical route, and emergency-stop coverage. Keep UVC Off during first route validation.

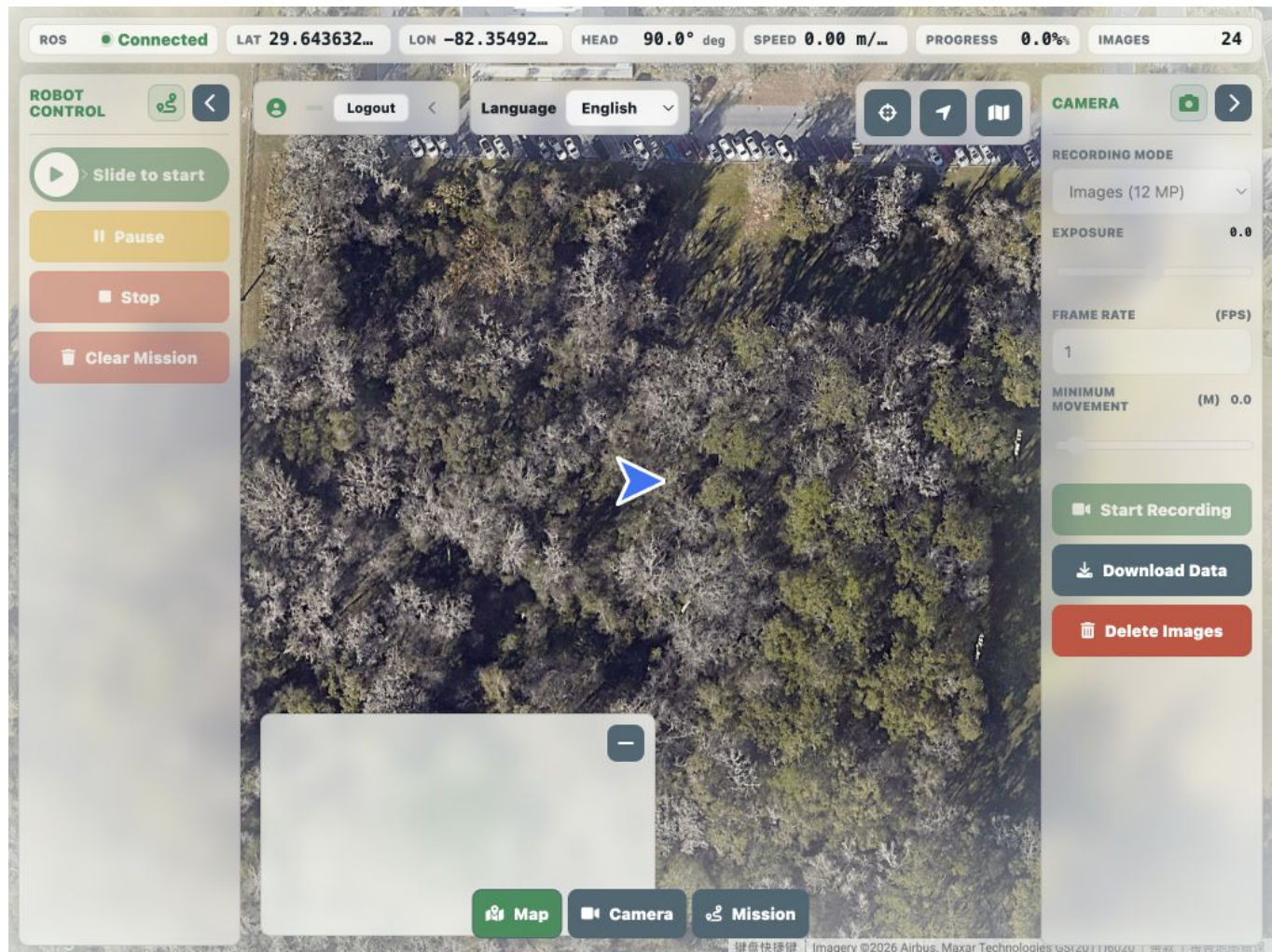


Figure 5.3: Map view: monitor position and route while Robot Control and Camera controls remain available.

Current web-interface behavior: these instructions were verified against the navigation workspace's pinned web release. The console is designed for landscape orientation. On a phone or tablet, rotate the device before operating.

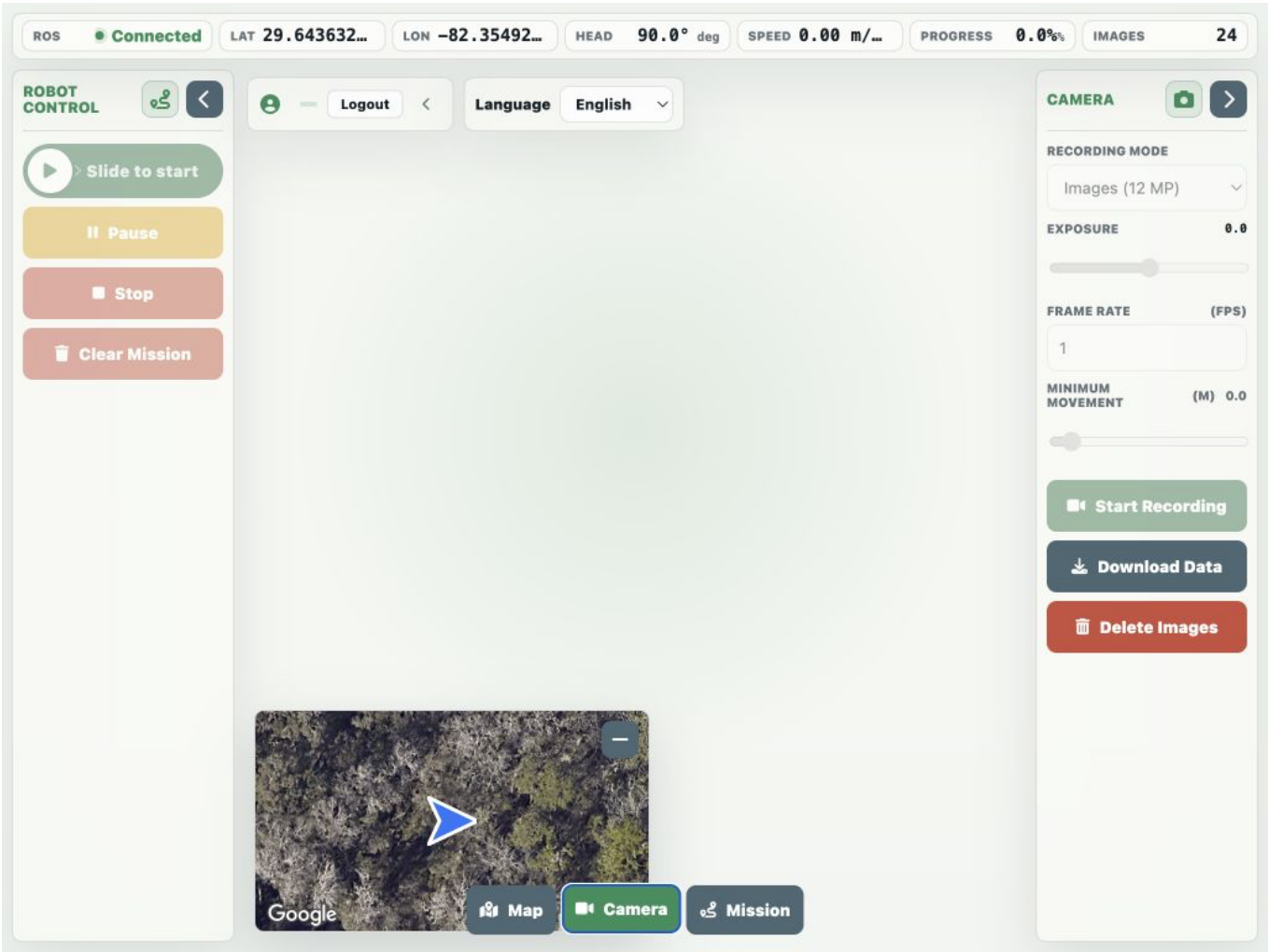


Figure 5.4: Camera view: inspect the live video workspace while retaining Robot Control and recording controls.

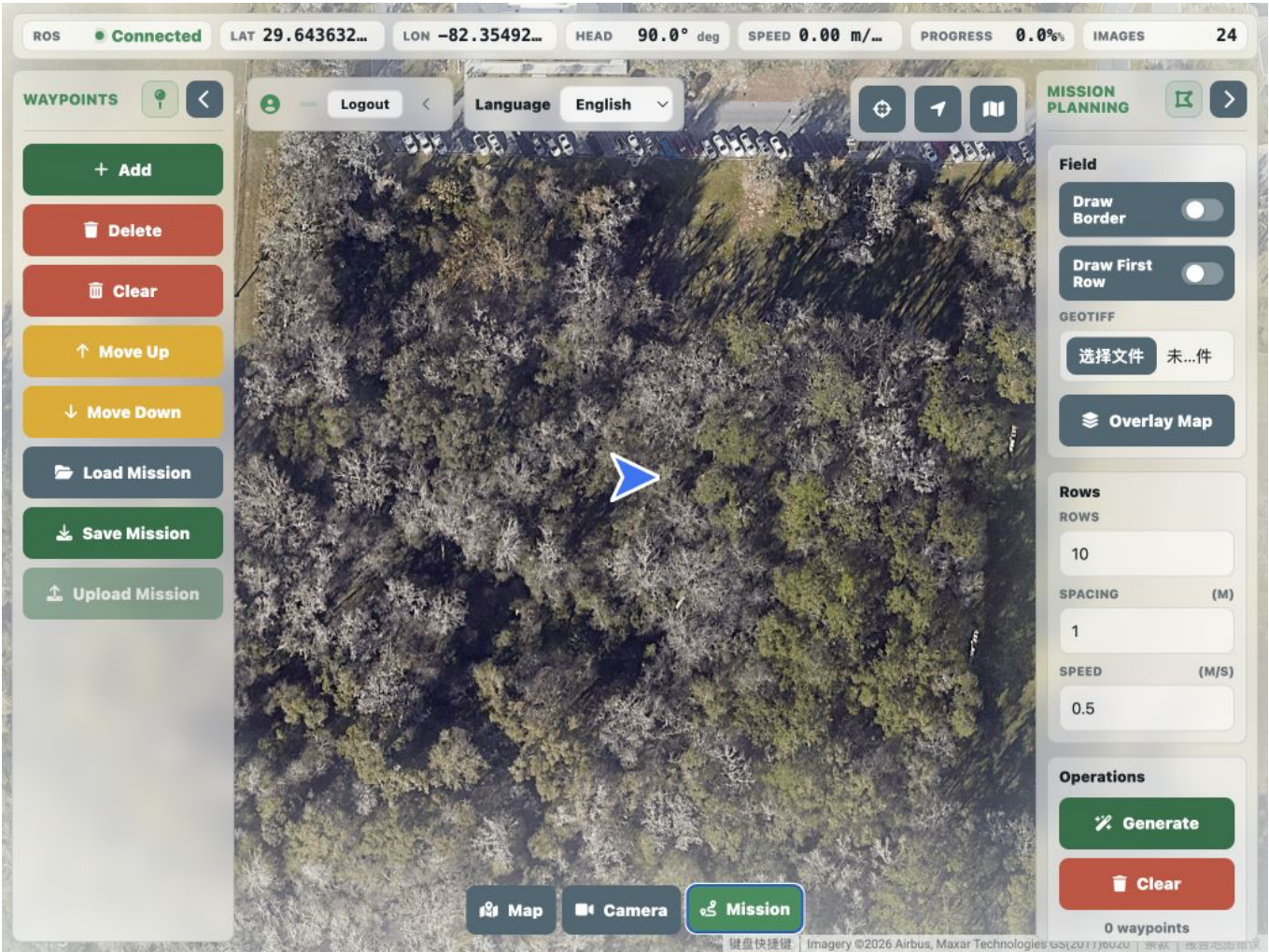


Figure 5.5: Mission view: edit Waypoints on the left and configure field rows and Mission Planning on the right.

5.4.1 Create a navigation web account (administrator)

Authorized administrators only. The navigation webpage has no Create Account button or self-registration. Create accounts from an authorized SSH session while the robot is in Hold and UVC is Off/Disarmed.

1. Connect to the navigation controller using the approved procedure in Navigation controller: Wi-Fi and SSH access, then sign in as the `mars` operating-system user.
2. Open the installed workspace: `cd /home/mars/mars_server`.
3. Select the account role. Use `viewer` for map-and-telemetry-only access. Use `operator` only for trained personnel who may plan missions and claim robot control.
4. Choose a unique username and a strong, site-approved password. The account script validates the role but does not enforce password strength. Do not reuse the Wi-Fi, SSH, or another operator's password.
5. Create the account with `/opt/robot-web-server/bin/python3 scripts/create_user.py <username> '<password>' <role>`. Replace `<role>` with exactly `operator` or `viewer`. Protect the terminal and command history according to site credential policy because the password is supplied as a command argument.
6. Require the confirmation `User '<username>' (<role>)` saved. The server reloads the account file for each login, so a web-service restart is not required.
7. If the username already exists, this command replaces that account's password and role. Coordinate the change and require the user to log out and sign in again; an existing browser session does not automatically receive a changed role.
8. Open `https://mars.local/` in a separate browser session, verify the new credentials and role badge, then log out. Do not claim control merely to test an account.
9. Run `exit` to close SSH and record the account creation or change in the controlled site log. Never record the password in the service report.

5.4.2 1 · Sign in and take operator control

1. Power the robot in Hold and allow the navigation controller to complete boot. The navigation services start automatically; operators do not need to launch them. Join the approved robot/navigation network and open `https://mars.local/` or the commissioned web address.
2. Rotate a phone/tablet to landscape. Sign in with the assigned account. There is no browser self-registration; an administrator creates accounts.
3. A **Viewer** sees the map and telemetry only; Camera, Mission and the control panels are hidden. An **Operator** can plan waypoints, but robot/camera commands remain disabled until that session claims control.
4. Select **Claim Control**. Only one operator session can hold control. Claiming immediately transfers control from another operator without asking them, so coordinate first.
5. Do not reuse one account on two devices: a newer sign-in with the same username signs out the earlier session and releases its control.
6. Verify **ROS Connected**, plausible latitude/longitude, heading, speed, robot marker, live video, current RTK/localization, and no navigation fault. ROS Disconnected blocks navigation and recording commands.

5.4.3 2 · Set up a waypoint mission

1. Select **Mission**. This keeps the map in the center and opens the editable Mission Planning and Waypoints panels.
2. For a generated row path, enable **Draw Border** and tap at least three boundary points. Drag the editable polygon if correction is needed. Keep it inside verified safe terrain and away from obstacles, ditches, public access, water and unverified turns.
3. Enable **Draw First Row**; this turns Draw Border off. Tap point A and then B to define row direction. Drag A/B or tap a marker and enter precise latitude/longitude when needed.
4. Enter **Rows** (at least 1), **Spacing** in metres, and **Speed**. The generator creates alternating row directions and clips row endpoints to the field polygon.
5. Optionally choose a georeferenced `.tif/.tiff` and select **Overlay Map**. Treat it only as a planning aid and verify its alignment against known field features.
6. Select **Generate**. If draft waypoints already exist, confirm their replacement. Inspect every generated endpoint, connector, turn, order and clearance.
7. For manual map points, turn both drawing toggles Off, then tap the map at each desired location. The **Add** button is different: it adds one waypoint at the robot's current position and refuses when the GPS fix is missing or stale.

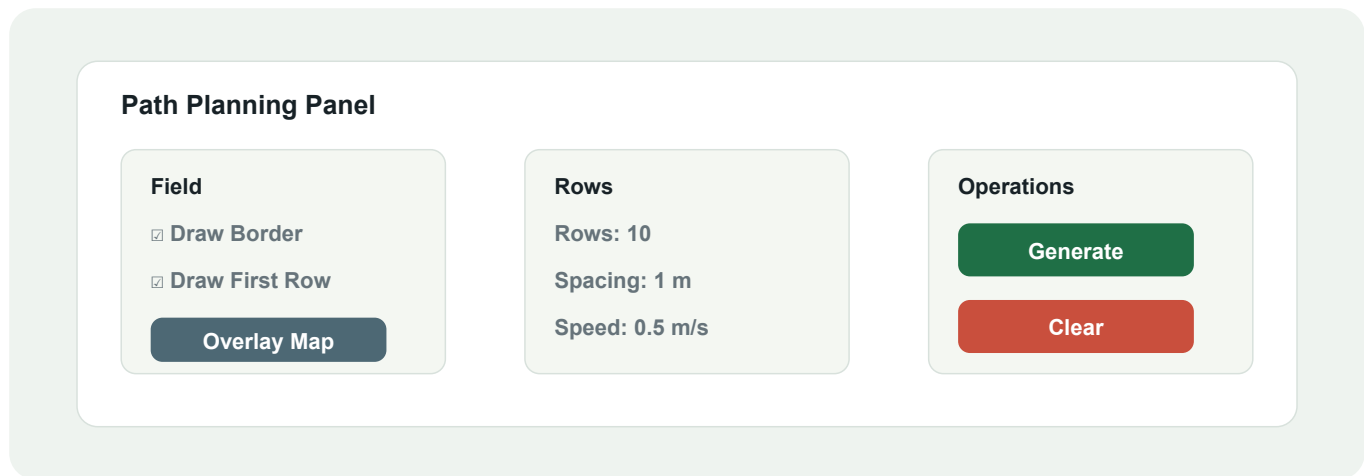


Figure 5.6: Current Mission Planning panel. Draw Border and Draw First Row are mutually exclusive drawing modes.

8. Drag waypoint markers only in Mission view. Tap one or more markers to select them. **Delete** removes selected points after confirmation; **Clear** removes all draft waypoints; **Move Up/Down** changes selected points in the route order.
9. The Waypoint Inspector edits every selected point. A blank field means the selected points have different values. Set latitude/longitude, speed 0.1–2.0 m/s, action, hold time, XY tolerance 0.01–5.0 m, and yaw tolerance 0.01–3.14 rad.
10. **None** passes the point without an action. **Take Photo** stops for the hold time and triggers a photo during the stop. **Stop & Wait** stops for the hold time. Choosing either action gives a new point a default 5-second hold unless changed.
11. Select **Save Mission** to download the current draft as `mission.waypoints` in QGC WPL 110 format. It preserves speed changes and action/hold commands but does not save the live field-border/geofence polygon.
12. **Load Mission** accepts `.mission` and `.waypoints`, validates coordinates, ID and 0.1–2.0 m/s speed, and loads the result into the editable Mission draft. Loading neither uploads nor starts the robot. Redraw and verify the border before enabling a geofence.

5.4.4 3 · Upload, switch to Auto, and start

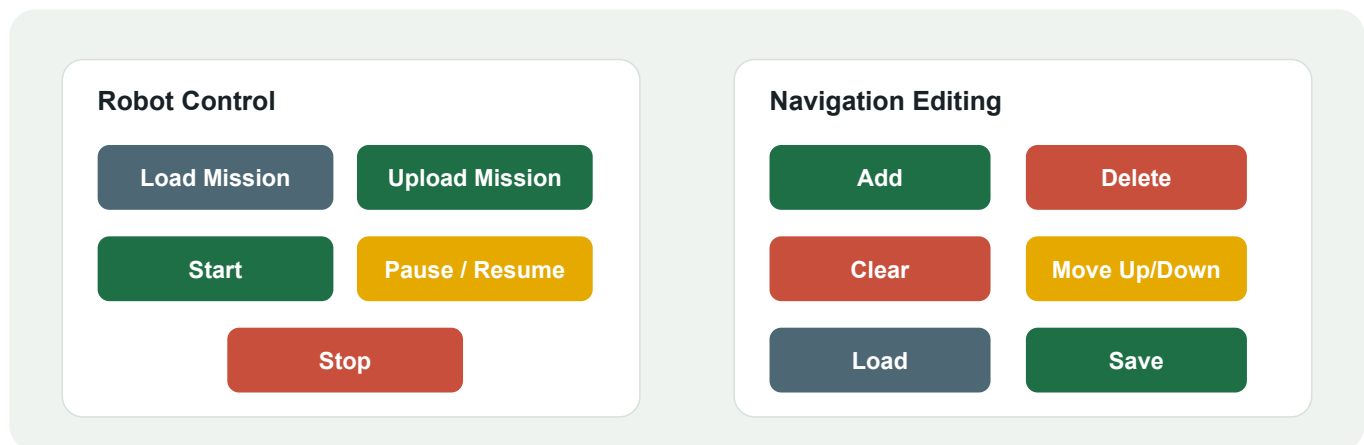


Figure 5.7: Upload transfers the reviewed draft and settings to the navigation node. It does not move the robot. Slide to start begins execution only after the robot is in Auto.

1. Keep the robot in Hold and stationary. Claim operator control; require ROS Connected and at least one valid draft waypoint.
2. Select **Upload Mission**. Review every listed point plus its coordinates, speed, action and hold. The dialog's default speed is only a fallback for a point without a valid individual speed.
3. Set default XY tolerance (0.01–5.0 m) and yaw tolerance (0.01–3.14 rad). A waypoint-specific inspector value overrides its corresponding upload default.
4. Leave **Enable Geofence** checked for field operation. A live Draw Border polygon with at least three points is required; saved/loaded mission files do not restore it. Cancel and redraw the boundary if the page says it is missing.

5. Confirm Upload and wait for the success message. The uploaded navigation route and active geofence become the non-editable Map view; draft editing remains in Mission view.
6. Walk or visually inspect the route, clear the area, and place a trained operator at the physical emergency stop.
7. **Single Joystick Remote Controller:** long-press Button 2. **Dual Joystick Remote Controller:** long-press SW6. Wait for the controller and robot to report **Auto**; otherwise the navigation action is rejected.
8. Drag **Slide to start** fully to the right. Verify Target becomes waypoint 1, distance/progress begin updating, and the first physical movement is slow and correctly directed.
9. Supervise continuously. The dashed target line, current/reached waypoint colors, driven track, distance, progress, RTK/localization, robot mode, CAN, speed and messages must agree with the physical robot.

5.4.5 4 · Pause, resume, stop, and return to Manual

Mode-change behavior: switching the robot from Auto to either **Manual** or **Hold** automatically stops active autonomous navigation. The webpage Stop button is an additional direct way to cancel the mission; it is not required before changing mode.

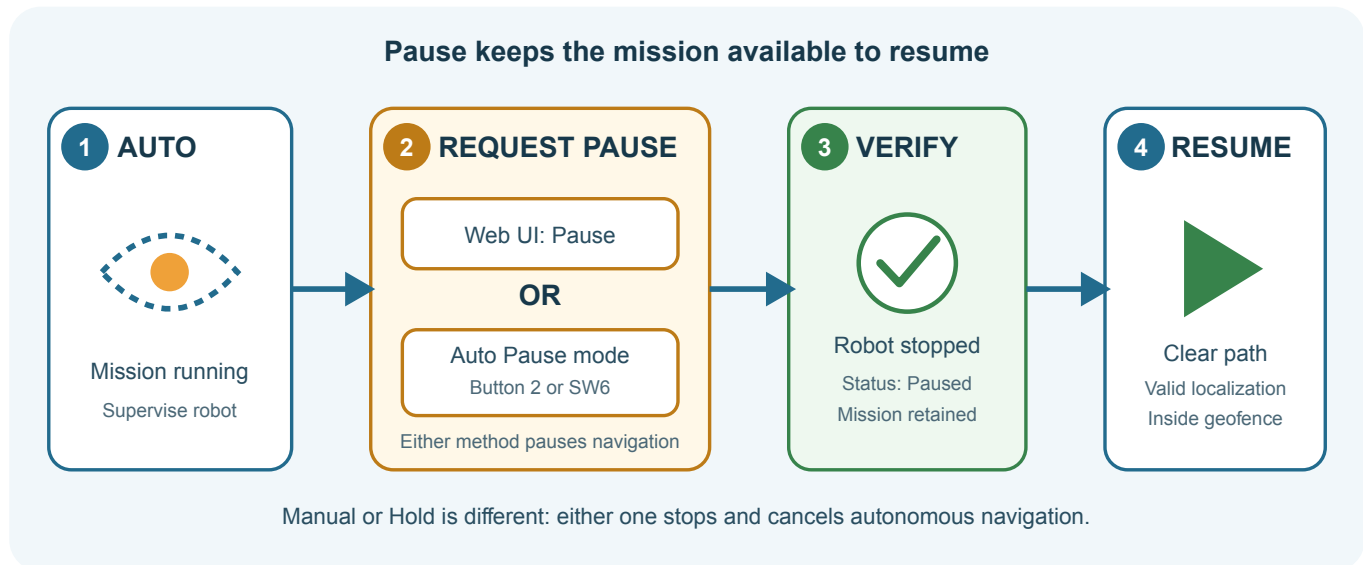


Figure 5.8: Selecting Auto Pause with Button 2 or SW6 pauses navigation just like the webpage Pause button. The mission remains available to resume after checks pass.

1. For a controlled pause that keeps the mission available to resume, either select webpage **Pause** or change the robot mode from Auto to **Auto Pause**: short-press Button 2 on Single Joystick Remote Controller or SW6 on Dual Joystick Remote Controller. Both methods pause autonomous navigation. Confirm the webpage button changes to Resume, the displayed robot mode is Auto Pause, the status reads Paused, and the robot physically stops.
2. Read the exact message. A geofence exit produces a high-visibility safety pause; Resume is rejected until the robot is back inside. Correct stale odometry, high position error or other reported causes before continuing.
3. Select **Resume** or use the approved handheld Auto-Pause toggle. A successful service response returns the console to Navigating. Verify the next motion physically.
4. To end autonomous operation from the webpage, select **Stop**; it publishes a cancel request and ends active execution. To end it from a handheld, request **Manual** or **Hold**; either mode change automatically stops autonomous navigation.
5. After any stop method, confirm the robot is physically stationary, commanded speed is zero, and the webpage no longer reports Navigating before approaching, editing, or handing over control. **Clear Mission** removes the uploaded navigation route only when navigation is inactive; the Waypoints-panel Clear controls the editable draft.
6. Other signed-in consoles receive operator, mission and live navigation state changes. Do not interpret a remote state change as permission to operate; identify the responsible operator.
7. To regain joystick control, request Manual with Button 1 or SW5, verify both the robot/controller and webpage show that Auto navigation has stopped, then move the joystick slowly. Select Hold instead when no manual movement is required.
8. Select **Release Control** or Logout at handover. Logout and normal page close attempt to release the operator lock; always

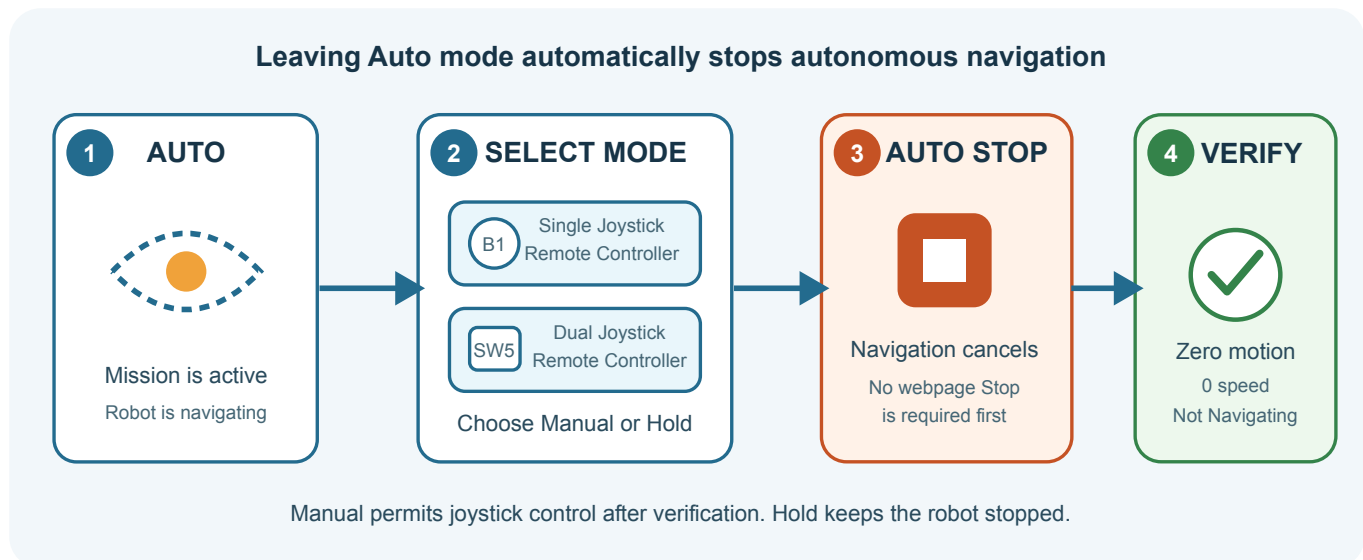


Figure 5.9: Changing from Auto to Manual or Hold automatically stops navigation. Always verify zero motion and that the console no longer says Navigating.

confirm the operator bar no longer shows your session as controller.

Intended automatic behavior: the robot follows waypoints in list order, applies each waypoint's speed/tolerance/action, shows the active target and driven track, pauses when commanded or when configured safety/localization conditions fail, rejects out-of-geofence start/resume, stops for an action's hold time, triggers Take Photo during that stop, and presents an audible/visible completion or failure dialog. The operator remains responsible for the surroundings and emergency stop.

5.5 Use robot Wi-Fi and cellular Internet together

Why both connections are needed: the navigation webpage and robot data are reached through the navigation controller's local Wi-Fi access point. The Google map tiles come from the Internet. A cellular phone or tablet must therefore remain connected to mars_xxxxxx by Wi-Fi while using 4G/LTE/5G cellular data for Internet traffic.

Before field use: an active SIM/eSIM, cellular-data plan and usable field signal are required. Cellular use may incur data or roaming charges. Complete setup while the robot is stationary in Hold and UVC Off/Disarmed.

5.5.1 Android phone or cellular tablet

1. Before joining the robot, turn **Airplane mode Off** and **Mobile data On**. Open an ordinary Internet page to confirm that cellular data works, then close that page.
2. Open **Settings -> Network & Internet -> Internet**, or **Settings -> Connections -> Wi-Fi** on Samsung devices. Select mars_xxxxxx and enter alphaaggech.
3. The robot AP does not provide Internet. If Android reports **No Internet**, choose **Stay connected, Use this network as is**, or **Connect anyway**. Do not allow the phone to disconnect from the robot AP.
4. On a Samsung Galaxy device, open **Settings -> Connections -> Wi-Fi -> menu -> Intelligent Wi-Fi** and enable **Switch to mobile data**. A SIM is required, and the option can vary by model or carrier. On other Android devices, enable the equivalent cellular-fallback or adaptive-connectivity option if available.
5. If the phone provides per-app network controls, allow the browser to use **Wi-Fi or mobile data**. Disable Data Saver for the browser during operation if it prevents map loading.

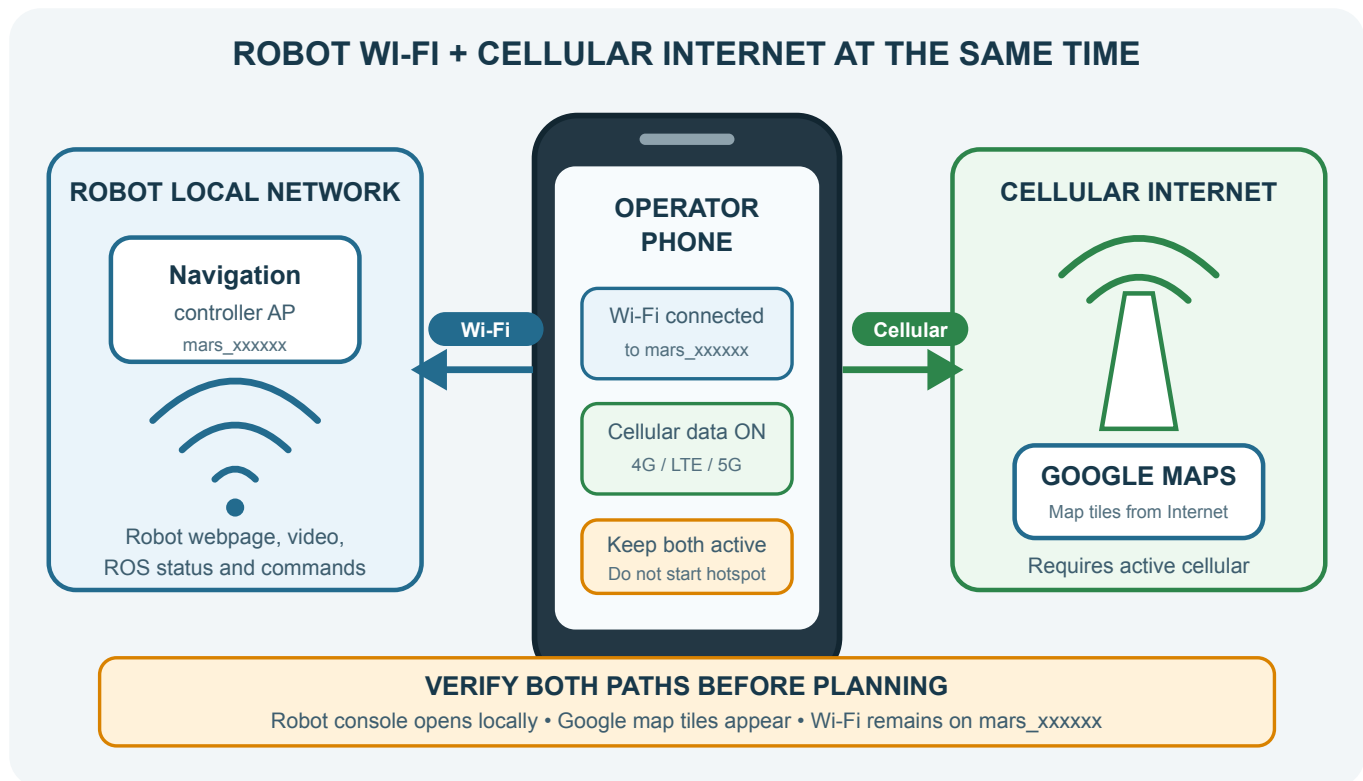


Figure 5.10: Keep both paths active: Wi-Fi carries the local robot webpage, video, status and commands; cellular data loads Google Maps. Do not enable the phone's hotspot.

- Return to Wi-Fi settings and confirm `mars_xxxxxx` still shows Connected, even if it also says No Internet. Do not switch to a different Wi-Fi network.

5.5.2 iPhone or cellular iPad

- Open **Settings** -> **Cellular** or **Mobile Data**. Turn Cellular Data On, select the correct data line if Dual SIM is used, and allow cellular data for the browser.
- On the same Cellular/Mobile Data page, scroll down and turn **Wi-Fi Assist** On. Wi-Fi Assist can use cellular data when the connected Wi-Fi has poor or no Internet service.
- Open **Settings** -> **Wi-Fi**, select `mars_xxxxxx`, and enter `alphaaggech`. Keep the blue checkmark on this network if iOS reports No Internet Connection.
- Return to the browser. Wi-Fi Assist works with foreground apps and may not operate while data roaming, so verify the map before starting a mission.

5.5.3 Verify the two data paths

- With Wi-Fi still connected to `mars_xxxxxx`, open `https://mars.local/` and sign in. Confirm the robot status, camera and ROS connection load. This proves the local Wi-Fi path.
- Open Map or Mission view. Pan or zoom slightly and confirm Google road or satellite tiles appear instead of a blank gray background. This proves the cellular Internet path.
- Recheck the phone status: Wi-Fi must remain connected to `mars_xxxxxx`, and the cellular indicator must show usable service. Do not begin planning until both paths work.
- Keep the browser in the foreground, the phone charged, and both connections enabled throughout navigation. Recheck after any loss of map, video, ROS or robot status.

5.5.4 Connection troubleshooting

Symptom	Likely cause	Action
Robot webpage works, but Google map is gray	Cellular data is Off, cellular fallback is disabled, browser cellular access is blocked, no field signal, data limit reached, or roaming prevents fallback	Keep robot Wi-Fi connected. Verify mobile data, signal, plan/roaming policy and browser permission; enable Switch to mobile data or Wi-Fi Assist; reload the page.
Google Maps works, but <code>mars.local</code> does not	The phone left <code>mars_XXXXXX</code> , selected another Wi-Fi network, or VPN/private DNS blocks local addressing	Reconnect to <code>mars_XXXXXX</code> and choose Stay connected. Disable automatic switching to better Wi-Fi. Pause an approved VPN/private-DNS service only if site policy permits, then retry.
Phone repeatedly drops the robot AP	The device rejects a Wi-Fi network without Internet or prefers another saved network	Select Connect anyway/Stay connected, disable automatic joining of nearby networks for the operation, and verify <code>mars_XXXXXX</code> before every command.
Turning on hotspot disconnects the robot	The phone cannot remain a Wi-Fi client while serving its own hotspot	Turn the hotspot Off and reconnect to <code>mars_XXXXXX</code> . Use cellular data directly on the operating device.
No dual-path option is available	Device, carrier or policy does not support simultaneous local Wi-Fi and cellular fallback	Use an approved cellular-capable phone/tablet that supports both paths. Do not operate with an unverified or blank map.

Phone menus vary by operating-system version, model and carrier. Platform references: Samsung Intelligent Wi-Fi and Apple Wi-Fi Assist.

5.6 Navigation webpage: current operator controls

Simulation notice: These interface screenshots are simulated examples created for this manual. Actual map imagery, robot position, status values, camera content, and interface appearance may differ during operation.

Area/control	Current behavior	Operator use
Orientation message	Blocks the portrait phone layout and requests landscape	Rotate before planning or operating.
Top status strip	ROS connection, latitude, longitude, heading, speed, mission progress and image count	Require Connected and plausible fresh values before commands.
Map	Main map plus small live-camera preview; shows robot, driven track, uploaded route, current target and active geofence	Supervise route and physical position together.
Center on robot	Centers the map on fresh robot GPS	Use when the robot marker is off-screen; a stale/missing fix is rejected.
Center on this device	Uses the phone/laptop browser geolocation	Allow location permission only when useful; it does not change robot localization.
Map-type button	Toggles roadmap and satellite map	Choose the view that best exposes field features; neither is a safety survey.
Camera	Main live WebRTC video plus a small robot-centered map preview	Inspect framing and use tap-to-focus/expose; never navigate from video alone.
Preview minus/restore	Minimizes or restores the small camera/map preview	Use to clear screen space without stopping either data path.
Mission	Shows the shared map with editable draft waypoints and the Mission Planning panel	Create/edit only while navigation is inactive.
Panel chevrons	Collapse/restore Robot Control, Waypoints, Camera and Mission Planning panels	Use for screen space; re-open the panel before its controls are needed.

Area/control	Current behavior	Operator use
Fullscreen	Requests browser fullscreen where supported	Useful on phones/tablets; browser/home-screen support varies.
Language	English, Chinese or Spanish; saved in that browser	Select a language the operator reads reliably.
Operator bar	Username, role, Claim Control, Release Control and Logout; can collapse	Verify who holds control before every command.
Robot Control readout	Target waypoint, distance, status and progress	Compare with the top strip, map and physical robot.
Load Mission	Loads <code>.mission</code> CSV or QGC WPL <code>.waypoints</code> into the editable Mission draft	Validate and review; load does not upload or start.
Save Mission	Current button downloads draft as <code>mission.waypoints</code> (QGC WPL 110)	Save after review. Redraw the field border after reload because the geofence is not stored.
Upload Mission	Requires operator control and ROS; reviews point properties/default tolerances and optional live-border geofence	Upload only after physical route review; disabled while navigating.
Slide to start	Sends the execute-mission action after upload	Use only with robot confirmed Auto and area clear.
Pause / Resume	The webpage Pause calls the pause service. Changing the robot from Auto to Auto Pause with Button 2 or SW6 also pauses navigation and retains the mission. Resume continues only after a successful response.	Confirm zero motion and Paused/Auto Pause. Correct a rejection, geofence, localization, or other pause cause before Resume.
Stop	Publishes mission cancellation	Directly cancels navigation. Changing robot mode from Auto to Manual or Hold also stops navigation automatically.
Clear Mission	Clears the uploaded navigation route when inactive	Different from Waypoints Clear, which clears the draft.
Add	Adds one draft point at the robot's current position; requires a fresh GPS fix	Use when physically placing waypoints with the robot.
Map tap in Mission	With both drawing toggles Off, adds a draft waypoint at the tapped location	Use for manual map-based route construction.
Delete / draft Clear	Deletes selected draft points or all draft points after confirmation	Confirm the count and route after removal.
Move Up / Down	Moves selected draft points one position while preserving multi-selection order	Recheck the red connecting line after every reorder.
Waypoint Inspector	Edits all selected points; blank means mixed values	Set coordinates, speed, action/hold and optional point tolerances.
Draw Border / First Row	Mutually exclusive map modes; border is editable/draggable, A/B markers are draggable	Required for row generation; border is also the live geofence source at upload.
Rows / Spacing / Speed / Generate	Builds alternating lines clipped to the polygon	Review every turn and row endpoint after generation.
GeoTIFF / Overlay Map	Loads local georeferenced TIFF as a planning overlay	Optional; verify projection and alignment.
Planning Clear	Removes draft border, A/B line, overlay and draft waypoints after confirmation	Does not substitute for Stop or Clear Mission.

Draft versus uploaded route: Mission view edits *planningWaypoints*. A successful Upload copies that reviewed draft into the active navigation route. Map view shows the active route. Saving stores draft waypoint commands but not the live border/geofence. Keep this distinction clear during handover.

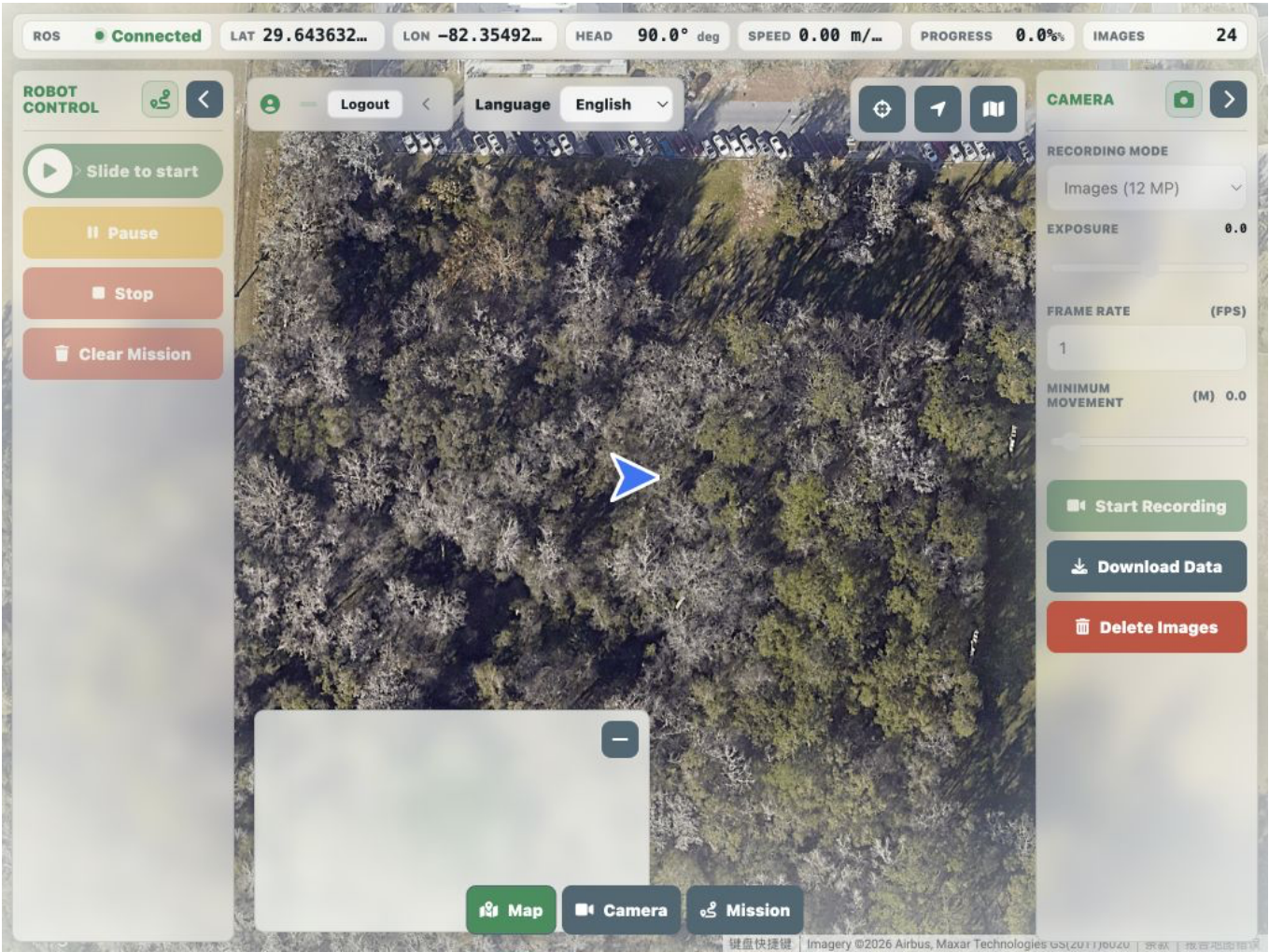


Figure 5.11: Map page with the top status strip, Robot Control panel, map controls, camera preview, and Camera panel.

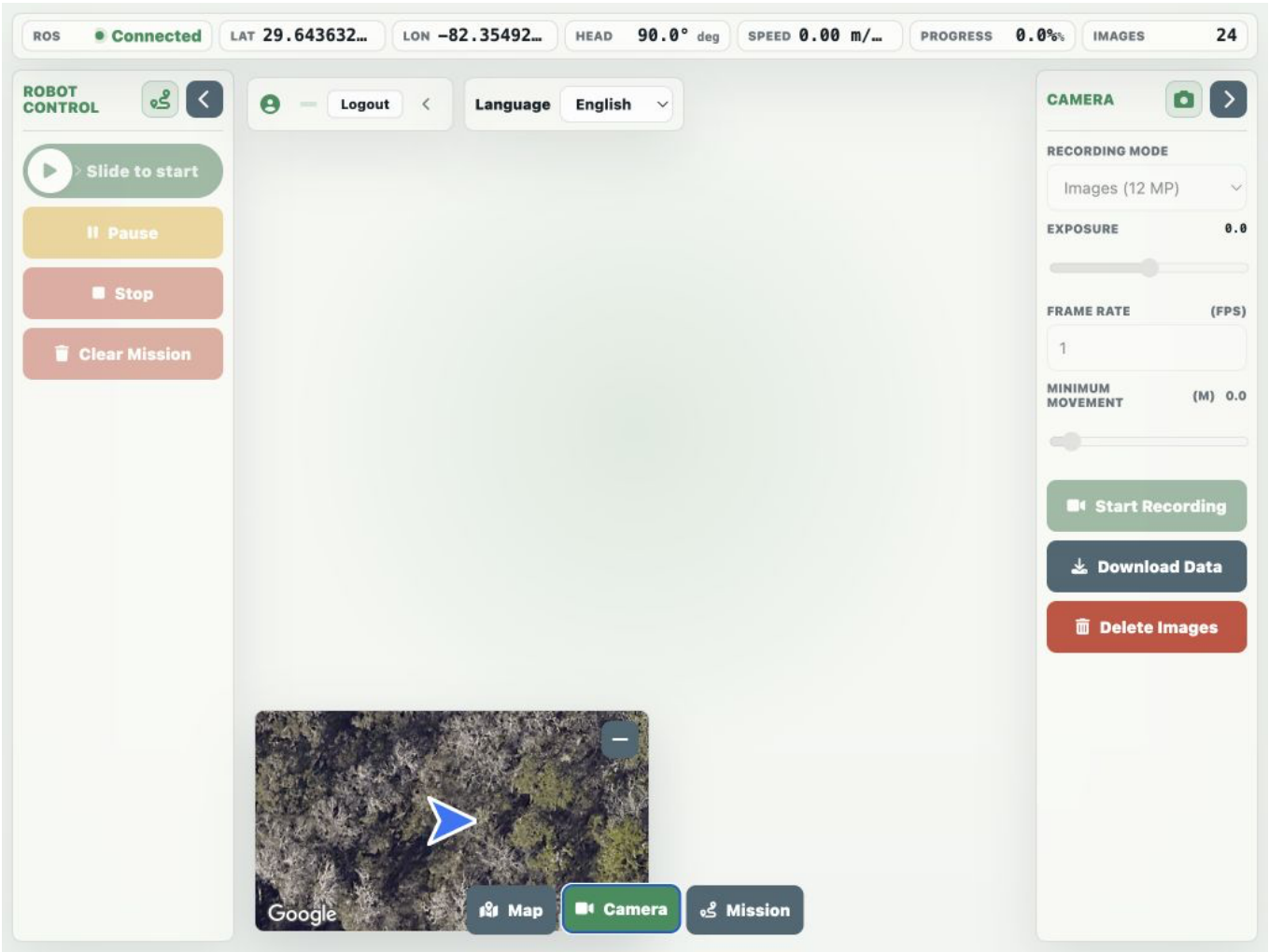


Figure 5.12: Camera page with the main video workspace, small map preview, Robot Control panel, and recording controls.

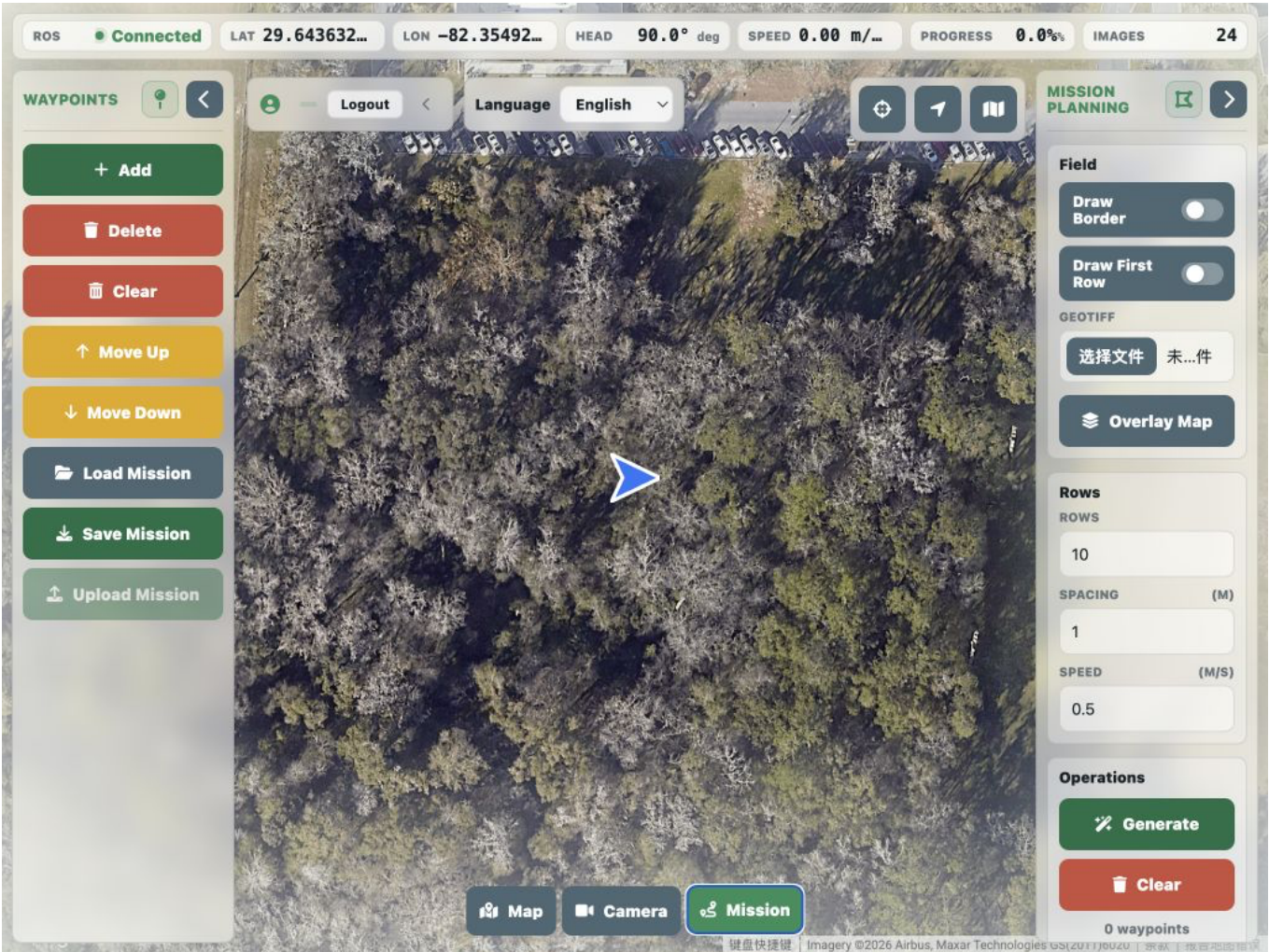


Figure 5.13: Mission page with Waypoints, editable map workspace, and Mission Planning controls. The console requires landscape orientation.

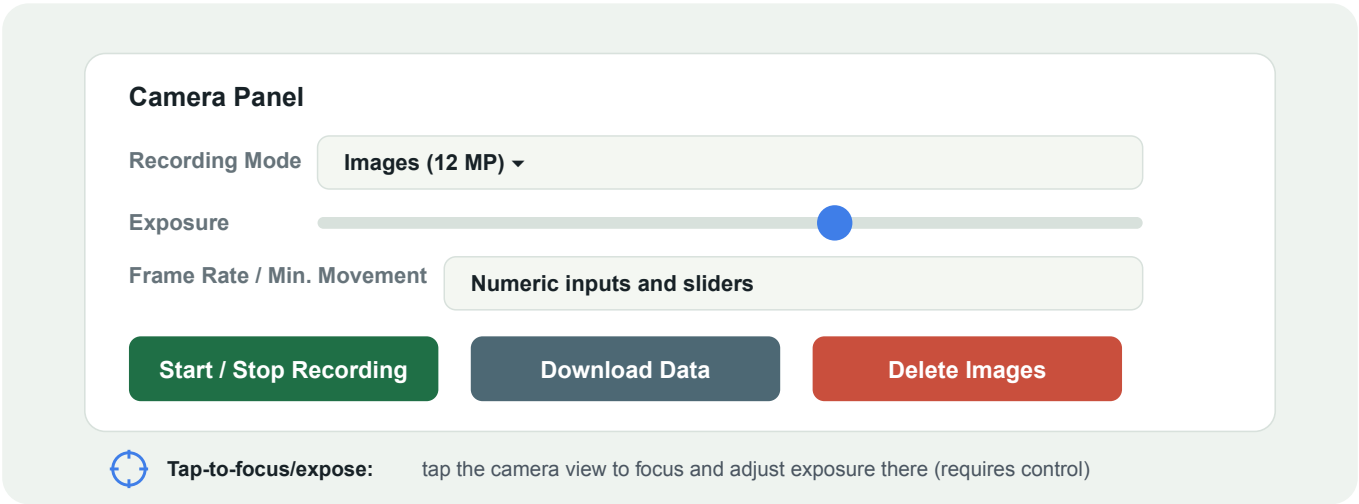


Figure 5.14: Current camera panel. Frame Rate and Min. Movement are image-only controls and become disabled in Video mode.

5.6.1 Record still images

1. Claim operator control and confirm ROS Connected, live video, available storage, clean lens and correct aim.
2. Select **Camera** so video is the main stage. In Recording Mode choose **Images (12 MP)**; stills are fixed at 4608 × 2592.
3. Set Exposure from -5 to +5 as required. Tap/click inside the actual video frame to request autofocus and auto-exposure at that point. The request is ignored in the small preview, without operator control, without ROS, or when tapping the letterbox outside the image.
4. Set Frame Rate from 0.1 to 2 images/s and Min. Movement from 0 to 1 m. A new still is gated by these settings; 0 m disables the movement separation.
5. Select **Start Recording**. Wait for a success message and the button to change to Stop Recording. Confirm Images count rises as expected; live preview may fall to about 14 fps during full-resolution still capture.
6. Select **Stop Recording** and wait for success before changing mode, downloading or shutting down.

5.6.2 Record video

1. Stop any current still recording, then select **Video (30 fps)**. Video records the 2304 × 1296 live stream as MP4; Frame Rate and Min. Movement are disabled because they do not apply.
2. Set exposure and verify the main live frame. Tap-to-focus/expose remains available with operator control and ROS.
3. Select **Start Recording**. Wait for success and verify the Stop Recording state.
4. Select **Stop Recording** before changing modes, leaving the job or removing power. Wait for the service success response so the MP4 can close correctly.

5.6.3 Download or delete recordings

1. Stop recording first and confirm the Stop Recording state.
2. Select **Download Data**. The picker lists collection sections with separate image/video counts. Select one section; Download stays disabled until a section is selected.
3. Select **Download**. The server prepares or reuses a ZIP and shows compression progress. Keep the page and network connection open until the browser download begins and completes.
4. Open the ZIP, compare the section name/counts and spot-check images and MP4 files before deleting robot data.
5. For authorized cleanup, select **Delete Images**. Despite the label, deletion selects one or more complete collection sections and removes their stored contents, including videos. Confirm the selected count carefully.
6. Wait for the success message. Deletion is not a backup and may not be recoverable.

Geofence behavior: the enforced polygon uses the Plan border expanded outward 2 m by default. Starting outside it is refused. Leaving it during navigation pauses the mission, and Resume is rejected until the robot returns inside.

5.7 Auto-navigation troubleshooting

Problem	What to do
camera.service inactive	Keep robot in Hold. Service staff check <code>systemctl status camera.service</code> and <code>journalctl -u camera.service -n 80 --no-pager</code> , then inspect power/network/storage/build.
Webpage opens, video shows 502 "No stream available on path cam"	Nginx cannot reach MediaMTX; check <code>mediamtx.service</code> . MediaMTX is running but camera node is not pushing. Check camera ribbon, permission, camera service, and logs.
Video connects but stays blank	Check UDP 8189 path between browser and robot.
Mission will not start	Check robot inside geofence, Auto mode, valid/current RTK odometry, CAN, uploaded mission, and emergency state.
Mission pauses	Read the exact message. Correct high covariance, stale odometry, CAN loss, or geofence exit before Resume.

5.8 Navigation controller: Wi-Fi and SSH access

Authorized field technicians only. Before connecting, select Hold, restrain the drive wheels for command-path testing, physically disconnect UVC power, and keep the emergency stop available. A terminal command can stop services or cause robot commands; begin with the read-only checks below.

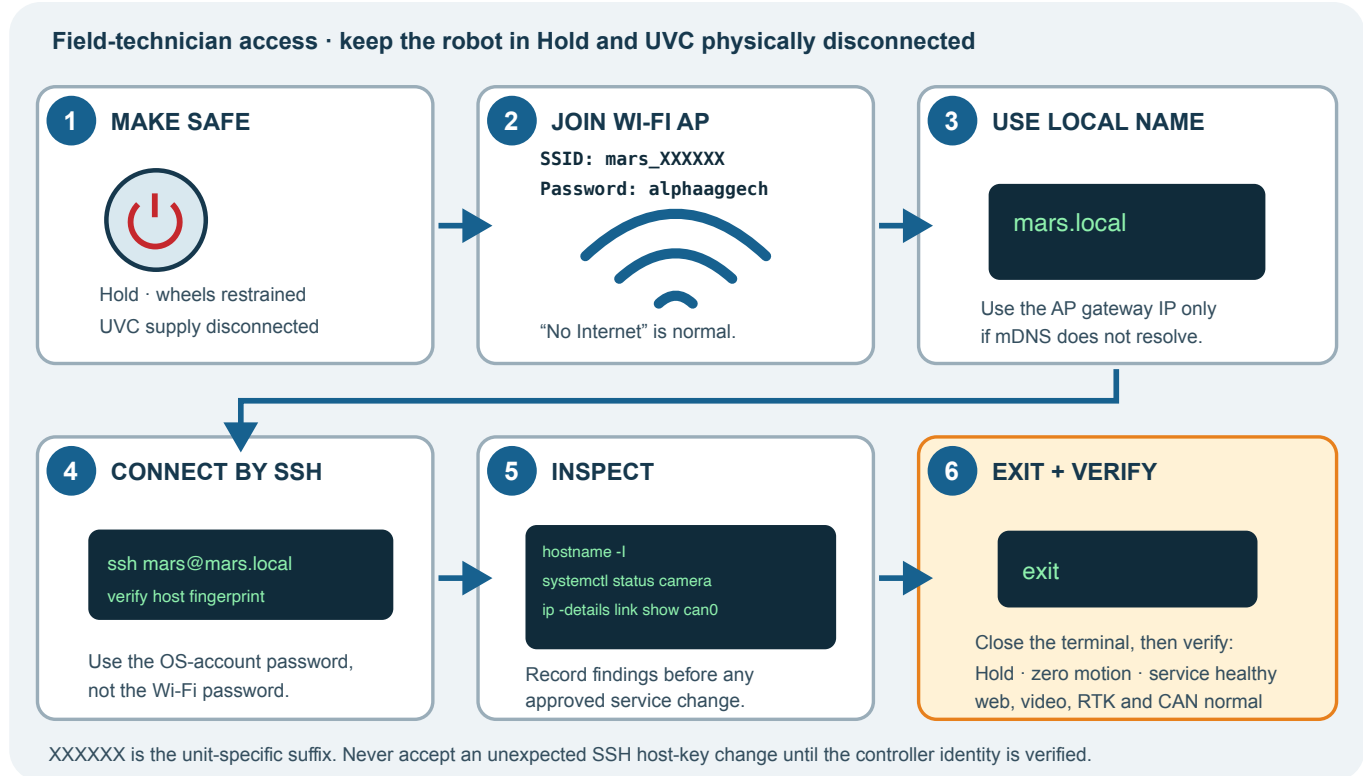


Figure 5.15: Technician access path: safe state -> join the correct AP -> use `mars.local` -> SSH as `mars` -> inspect -> exit and recommission.

5.8.1 Access credentials and addressing

Item	Installed/default value	Field note
Navigation-controller Wi-Fi AP	<code>mars_XXXXXX</code>	XXXXXX is the unit-specific suffix. Confirm the label/serial number so a technician does not join another nearby robot.
Navigation-controller AP password	<code>alphaaggech</code>	This navigation-controller value is spelled with ggech . Enter it exactly as shown.
General device AP default	<code>alphaagtech</code>	Use for other robot-system device APs unless that component's table or site record gives a device-specific value. It is different from the navigation-controller password above.
SSH hostname	<code>mars.local</code>	Primary technician target while connected to the navigation-controller network.
SSH user	<code>mars</code>	The deployed service runs as user/group <code>mars</code> and the workspace is under <code>/home/mars</code> .

Item	Installed/default value	Field note
SSH password	Site-issued operating-system account password	The Wi-Fi AP password is not automatically the SSH password. Obtain the OS credential through the site's credential record; do not guess or place it in a service report.
IP fallback	AP default-gateway address or commissioned address	Use only when <code>mars.local</code> does not resolve. Read the connected Wi-Fi adapter's default gateway and record it as <code><NAV_IP></code> .

Password precedence: the specific navigation-controller AP credential above overrides the general device default. If a commissioned unit no longer accepts a default, use the site credential record; do not factory-reset it in the field merely to restore a published password.

5.8.2 Connect a technician laptop to the navigation-controller AP

1. Keep the robot in Hold, verify zero motion, disconnect UVC power, and tell the responsible operator that technician access is starting.
2. On the laptop, open Wi-Fi networks and select the exact `mars_XXXXXX` matching the robot being serviced.
3. Enter `alphaaggech`. Remain connected if the laptop reports **No Internet**; this is a local service network.
4. Use `mars.local` as the navigation-controller hostname. Confirm the Wi-Fi SSID is still the intended robot before opening a browser or terminal.
5. If `mars.local` does not resolve, find the AP default gateway. On Windows run `ipconfig`; on Linux run `ip route`; on macOS run `route -n get default`. Record the gateway as `<NAV_IP>` for the fallback SSH command.

5.8.3 Open an SSH session

1. Open Windows Terminal/PowerShell, macOS Terminal, or a Linux terminal.
2. Run `ssh mars@mars.local`. If the local name does not resolve, run `ssh mars<NAV_IP>` using the AP gateway or commissioned IP found above.
3. On the first connection, compare the displayed SSH host-key fingerprint with the commissioned fingerprint or maintenance record. Accept it only when the robot identity and fingerprint are verified. An unexpected key change can indicate a replaced controller, reinstalled OS, wrong robot, or security problem.
4. Enter the site-issued `mars` operating-system password when prompted. The terminal does not display dots or characters while the password is typed; this is normal.
5. Confirm the prompt identifies the intended unit. Run `hostname`, `hostname -I`, and `whoami`. Expected hostname is `mars` and expected user is `mars`; compare the addresses with the unit record.

5.8.4 Read-only first inspection

```
hostname
hostname -I
whoami
uptime
git -C /home/mars/camera_catkin_ws branch --show-current
git -C /home/mars/camera_catkin_ws rev-parse --short HEAD
systemctl status camera.service --no-pager
journalctl -u camera.service -n 80 --no-pager
ip -4 addr show eth0
ip -details link show can0
df -h /home/mars
vcgencmd get_throttled
vcgencmd measure_temp
```


Expected results are the approved hostname/IP, user `mars`, branch `video-stream`, recorded commit, `camera.service` active, `eth0` configured, `can0` UP at the commissioned 500 kbit/s, adequate free storage, no active undervoltage/throttling, and acceptable temperature. Record the output before changing configuration or restarting a service.

5.8.5 End the session and return the system to service

1. Do not leave an unattended administrator terminal connected. Finish the approved work and save the service record.
2. Run `exit` and confirm the SSH session closes. Disconnect the laptop from `mars_XXXXXX`.
3. Complete post-service commissioning: verify the production service, web console, live video, RTK localization, CAN, Hold/Auto gating, Pause/Resume, geofence, and physical emergency stop.
4. Remove wheel restraint and restore UVC power only after the responsible operator accepts the complete system and every cover/connection is restored.

5.8.6 SSH troubleshooting

Message/symptom	Likely cause	Technician action
AP is not listed	Navigation controller has no PoE power, is still booting, AP service failed, or SSID was changed	Keep Hold; check robot PoE-switch power/link, cable, boot time, unit label, and commissioned network record.
Wi-Fi password rejected	Typing error, wrong nearby unit, or commissioned password changed	Verify exact SSID and the specific spelling <code>alphaaggech</code> ; then use the controlled credential record. Do not reset the unit.
Could not resolve hostname <code>mars.local</code>	mDNS unavailable or blocked on the technician laptop	Confirm the correct AP, then use the AP default-gateway or commissioned IP: <code>ssh mars@<NAV_IP></code> .
Connection timed out / no route	Wrong network/IP, laptop changed networks, AP isolation, or controller unavailable	Recheck the connected SSID, adapter address and default gateway; verify PoE/link; test the recorded address.
Connection refused	Host reachable but SSH service not listening	Use local/authorized recovery access to inspect the SSH service; do not repeatedly reboot an operating robot.
Permission denied	Wrong OS password, wrong user, locked account, or key policy	Use user <code>mars</code> and the site-issued OS credential. Escalate to the system administrator instead of guessing.
Host-key warning	Wrong robot, address reused, OS reinstalled, controller replaced, or possible interception	Stop and verify the physical unit and commissioned fingerprint. Do not bypass the warning until the change is explained and recorded.
SSH works but web/ROS is down	Application service, storage, network, camera, ROS, or CAN fault	Run the read-only inspection and follow navigation troubleshooting and field-engineer instructions.

5.9 Navigation controller: field-engineer instructions

Engineering work can cause motion. Mechanically restrain the drive wheels, disconnect UVC power, keep the physical emergency stop staffed, and select Hold before ROS or CAN tests.

5.9.1 Deployed baseline

Item	Value	Purpose
Workspace	<code>/home/mars/camera_catkin_ws</code>	Production catkin workspace

Item	Value	Purpose
Branch	video-stream	Current navigation/WebRTC implementation
Service	camera.service, user/group mars	Production startup and restart
Configuration	scripts/config_default.yaml	ROS master, node name, frame, launch path
Main launch	launch/camera.launch	Top-level deployed launch
ROS master	127.0.0.1:11311	Local ROS graph
ROS IP source	First IPv4 on eth0	Set by start_camera.py
Robot bus	can0	CANopen robot communication
Images	/home/mars/images	Capture output

5.9.2 Startup chain

1. Systemd waits for network-online and starts camera.service.
2. The service enters the workspace, sources devel/setup.bash, and runs scripts/start_camera.py.
3. The script reads config_default.yaml, sets ROS_MASTER_URI/ROS_IP, then launches camera.launch with --wait.
4. The launch starts camera hardware management, mars_bringup, rosbridge, camera streaming, and camera trigger.
5. mars_bringup starts dual-GNSS localization and navigation plus the base_footprint -> base_link transform.
6. The camera publishes H.264 to MediaMTX; nginx exposes secure web, WebSocket, and WHEP endpoints.

5.9.3 Read-only inspection commands

```
cd /home/mars/camera_catkin_ws
git branch --show-current
git rev-parse --short HEAD
sudo systemctl status camera.service
journalctl -u camera.service -n 80 --no-pager
ip -4 addr show eth0
ip -details link show can0
df -h /home/mars
vcgencmd get_throttled
vcgencmd measure_temp
```

Expected: video-stream; service Active; intended IPv4 on eth0; can0 UP; adequate storage; no active undervoltage/throttling; acceptable temperature.

5.9.4 Controlled update and build

1. Record the commit/configuration and approved rollback point; copy images/logs.
2. Restrain the robot, select Hold, and physically disconnect UVC power.
3. Stop with `sudo systemctl stop camera.service`.
4. Select the approved video-stream revision through the controlled Git procedure.
5. If manifests changed, run `rosdep install --from-paths src --ignore-src -r -y`.
6. Run `catkin_make` at workspace root. Do not restart after a failed build.
7. Start with `sudo systemctl start camera.service` and follow the journal.
8. Complete commissioning below; roll back if a safety function fails.

5.9.5 ROS and navigation interfaces

Interface/parameter	Deployed detail
/odometry/map	Fused map-frame odometry input
/mars/robot_mode	Robot mode used for Auto gating
/mars/cmd_vel	Velocity output: linear X and angular Z

Interface/parameter	Deployed detail
/take_photo	Mission photo service
Control frequency	10 Hz
Max acceleration / linear speed	0.5 / 1.0 m/s from top-level launch
Look-ahead / wheelbase	1.0 m / 0.5 m
Max angular velocity	1.0 rad/s
Odometry timeout	1.0 s
Localization check	Enabled by default; do not disable for field operation

5.9.6 CAN bridge details

mars_canopen uses master node 11 and slave node 10. Velocity values are multiplied by 1000 and encoded as signed 16-bit linear/turning values. Default COB-IDs: status $0x200 + \text{node}$, master battery $0x211 + \text{node}$, slave battery $0x200 + \text{slave}$, velocity $0x180 + \text{node}$.

- Mode is polled from SDO $0x200C$ at 1 Hz; allow about one second after a change.
- In Auto, missing cmd_vel for 0.5 s sends one zero-velocity PDO.
- In Auto Pause, the bridge sends no velocity PDO, including zero; verify complete motor behavior during commissioning.

5.9.7 Video and network path

Stage	Address	Meaning
H.264 ingest	RTSP TCP 127.0.0.1:8554/cam	Camera node -> MediaMTX
WHEP signaling	:8889; nginx / cam/whep	Browser session setup
WebRTC media	UDP :8189	Direct browser<->robot media
Rosbridge	127.0.0.1:9090	Local WS; nginx supplies browser TLS
Camera encoding	4 Mbit/s; JPEG quality 92	Top-level launch values

5.9.8 Post-service commissioning

1. Verify service recovery after a controlled navigation-controller reboot.
2. Verify eth0, can0, ROS nodes, transforms, and fresh /odometry/map.
3. Verify live video, still capture, WHEP signaling, UDP media, and image storage.
4. With wheels raised, verify navigation velocity reaches CAN only in Auto.
5. Interrupt velocity input in Auto and verify the 0.5-second timeout response.
6. Test Upload, Execute, Pause, Resume, completion, and photo capture inside a visible geofence.
7. Induce a safe localization-quality failure and verify pause; restore valid localization before Resume.
8. Verify outside-geofence start refusal and geofence-exit pause in simulation or restrained test.
9. Test physical emergency stop independently of ROS and the webpage.

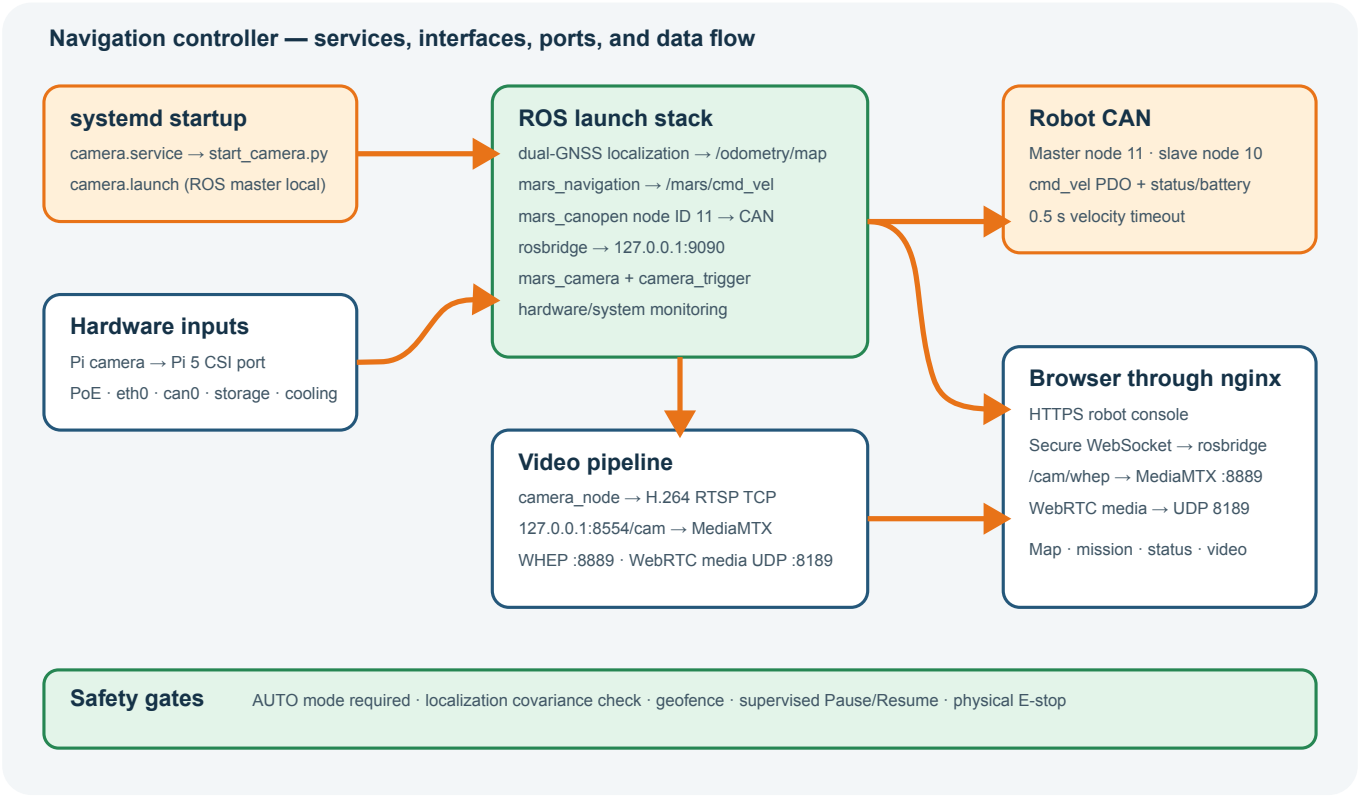


Figure 5.16: Deployed navigation-controller architecture from the video-stream branch. Follow the arrows to isolate startup, ROS, CAN, web, and video faults.

Operations

6.1 Operate the complete robot system

This chapter brings the component instructions together into three field procedures. Use **Manual driving** to position or transport the robot, **UVC operation** to perform a controlled treatment, and **Autonomous navigation** to execute a planned route. Read the linked component chapter whenever a control, status message, or connection is unfamiliar.

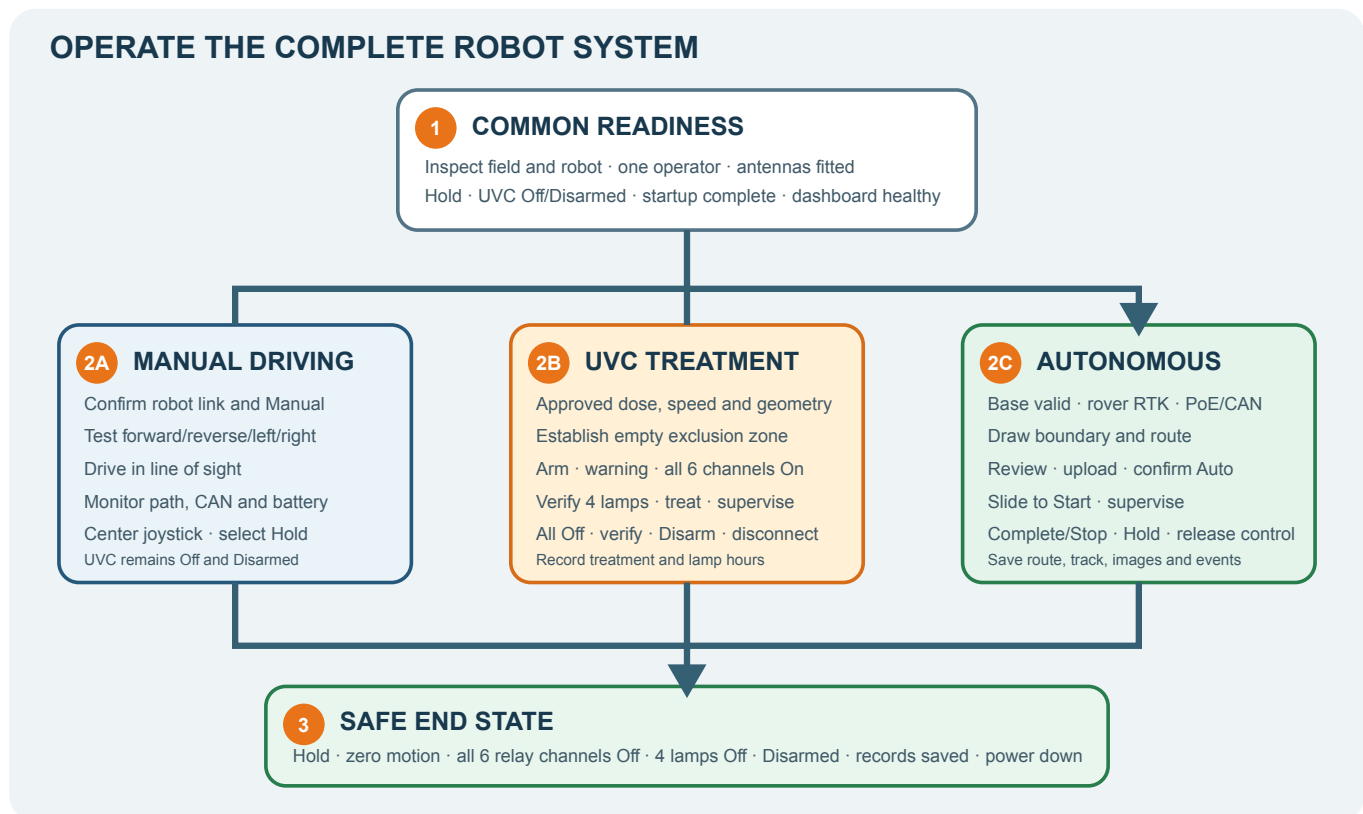
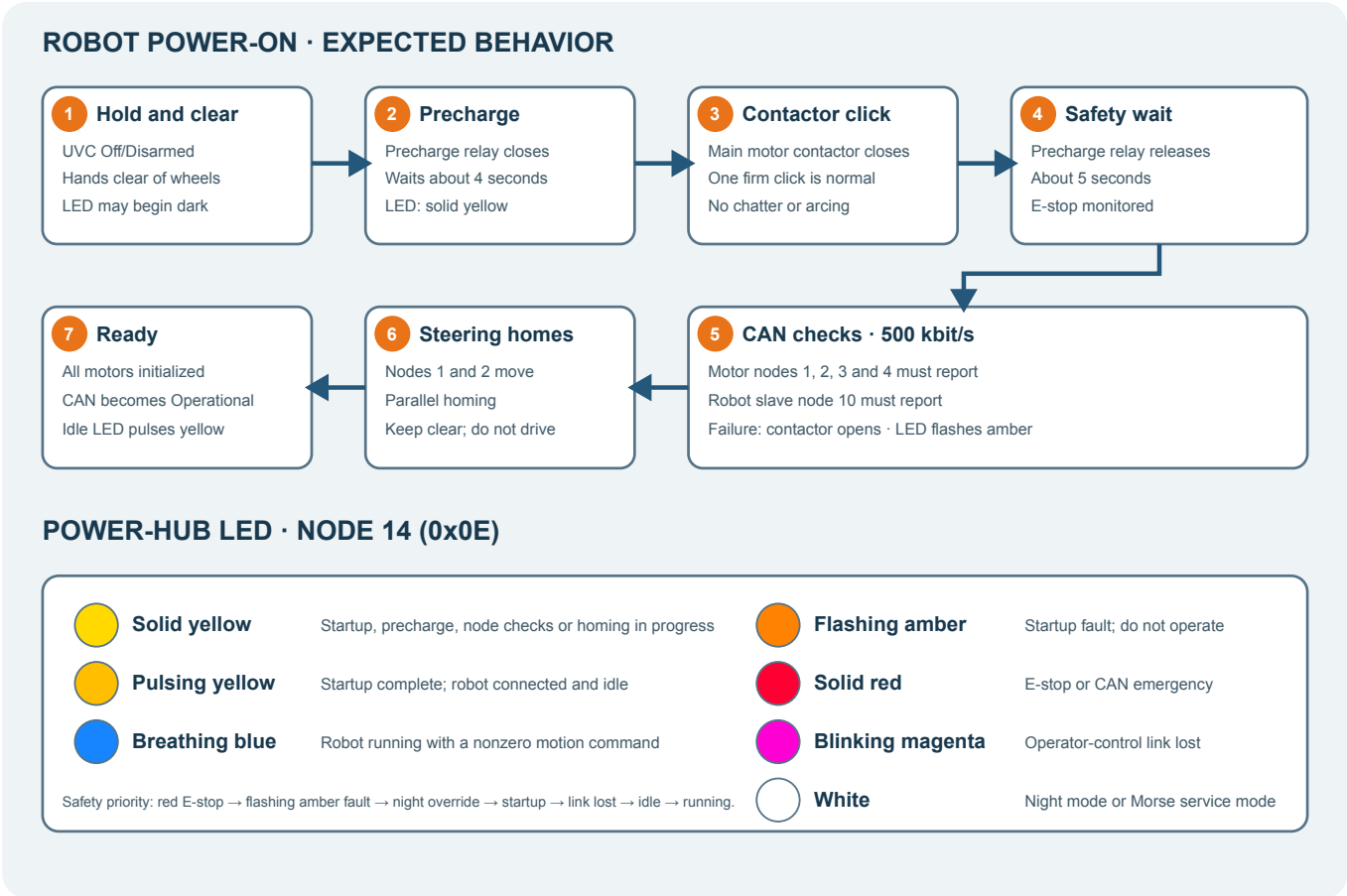


Figure 6.1: Every path begins with inspection and Hold, requires a healthy robot, and ends with a verified stop. UVC and autonomous operation add their own readiness gates.

The physical emergency stop overrides every normal procedure. Use it immediately for unexpected motion, loss of control, a person or animal entering the UVC exclusion zone, a lamp that remains on after an Off command, smoke, arcing, collision risk, or any condition the operator cannot positively identify.

6.1.1 Common readiness procedure

1. **Choose one responsible operator.** Identify who holds the handheld, who watches the work area, and who can reach the physical emergency stop. Do not allow two handhelds or two web operators to issue commands at the same time.
2. **Inspect the field and robot.** Check terrain, slopes, ditches, people, animals, crop clearance, wheels, steering, covers, batteries, antennas, GNSS mounts, navigation-controller window, UVC fixtures, cables, eight side ports, power-hub LED, warning devices, and physical emergency stop.
3. **Make the initial state safe.** Center both joysticks. Select Hold. Confirm UVC is Disarmed, all six relay channels are Off, all four lamps are physically Off, and the UVC disconnect is open unless the UVC procedure specifically requires power later.



4. **Attach and identify antennas.** Fit every required antenna before power-on. On the Dual Joystick Remote Controller: left is RTK LoRa, middle is GNSS, and right is robot-control LoRa. Keep the RTK radio plan different from the RC radio plan.
5. **Power the robot in Hold.** Keep clear of the steering wheels. Expect motor-driver precharge, one firm contactor sound, automatic steering homing, then a pulsing-yellow ready/idle power-hub indication. Stop for repeated contactor noise, abnormal steering, solid red, flashing amber, smoke, odor, or heat.
6. **Verify the dashboard.** Require no emergency or startup fault; motor nodes 1–4 online; slave node 10, power hub node 14, relay node 15, and rover node 20 online when required; batteries acceptable; CAN at 500 kbit/s; and plausible command/feedback values.
7. **Select the operation below.** Do not enter Manual, arm UVC, or start Auto until every readiness item for that operation is satisfied.

6.1.2 Operation 1 · Manual driving

Use Manual for direct, line-of-sight driving. The operator must continuously see the robot and travel path. Do not drive from the camera image alone, and keep UVC Off and Disarmed unless following the separate UVC procedure.

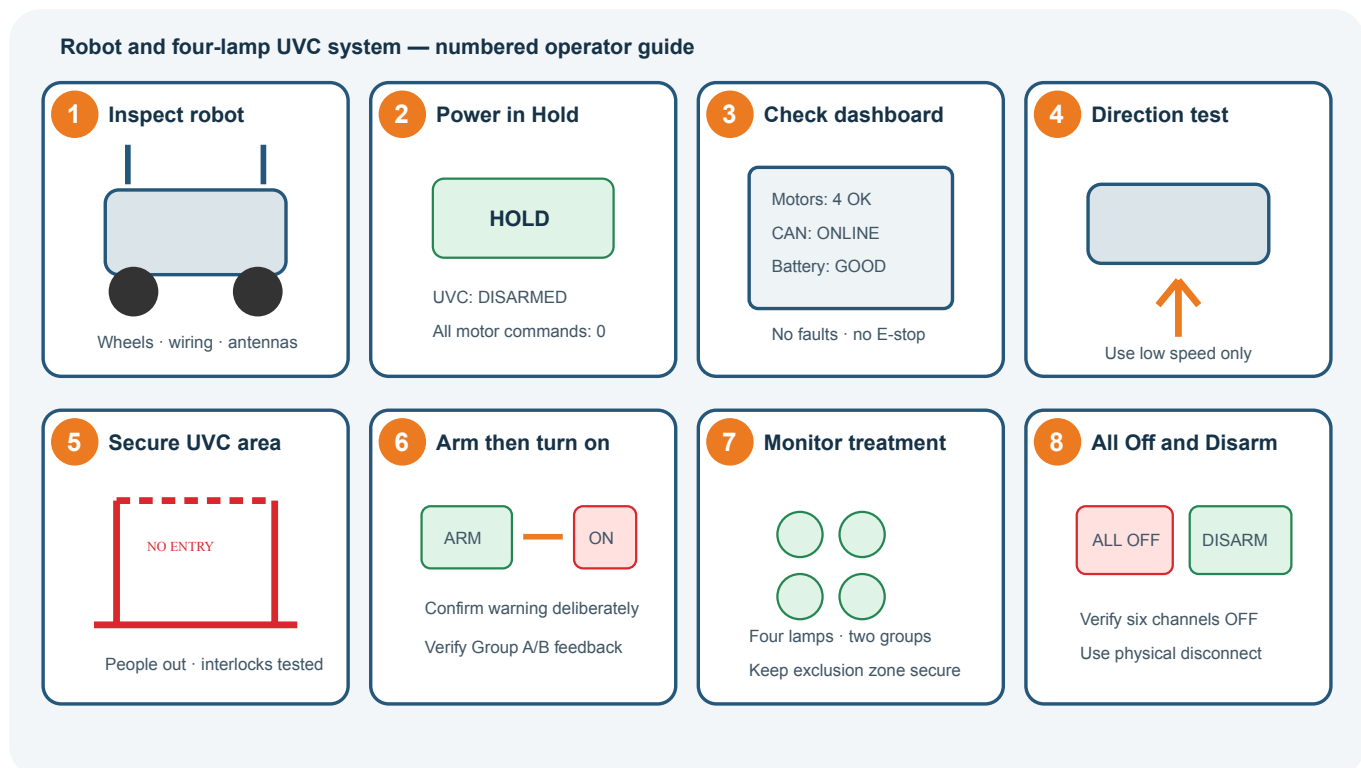
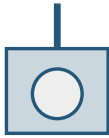


Figure 6.3: Use this system sequence together with the correct handheld picture below.

1. Complete the **Common readiness procedure**. Stand where the complete robot and intended path remain visible. Keep bystanders clear and the emergency stop accessible.
2. Confirm the handheld reports a healthy robot link and **Hold**. “Robot Lost,” Offline, stale data, or an emergency overlay prohibits driving.
3. Request **Manual**: short-press Button 1 on Single Joystick Remote Controller or press SW5 on Dual Joystick Remote Controller. Wait until the handheld and robot dashboard both report Manual.
4. With open space around every wheel, apply a very small forward command. Verify both drive wheels move the robot forward and feedback follows the command. Release the joystick and confirm zero motion.
5. Repeat at low speed for reverse, left steering, and right steering. Stop immediately if direction, steering angle, sound, current, or feedback is wrong.
6. Drive with smooth joystick inputs and walking-speed supervision until the route is proven. Reduce speed before turns, slopes, loose soil, row ends, obstacles, or poor visibility.

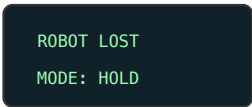
Single Joystick Remote Controller — match each picture to the numbered instruction

1 Inspect and connect



Antenna fitted
Stick centered
No damage

2 Power on in Hold



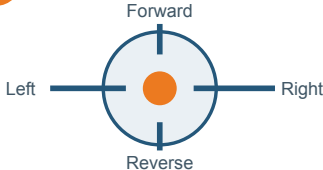
3 Wait for connection



4 Select Manual



5 Test directions slowly



6 Stop and select Hold



7 Open Wi-Fi settings when needed



Network: mars_rc_XXXXXX
Password: alphaagtech
Open: 192.168.10.1

8 Emergency: use physical E-stop

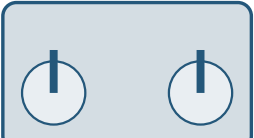


Release the joystick
Press robot E-stop
Do not approach moving robot

Figure 6.4: Single Joystick Remote Controller: Button 1 changes between Hold and Manual; the joystick commands motion. Confirm the displayed mode after every press.

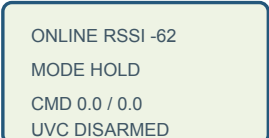
Dual Joystick Remote Controller — numbered operator guide

1 Prepare controls



Left RTK · Middle GPS · Right RC
J1/J2 centered

2 Read Home screen



3 Drive with J2



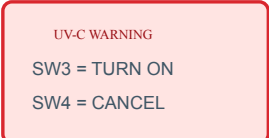
Manual / Hold

4 Navigate pages



Previous / Next
SW3 Select · SW4 Back

5 Confirm UVC warning



6 Use Wi-Fi setup



RC-Controller
rccontroller
192.168.4.1
Keep RC and RTK radio settings separate

Figure 6.5: Dual Joystick Remote Controller: SW5 changes between Hold and Manual. Keep the UVC controls untouched during ordinary driving.

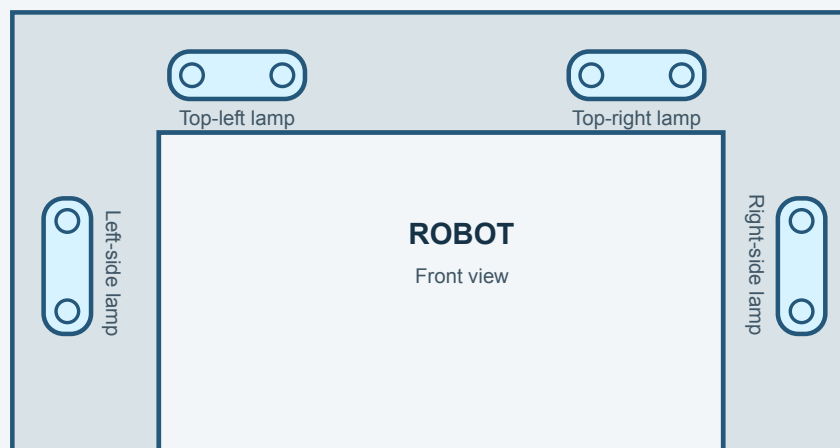
7. Continuously monitor robot link, battery, CAN, motor status, power-hub indication, surrounding people/animals, wheel clearance, and the actual path. A nonzero motion command normally changes the hub indication from pulsing yellow to breathing blue.
8. For a normal stop, center/release the joystick, confirm zero motion, then press Button 1 or SW5 to request **Hold**. Verify Hold on the display and verify the robot is physically stationary.
9. For an unsafe stop, press the physical emergency stop. Do not rely on a webpage, radio command, mode label, or status LED as the only emergency action.
10. When manual work is complete, park on stable ground, select Hold, verify UVC Off/Disarmed, and follow System shut-down.

Manual operation is complete when: the robot is stationary in Hold, the joystick is centered, motion feedback is zero, no new fault is present, and the next operator understands the machine state.

6.1.3 Operation 2 · UVC treatment

UV-C can injure eyes and skin. No person or animal may remain inside the established treatment/exclusion zone. PPE does not permit routine occupancy. All six relay channels are commanded together by the robot; four channels feed the four installed lamp/ballast circuits.

Four UVC lamps · two power branches · all six relay channels switch together



Relay Group A · LEFT

left-side + top-left

Relay Group B · RIGHT

right-side + top-right

Groups A/B identify physical wiring branches; the installed command switches all six relay channels together.

Figure 6.6: Group A contains the left-side and top-left lamps. Group B contains the right-side and top-right lamps. The groups identify the physical power branches even though the installed robot command switches all six relay channels together.

1. **Confirm treatment authority.** Use the approved crop/treatment protocol, target dose, lamp-to-plant distance, robot speed, pass spacing, number of passes, and allowable environmental conditions. The speed calculator is only a planning aid; use radiometer and field validation.
2. **Inspect the UVC hardware with power isolated.** Check four lamps, guards, sockets, ballast wiring, relay connections, two inverters, branch protection, warning indicator, labels, and physical disconnect. Confirm the relay electronics use a power-hub 5 V CAN port.
3. **Identify all four lamp positions.** Confirm left-side, top-left, right-side, and top-right labels and their ballast/relay branches. Reject any missing, cracked, loose, blackened, wet, or incorrectly labelled lamp assembly.
4. **Establish the exclusion zone.** Use the site UV-C assessment for lamp height/orientation, measured irradiance, reflections, crop canopy, travel path, slopes, and nearby access. Remove people and animals; place signs, cones, flagging, gates, or

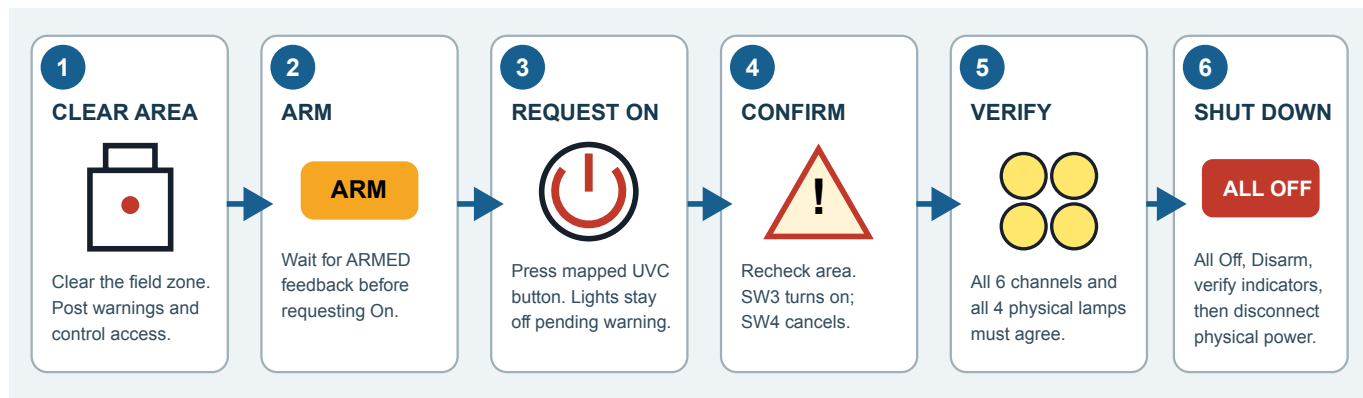


Figure 6.7: Never omit the final physical verification: all four lamps must be visibly Off before anyone approaches.

spotters as required.

5. **Position the robot.** Use Manual driving with UVC Off/Disarmed. Park at the treatment start in Hold, center the joystick, and verify zero motion.
6. **Energize the UVC supply without turning lamps on.** From outside the exclusion zone, close the approved physical UVC supply path. Confirm both inverters are normal, relay node 15 is online, all six applied relay bits are Off, and all four lamps remain physically Off.
7. **Arm deliberately.** Command Arm and confirm Armed. Arming alone must not illuminate any lamp. If a lamp starts, use the physical UVC disconnect and remove the system from service.
8. **Request UVC On.** On Dual Joystick Remote Controller, read the UV-C warning and press SW3 within 10 seconds to confirm; press SW4 to cancel. The robot commands all six relay channels together. Verify the requested and applied status agree.
9. **Verify illumination remotely.** From outside the zone, confirm the left-side, top-left, right-side, and top-right lamps are operating and the external warning indication is active. Never approach a dark-looking lamp while other lamps are energized.
10. **Perform the treatment.** Drive only at the validated speed and geometry. Maintain the specified plant distance and pass spacing. Continuously supervise the exclusion boundary, travel path, speed, robot link, CAN heartbeat, relay feedback, inverter status, lamp condition, and warning indicator.
11. **Stop for any abnormal condition.** Command All Off and use the physical disconnect for entry, unexpected access, robot deviation, loss of link/heartbeat, relay disagreement, inverter alarm, lamp flicker/breakage, obstruction, uncertain speed/dose, or a lamp that does not follow the command.
12. **Finish the pass.** Center the joystick and select Hold. Command All Off; verify all six applied bits Off and all four lamps physically Off. Then Disarm.
13. **Make entry safe.** Open the physical UVC disconnect, apply lockout when inspection is required, wait the site-required cooling/ventilation time, and only then enter with the task-required PPE.
14. **Record the treatment.** Log field/row, date, operator, crop, dose basis, speed, distance, passes, lamp hours, active lamp positions, alarms, interruptions, and any maintenance needed.

UVC operation is complete when: the robot is in Hold, all six relay channels are verified Off, all four lamps and warning indicators are physically Off, UVC is Disarmed, the physical supply is disconnected, and the treatment record is complete.

6.1.4 Operation 3 · Autonomous navigation

Simulation notice: The navigation-console image below is a simulated example. Actual map imagery, robot position, status values, camera content, and interface appearance may differ during operation.

Auto is supervised operation. A trained operator must monitor the physical robot, navigation console, RTK quality, route, and emergency stop throughout the mission. Keep UVC Off/Disarmed during route commissioning; combine UVC with Auto only under an approved, validated site procedure.

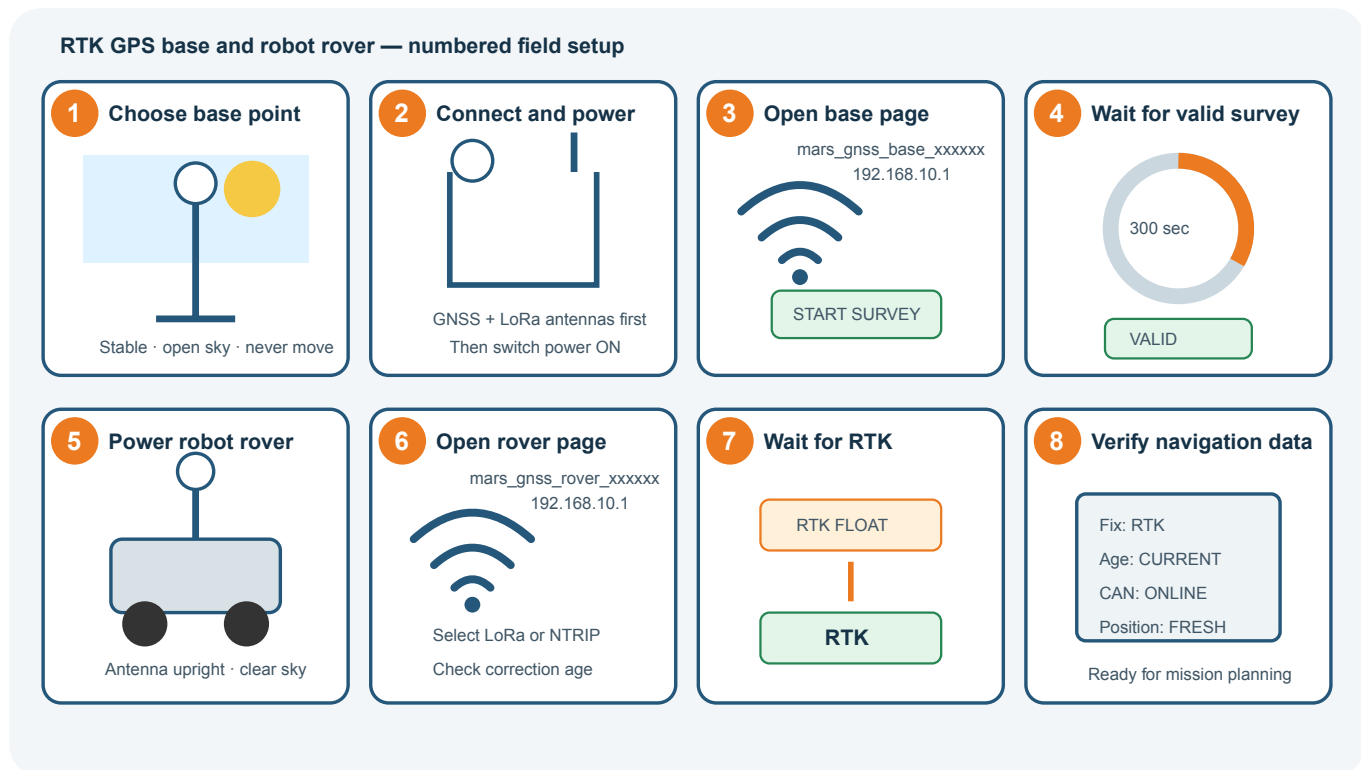


Figure 6.8: Autonomous navigation begins with a valid base, fresh corrections, and a verified rover solution.

A · Prepare localization and system

- Place the RTK base at its recorded point with clear sky. Attach its GNSS and RTK LoRa antennas, power it, and open `mars_gnss_base.local`. Verify Fixed Location or complete the approved survey; do not move the base afterward.
- Verify the base correction output. The base and rover RTK LoRa settings must match each other and must remain separated from the RC LoRa settings.
- Complete the **Common readiness procedure**. Confirm the rover receives 5 V and CAN from its power-hub 5 V CAN port and reports an approved RTK fix, fresh correction age, stable heading, acceptable accuracy, and solution status OK.
- Confirm the navigation controller receives operating power/Ethernet from the PoE switch and uses its assigned power-hub 5 V CAN connection. Configure the operating phone for simultaneous robot Wi-Fi and cellular Internet according to Robot Wi-Fi + cellular Internet. Require web access, loaded Google map tiles, live camera, current map position, and no service fault.
- Manually position the robot at a safe mission start if needed. Return to Hold, center the joystick, verify zero motion, and keep UVC Off/Disarmed.

B · Create and review the mission

- Open the approved navigation address, sign in as Operator, and select **Claim Control**. Confirm no other operator is issuing commands.
- Select **Mission**, then **Draw Border**. Tap at least three points around the permitted operating area. Keep the boundary inside safe terrain and away from ditches, roads, people, structures, water, steep slopes, and unverified crop clearance.
- Select **Draw First Row** and choose points A and B to set the row direction. Enter row count, spacing, and a conservative default speed, then select **Generate**.
- Inspect every straight segment, turn, row transition, waypoint order, start direction, and final stop. Drag or edit points only while navigation is inactive. Add actions such as Take Photo or Stop & Wait only where the physical surroundings allow them.
- Open each critical waypoint and verify latitude, longitude, speed, XY tolerance, yaw tolerance, action, and hold time. Save the approved mission file for records.
- Select **Upload Mission**. Review the confirmation values and keep the geofence enabled. Confirm the robot marker begins inside the permitted boundary and the uploaded waypoint count/order matches the reviewed plan.

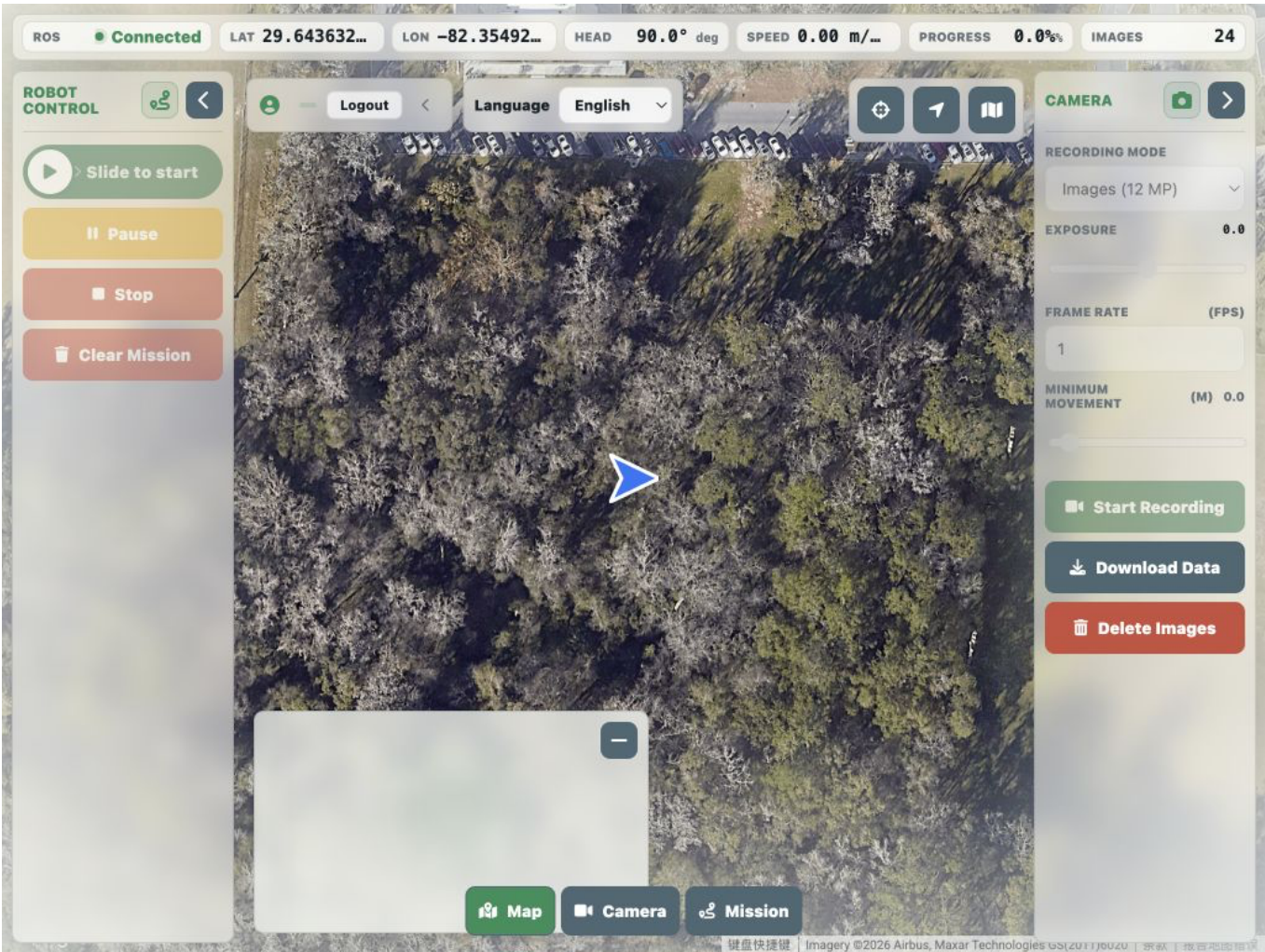


Figure 6.9: Before planning, require ROS Connected, plausible position and heading, current RTK localization, live camera, CAN communication, and a stationary robot in Hold.

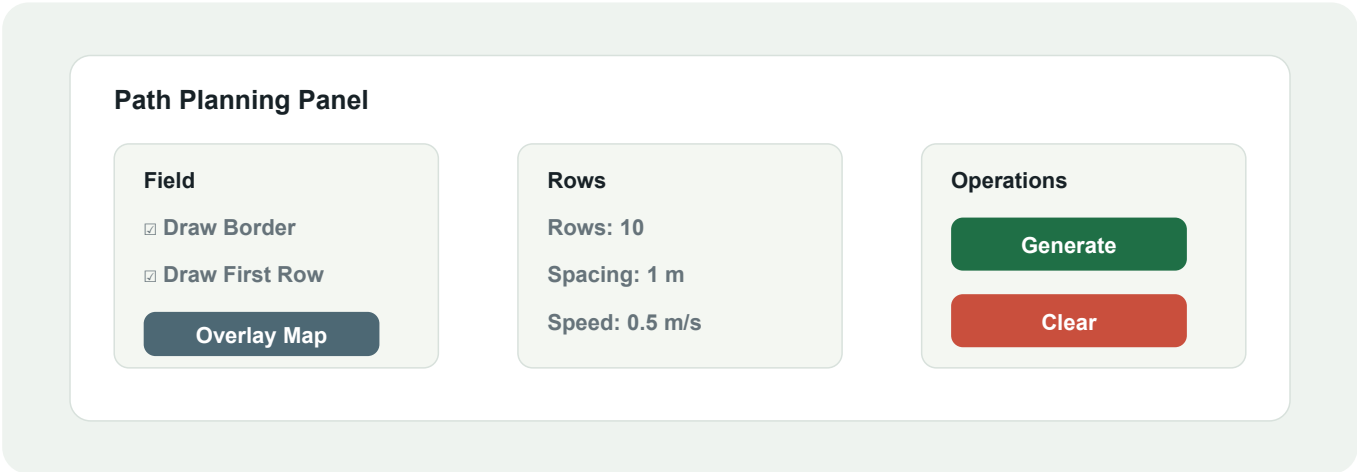


Figure 6.10: Draw the permitted field first, then define row direction and generate or manually add waypoints.

C · Start, supervise, pause, and finish

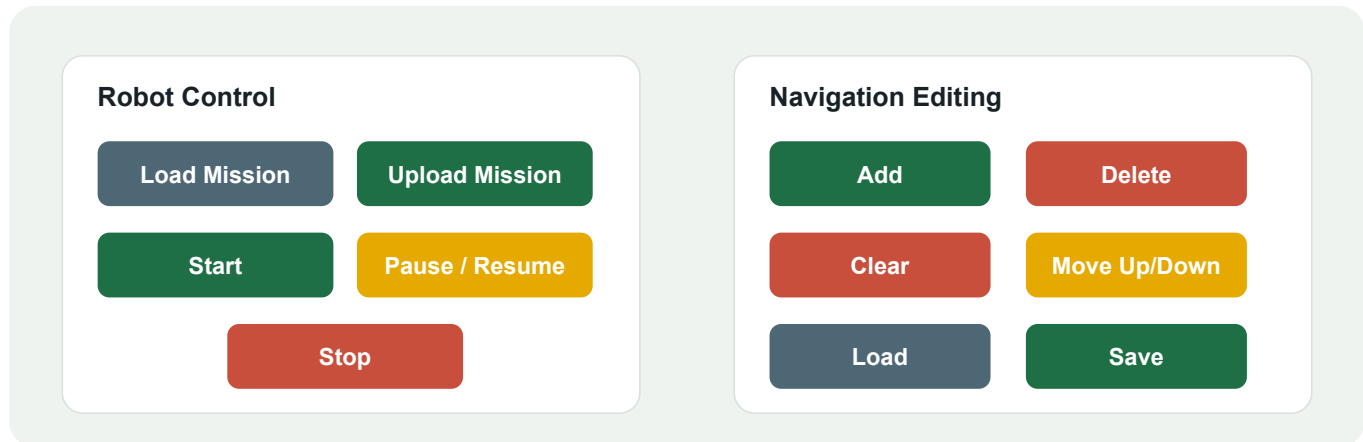


Figure 6.11: Uploading does not move the robot. Motion begins only after the robot reports Auto and the deliberate Slide to Start gesture is completed.

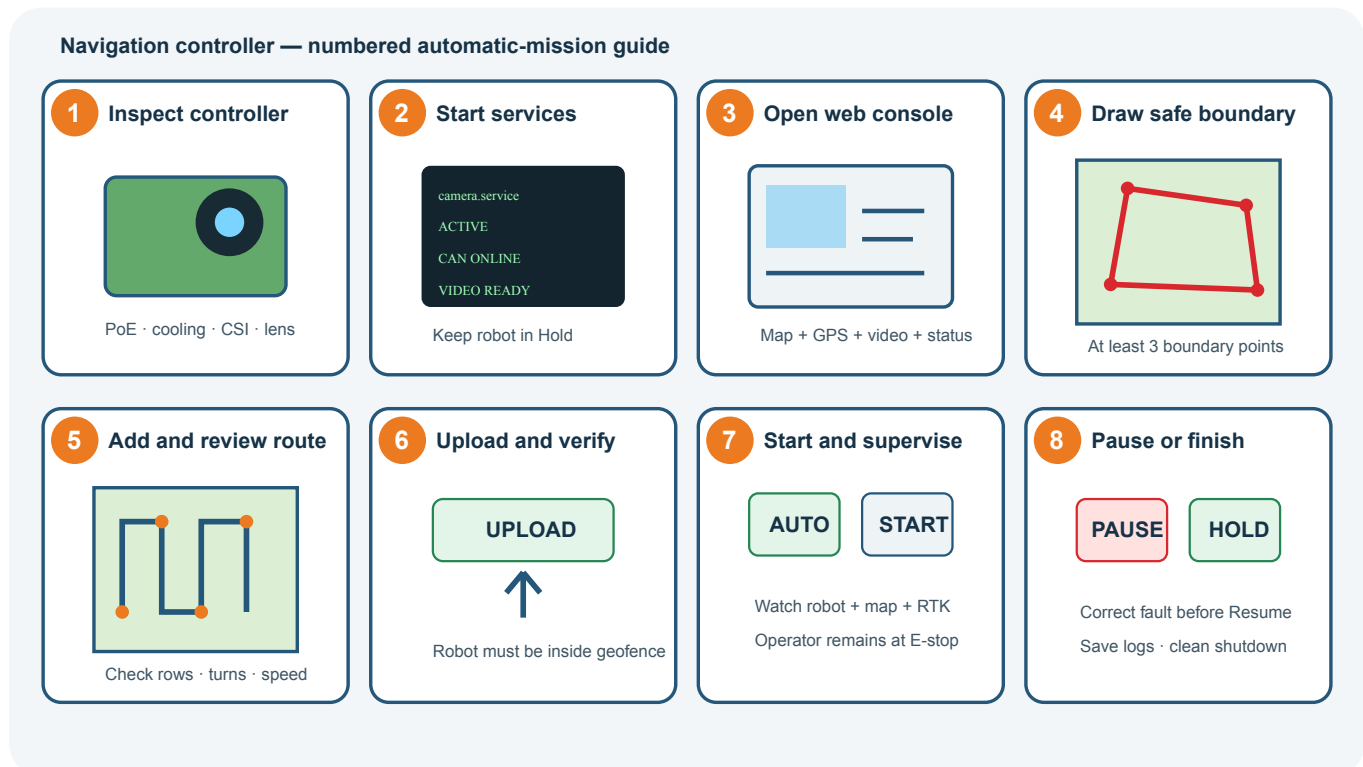


Figure 6.12: Monitor the physical robot and the console together; neither view is sufficient by itself.

12. Walk or otherwise inspect the route, clear the operating area, brief spotters, and place the emergency-stop operator where the robot and access points remain observable.
13. Change the robot mode to **Auto** using the selected remote controller. Follow the applicable Single Joystick Remote Controller or Dual Joystick Remote Controller manual for its physical controls. Wait for both the remote controller and webpage to report Auto; the displayed mode is the confirmation.
14. On the webpage, complete **Slide to Start**. Verify the first movement is slow and directed toward waypoint 1. Press the emergency stop for movement in the wrong direction or any immediate hazard.
15. Continuously compare the physical robot with the map marker, active waypoint, driven track, heading, speed, RTK state/correction age, covariance/localization status, CAN state, battery, camera, geofence, and messages.
16. For a controlled pause, select webpage **Pause** or change the robot mode to **Auto Pause** using the selected remote controller. Confirm the robot physically stops, the webpage reports Paused, and the displayed robot mode is Auto Pause. Identify and

correct the pause reason before continuing.

17. Resume only when the path is clear, localization is valid, the robot is inside the geofence, and the cause has been corrected. Select webpage **Resume** or change the robot mode from Auto Pause back to **Auto** using the selected remote controller; then verify Auto and the next physical motion.
18. End autonomous navigation in either of two ways: select webpage **Stop**, or change the robot from Auto to **Manual** or **Hold** with the handheld. A change to Manual or Hold automatically stops the active navigation mission.
19. After the mode change or webpage Stop, verify physical motion and commanded speed are zero and the console no longer reports Navigating. For joystick control, confirm Manual before moving the joystick; choose Hold when no manual movement is required.
20. At the final waypoint, verify the webpage reports completion, commanded speed is zero, and the robot is physically stopped. Select Hold and release web control or log out.
21. Save the mission result, images/videos, driven track, alarms, pauses, RTK quality, and any route corrections. Then follow System shutdown.

Autonomous operation is complete when: mission execution is stopped or completed, speed command and physical motion are zero, the robot is in Hold, web control is released, UVC remains Off/Disarmed unless separately authorized, and mission records are saved.

6.1.5 System shutdown

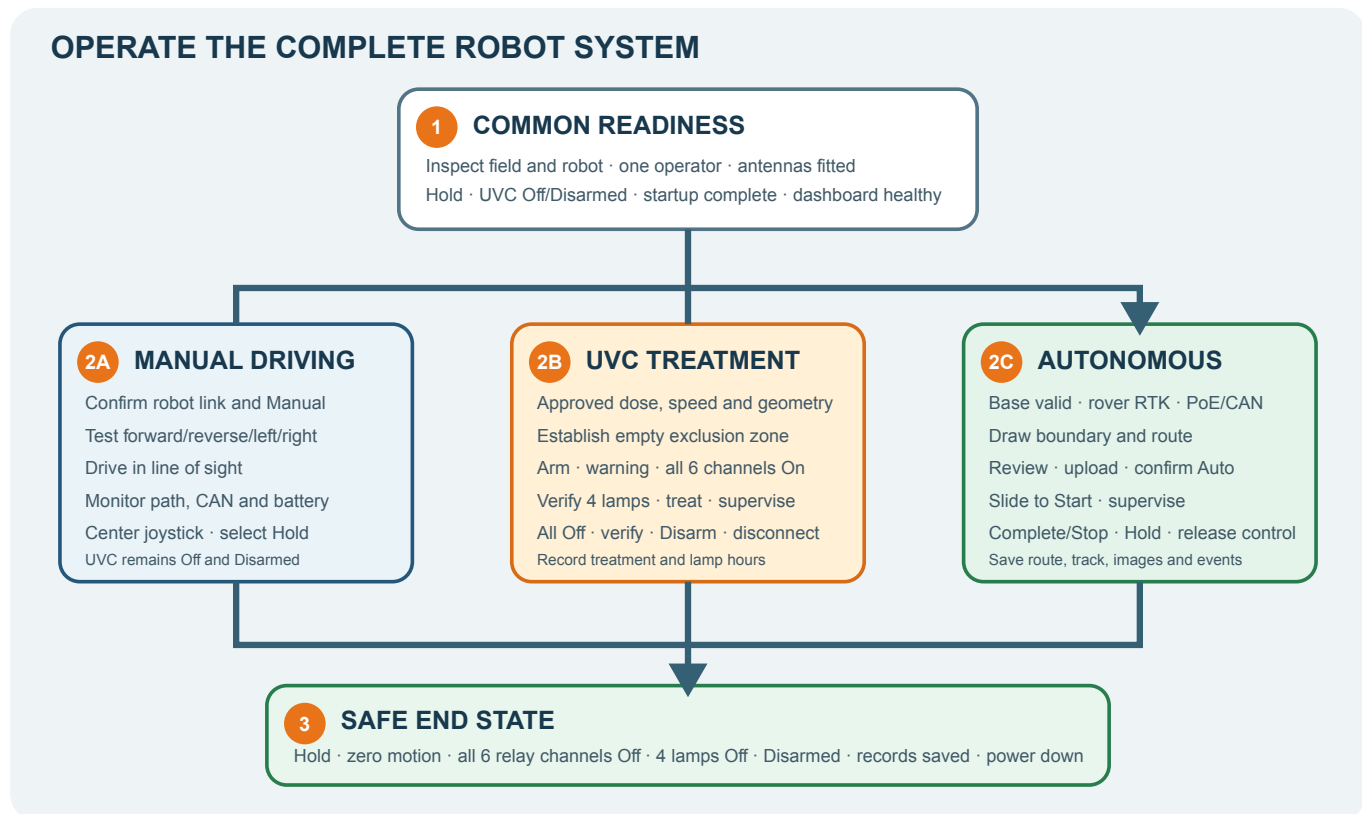


Figure 6.13: Every operation returns through the same safe end state before power is removed.

1. Park on stable ground away from traffic, people, animals, standing water, and combustible material.
2. Stop the mission if active, center the joystick, select Hold, and verify zero command and zero physical motion.
3. Command UVC All Off, verify all six relay channels and four physical lamps Off, Disarm, and open the physical UVC disconnect.
4. Release navigation web control and save required mission, treatment, image/video, fault, and maintenance records.
5. Switch off the robot using the approved order. Remember that the main contactor and stored electrical energy require the specified discharge/lockout procedure before service.
6. Switch off handhelds, base equipment, and accessories as the field plan requires. Protect antennas/connectors and store or charge batteries according to the maintenance procedure.

7. Walk around the stopped system. Check for heat, odor, damage, loose cables, crop entanglement, lamp damage, fluid ingress, or a fault requiring tag-out.

Maintenance

7.1 System maintenance schedule

When	Work
Before every use	Inspect cases, controls, the three Dual Joystick Remote Controller antennas, wheels, steering, batteries, eight side ports, cables, emergency stop, GPS mounts, navigation controller/camera/PoE, lamps, guards, warning indicators and every installed safety device; verify Hold and UVC Off.
After every use	Hold, All Off, Disarm, physical disconnect, approved power-down, clean lenses/displays, record faults, store/charge batteries correctly.
Weekly or after transport	Check joystick calibration, fasteners, antenna cables, contactor, steering home, PoE switch, navigation-controller cooling/storage, camera focus, base mount, rover correction, and the UVC relay's 5 V power/CAN connection and fail-safe.
After software/configuration change	Record new version/settings; restrain robot; test emergency stop, Hold, directions, link loss, CAN, GPS, mission Pause/Resume, geofence, video, All Off, Disarm, and heartbeat shutdown.

7.2 Emergency actions

Unsafe robot movement: release controller, press the physical emergency stop, keep people away, and disconnect power only when safe.

Possible UV-C exposure: do not enter immediately. Command All Off if available, use the physical UVC disconnect, follow site exposure/emergency procedures, and have the system inspected before reuse.

Lost radio or webpage: do not assume the robot or lamps are off. Use physical safety controls and verify the machine directly from a safe location.

1. Protect people first; use the physical emergency control.
2. Do not reset or re-arm until the cause is identified.
3. Inspect actual motor, relay, lamp, and indicator state.
4. Record the event and complete a restrained functional test before returning to service.

7.3 Entire robot system: field-engineer commissioning

Commission from the physical layer upward. Do not troubleshoot application software until power, grounding, antenna, termination, and bus wiring are verified.

Stage	Acceptance criteria
1 · Mechanical	Fasteners secure; wheels/steering free; antennas, camera, and GPS mounts rigid; four UVC fixtures, shielding, and access controls complete.
2 · Power	Correct polarity/voltage; batteries healthy; four battery-power ports and four CAN ports labeled; contactor and branch protection correct; PoE switch powered; navigation controller has no undervoltage; UVC supply independently disconnectable.
3 · Safety	Physical E-stop stops motion independently; both UVC groups Off/Disarm; PPE, interlocks, and indicators verified.
4 · CAN	Every side CAN port verified as 12 V, GND, CANH, CANL; 500 kbit/s ; correct termination; unique node IDs; controllers, motors, batteries, GPS, navigation controller, and relay online without bus errors.
5 · Radio	RC endpoints match each other; RTK endpoints match each other; RC and RTK settings are separated to prevent interference; normal RSSI and safe link-loss response.
6 · Manual motion	Hold at startup; centered zero; correct drive/steering direction; feedback follows low-speed command.
7 · GPS	Base valid/fixed; fresh corrections; rover RTK; position delivered through CAN and ROS.
8 · Navigation	PoE switch/module, navigation-controller services, ROS, CAN, CSI camera, and video healthy; mission/geofence/Pause/Resume/photo and safety gates verified.
9 · UVC	Left-side/top-left Group A and right-side/top-right Group B labels verified; requested/applied/physical states agree; heartbeat/CAN emergency shutdown and warning confirmation verified.
10 · Records	Firmware, commits, node IDs, port pinout, radio/group mapping, lamp hours, base coordinates, calibration, test results, and backups recorded.

7.3.1 Engineering handover package

- Controller, robot, relay, base, and rover firmware versions and approved binaries.
- Navigation-controller OS image, Raspberry Pi 5/CSI/PoE hardware record, `video-stream` commit, build/dependency record, and installed service files.
- Robot-side power/CAN port schedule, contactor and protection diagram, **500 kbit/s** CAN setting, node/COB-ID schedule, bus diagram, and termination locations.
- UVC Group A/B channel schedule, fixture/ballast diagram, four lamp positions, hours, approved dose/speed table, and radiometer calibration record.
- Separate RC and RTK radio settings/channel plan; store passwords separately under site policy.
- Base coordinates/survey method, antenna offsets, and navigation geometry parameters.
- Safety-validation record for E-stop, PPE, interlocks, UVC fail-safe, link loss, command timeout, geofence, and localization pause.